

The Weil Conjecture. II

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English translation

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Introduction

In [1], cited below as I, we proved the Weil conjecture giving the complex absolute value of the eigenvalues of Frobenius acting on the cohomology of a projective smooth variety defined over a finite field. Here we study cohomology with values in a sheaf: the point is to pass from pointwise properties of the sheaf to corresponding properties of its cohomology.

Let therefore X_0 be a scheme of finite type over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a $\mathbb{Q}_\ell$ -sheaf on X_0 . We choose an algebraic closure $\overline{\mathbb{F}}_q$ of $\mathbb{F}_q$, and we indicate by suppressing the subscript 0 the extension of the base field from $\mathbb{F}_q$ to $\overline{\mathbb{F}}_q$; see (0.7). For $x_0 \in |X_0|$ a closed point of X_0 , and $x \in X(\overline{\mathbb{F}}_q)$ above it, one has the Frobenius automorphism F_{x_0} of the fibre $\mathcal{F}_x$ of $\mathcal{F}$ at x (I, (1.11)–(1.13), or (1.18)). We shall call its eigenvalues the eigenvalues of F_{x_0} on $\mathcal{F}_0$. We say that $\mathcal{F}_0$ is pointwise pure of weight n if, for every $x_0 \in |X_0|$, the eigenvalues of F_{x_0} on $\mathcal{F}_0$ are algebraic numbers all of whose complex conjugates have absolute value $N(x_0)^{n/2}$. We say that $\mathcal{F}_0$ is mixed if it is an iterated extension of pointwise pure sheaves; the weights of these pure constituents are the weights of $\mathcal{F}_0$. Our essential result is the following.

Theorem 1 (Theorem 1, (3.3.1)). *Let $f : X_0 \rightarrow S_0$ be a morphism of schemes of finite type over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a mixed sheaf of weights $\leq n$ on X_0 . Then, for every i , the sheaf*

$$R^i f_! \mathcal{F}_0$$

on S_0 is mixed of weights $\leq n + i$.

For $S_0 = \text{Spec}(\mathbb{F}_q)$, this theorem says that, for every eigenvalue α of Frobenius on $H_c^i(X, \mathcal{F})$, there is an integer $m \leq n + i$ (the weight of α) such that all complex conjugates of α have absolute value $q^{m/2}$.

Poincaré duality sometimes makes it possible to complement these upper bounds by lower bounds (3.3.5). For example, if X_0 is proper and smooth, and if $\mathcal{F}_0$ is lisse (“twisted constant”, in another terminology) and pointwise pure of weight n , then the eigenvalues of Frobenius on $H^i(X, \mathcal{F})$ are all of weight $n + i$; we shall also express this by saying that $H^i(X, \mathcal{F})$ is pure of weight $n + i$.

For $\mathcal{F}_0 = \mathbb{Q}_\ell$ (of weight 0), one recovers the main result of I, with the hypothesis “projective and smooth” replaced by “proper and smooth”.

A simple reduction, parallel to the proof of the finiteness theorem for the $R^i f_!$ (cf. SGA 4, XIV), reduces Theorem 1 to the following theorem, together with a local study at infinity of lisse pointwise pure sheaves on a curve.

Theorem 2 (Theorem 2, cf. (3.2.3)). *Let X_0 be a proper smooth curve over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be the inclusion of a dense open subset, and let $\mathcal{F}_0$ be a lisse pointwise pure sheaf of weight n on U_0 . Then $H^i(X, j_* \mathcal{F})$ is pure of weight $n + i$.*

Here are the main lines of the proof.

A. Cleanups.

- (i) If $u : X'_0 \rightarrow X_0$ is a finite surjective morphism whose source is a proper smooth curve, and if primes denote base change by u , then the groups

$$H^i(X, j_* \mathcal{F})$$

are direct factors of

$$H^i(X', j'_* \mathcal{F}').$$

This argument allows one to reduce to the case where the local monodromy of $\mathcal{F}$ at the points of $X - U$ is unipotent.

- (ii) A duality theorem asserts that $H^i(X, j_* \mathcal{F})$ and

$$H^{2-i}(X, j_*(\mathcal{F}^\vee))$$

are in perfect duality, with values in $\mathbb{Q}_\ell(-1)$. This reduces the problem to verifying that the complex conjugates α' of the eigenvalues α of Frobenius on $H^i(X, j_* \mathcal{F})$ have absolute value bounded in the required way. The difficult case is H^1 .

B. Complex embeddings. Let $\alpha \in \overline{\mathbb{Q}_\ell}$ be an eigenvalue of Frobenius on $H^i(X, j_* \mathcal{F})$. It is convenient to reformulate the estimates to be checked as follows: for every isomorphism

$$\iota : \overline{\mathbb{Q}_\ell} \xrightarrow{\sim} \mathbb{C},$$

one must prove an inequality of the form

$$|\iota(\alpha)| \leq q^{(i+n)/2}.$$

In the proof, each isomorphism ι will be treated separately. This leads one to introduce the notions of pointwise ι -pure and ι -mixed sheaves: one looks not at all complex conjugates of the eigenvalues, but only at their images under ι . It is also convenient to allow the weights to be arbitrary real numbers. See (1.2.6). We shall prove Theorem 2 with “pure” replaced by “ ι -pure”. I refer to (1.2.11) the reader who, like me, dislikes the axiom of choice implicit in the use of isomorphisms between $\overline{\mathbb{Q}}_\ell$ and $\mathbb{C}$.

C. Local monodromy of ι -pure sheaves. Put $S_0 = X_0 - U_0$. One begins by proving that, if $\mathcal{F}_0$ is lisse and pointwise ι -pure of weight $\beta \in \mathbb{R}$, the ι -weight

$$w_q(\alpha) = \frac{2 \log |\iota(\alpha)|}{\log q}$$

of an eigenvalue α of F_{x_0} on $j_*\mathcal{F}_0$, for $x_0 \in S_0$, is of the form $\beta - m$, with m an integer ≥ 0 , and one determines m in terms of the local monodromy of $\mathcal{F}_0$ at x_0 (1.8.4). More generally, one determines the corresponding weights of the eigenvalues of F_{x_0} on $\mathcal{F}_0$ in the sense of (1.10.2).

The first step is to prove the necessary upper bound. This is done by using Grothendieck’s formula for the function $Z(U_0, \mathcal{F}_0, t)$: after applying ι , one finds on the left an infinite product converging in the appropriate half-plane, and on the right a rational function whose numerator is essentially

$$\det(1 - Ft, H_c^1(U, \mathcal{F}));$$

one uses the fact that the fibres $(j_*\mathcal{F})_x$ for $x \in S$ contribute to $H_c^2(U, \mathcal{F})$. The second step is to apply this result to tensor powers of $\mathcal{F}_0$ and of its dual, while taking account of the known structure of local monodromy.

Once this is known, it is permissible, and convenient, to study $j_!\mathcal{F}_0$ rather than $j_*\mathcal{F}_0$: if $\mathcal{F}_0$ is lisse and pointwise ι -pure of weight $\beta \in \mathbb{R}$, the point is to show that the eigenvalues α of Frobenius on

$$H_c^1(U, \mathcal{F}) = H^1(X, j_!\mathcal{F})$$

have weight $w_q(\alpha) \leq \beta + 1$. To simplify the notation, we assume $\beta = 0$; one reduces to this case by twisting (1.2.7).

D. Passage to $X_0 \times X_0$. The main geometric idea is to pass from $(U_0, \mathcal{F}_0)$ to

$$(U_0 \times U_0, \mathcal{F}_0 \boxtimes \mathcal{F}_0)$$

and to analyze

$$H^2(U \times U, \mathcal{F} \boxtimes \mathcal{F}) = H^2(X \times X, j_!\mathcal{F} \boxtimes j_!\mathcal{F})$$

by means of a pencil of hyperplane sections of $X_0 \times X_0$, assumed to be in general position, for a suitable projective compactification of $X_0 \times X_0$. We assume that the cleanup A(i) has been carried out.

The aim is to prove that the weight of an eigenvalue α of Frobenius on

$$H^2(X \times X, j_!\mathcal{F} \boxtimes j_!\mathcal{F})$$

satisfies $w_q(\alpha) \leq 2 + \varepsilon$ in the inductive form of the argument. If α is now an eigenvalue of Frobenius on $H^1(X, j_!\mathcal{F})$, the Kunneth formula asserts that α^2 is an eigenvalue of Frobenius on the preceding H^2 , and one obtains an improved bound for $H^1(X, j_!\mathcal{F})$. Iterating this procedure replaces the use of large Cartesian powers in I.

Take a sufficiently general pencil of hyperplane sections of $X_0 \times X_0$. The hyperplane sections of the pencil are the fibres of a morphism

$$f : Y_0 \longrightarrow \mathbb{P}_0^1,$$

where Y_0 is obtained from $X_0 \times X_0$ by blowing up finitely many points. Let $\mathcal{G}_0$ be the inverse image of $j_! \mathcal{F}_0 \boxtimes j_! \mathcal{F}_0$ on Y_0 . The Leray spectral sequence for f reduces, in essence, the study of

$$H^2(X \times X, j_! \mathcal{F} \boxtimes j_! \mathcal{F})$$

to that of

$$H^1(\mathbb{P}^1, R^1 f_* \mathcal{G}).$$

The generalization (1.5.1) of the fundamental estimate I (1.3.2) shows that, over a certain open subset $V_0 \subset \mathbb{P}_0^1$ on which the sheaf $R^1 f_* \mathcal{G}_0$ is lisse, it is a successive extension of lisse pointwise ι -pure sheaves. The hypotheses left from I (1.3.2) are replaced by the requirement that $R^1 f_* \mathcal{G}_0$ be, on V_0 , a direct factor of a lisse ι -real sheaf, that is, a sheaf $\mathcal{H}_0$ such that the polynomials

$$\det(1 - F_x t, \mathcal{H}_0)$$

have real coefficients. The results of Section 2 (see E below) let one deduce this from the fact that $\mathcal{F}_0$ itself is a direct factor of an ι -real sheaf, namely $\mathcal{F}_0 \oplus \mathcal{F}_0^\vee$: because $\mathcal{F}_0$ has weight 0, its dual plays the role of a complex conjugate.

How are the weights of $R^1 f_* \mathcal{G}_0|_{V_0}$ computed? In I (1.3.2), one hypothesis ensured that the sheaf considered had no nontrivial lisse subsheaf, hence was pointwise ι -pure by (1.5.1), and another hypothesis made it possible to compute its weight. In the application I (1.6.3), those hypotheses came from the theory of Lefschetz pencils, more precisely from the conjugacy theorem for vanishing cycles. Here $R^1 f_* \mathcal{G}_0$ has ramification points of three geometrically distinct types, so another argument is needed.

The theory of vanishing cycles makes it possible to control the jump of $R^1 f_* \mathcal{G}_0$ at a point of $\mathbb{P}_0^1 - V_0$ from local information on $X_0 \times X_0$, namely the groups of vanishing cycles. A local calculation, together with the results in C applied to $\mathcal{F}_0$, determines the weights of the Frobenius eigenvalues on those groups of vanishing cycles. These weights turn out to be integers. Applying the results of C to the successive quotients of $R^1 f_* \mathcal{G}_0|_{V_0}$ for a suitable filtration, one shows that $R^1 f_* \mathcal{G}_0$ is a successive extension of lisse sheaves on V_0 which either

- (a) are lisse on all of $\mathbb{P}_0^1$ (with weights not detected by vanishing cycles), or
- (b) have restriction to V_0 pointwise ι -pure of integral weight.

Fortunately, the sheaves in case (a) do not contribute to the group

$$H^1(\mathbb{P}^1, R^1 f_* \mathcal{G})$$

which interests us, since $\mathbb{P}^1$ is simply connected and $H^1(\mathbb{P}^1, \mathbb{Q}_\ell) = 0$.

Which integral weights m appear? If one already knows the theorem with an error $\leq \delta < 1$ and applies it to the fibres of f , it gives an estimate $m < 1 + \delta$; since m is an integer, one even has $m \leq 1$. Applying the theorem, with the same error, to $\mathbb{P}^1$ and to the successive quotients of $R^1 f_* \mathcal{G}_0$ for the filtration introduced above, one obtains the desired estimate for the eigenvalues of Frobenius on $H^1(\mathbb{P}^1, R^1 f_* \mathcal{G})$. At the start, however, one only has the cruder bound, and allowing weight 2 ruins the argument.

E. The weight of an H^1 is < 2 . We prove directly that, for $\mathcal{F}_0$ lisse and pointwise ι -pure of weight 0, the eigenvalues α of Frobenius on $H^1(X, j_* \mathcal{F})$ have weight $w_q(\alpha) < 2$. Applied to the fibre of f at a point of V_0 , this is enough to start the proof. It also separates the Frobenius eigenvalues on $H^1(X, j_* \mathcal{F})$, of weight < 2 , from those on $H^2(X, j_! \mathcal{F})$, of weight 2. Applied to the fibres of f , it is used in proving that $R^1 f_* \mathcal{G}_0$ is, on V_0 , a direct factor of a lisse ι -real sheaf. The proof is inspired by Mertens' proof of the Hadamard–de la Vallée Poussin theorem, namely the nonvanishing of $\zeta(s)$ on the line $\Re(s) = 1$.

From E one can deduce that, if $\mathcal{F}_0$ is a lisse pointwise ι -pure sheaf on a smooth curve U_0 , the fibre $\mathcal{F}_u$ of $\mathcal{F}$ at $u \in |U|$ is a completely reducible representation of the geometric monodromy

group; see (3.4.1)(iii) and (3.4.14). Via the heuristic dictionary of [2], I, this corresponds to the semisimplicity theorem [2], II (4.2.6) in the theory of variations of Hodge structure. A particular case shows that if X is a nonsingular projective variety over an algebraically closed field k , if $(H_t)_{t \in \mathbb{P}^1}$ is a Lefschetz pencil of hyperplane sections of X , with H_t singular for $t \in S$, and if $u \in \mathbb{P}^1 - S$, then the $H^i(H_u, \mathbb{Q}_\ell)$ are completely reducible representations of $\pi_1(\mathbb{P}^1 - S, u)$. A specialization argument reduces one to the case $k = \overline{\mathbb{F}}_q$; then X and the pencil are defined over a finite field, and the required purity hypothesis follows from the Weil conjecture for the H^i . As is well known, the original arguments of Lefschetz allow one to deduce from this complete reducibility the hard Lefschetz theorem for iterated cup product with the cohomology class of a hyperplane section (4.1.1). We also give a variant for cohomology with values in a sheaf, even in a complex (6.2.13).

Other purely geometric statements result from the existence of a formalism of weights: the local invariant-cycle theorem (3.6.1), for instance, and the results announced in [3] concerning the existence of a grading by weight on $\mathbb{Q}_\ell$ -homotopy types (5.3.4). Parallel results are available in Hodge theory, where one can also speak of weights: [12], [9].

Theorem 1 belongs to a formalism that is often far easier to use than I. In (3.7) we show how it simplifies the proofs of the applications given in (1.8). For instructions on its application to estimates of trigonometric sums, I refer to SGA 4 $\frac{1}{2}$ [Sommes trigonometriques], especially nos. 1, 2, 3, and 7.

The essential results of the article concern schemes of finite type over a finite field, with two “buts”. First, they have geometric consequences for a scheme of finite type over the algebraic closure of a finite field, and specialization arguments allow one to pass from there to the case of an arbitrary algebraically closed base field. Second, some results hold for schemes of finite type over $\mathbb{Z}$. They follow immediately from the proofs of the analogous results over a finite field, but not from their statements. We shall omit them from the section-by-section review below.

In (1.1) we recall what a constructible $\mathbb{Q}_\ell$ -sheaf is, with a few variants. We also give a formalism for a corresponding derived category. In (1.2) we define pointwise pure, ι -pure, mixed sheaves, and so on, and state in a more precise form a conjecture according to which, on a scheme of finite type over $\mathbb{F}_q$, every constructible $\mathbb{Q}_\ell$ -sheaf is ι -mixed. Sections (1.3) to (1.5) are devoted to the announced generalization of I (1.3.2). In (1.3), Grothendieck’s theorem I (1.3.8) is used to define the determinantal weights of a lisse sheaf $\mathcal{F}_0$ on a smooth curve U_0 . They depend only on the highest exterior powers of the constituents of $\mathcal{F}_0$, and control the H^1 (and hence the H^2) of every sheaf obtained from $\mathcal{F}_0$ by passing to a tensor space. Proposition (1.3.13), innocent in appearance, will play the role played in I (1.3.2) by the classical invariant theory of the symplectic group.

Sections (1.6) to (1.8) are devoted to point C above in the proof. Section (1.6) contains algebraic preliminaries: the theory of a nilpotent endomorphism of a vector space. Section (1.7) contains geometric preliminaries: how local monodromy supplies nilpotent endomorphisms. The multidimensional end of (1.7) will be used only in (1.9), which itself is unnecessary for the rest of the article.

Section (1.9) is an application of (1.8) to the local monodromy of a mixed lisse sheaf on the complement of a normal-crossing divisor along which it is tamely ramified. An analogous result for variations of Hodge structure has just been obtained by Cattani and Kaplan.

In (1.10), the methods of (1.6)–(1.8) are applied to non-archimedean absolute values. The semicontinuity result (1.10.7) has made it possible to prove that, for a general complete intersection in characteristic p , the Newton polygon (defined by the slopes of crystalline cohomology) coincides with the Hodge polygon. That this problem, of p -adic nature, could be solved only by ℓ -adic methods is due to the present absence of a good theory of vanishing cycles in p -adic cohomology.

Section (1.11), on specialization of monodromy, is technical and contains no surprises.

Section 2 contains part E above of the proof. In (3.5), by known arguments, one deduces

from it an equidistribution theorem, of which one special case is the analogue, for a function field, of the Sato–Tate conjecture concerning the distribution of Frobenius angles for an elliptic curve over K .

Section (3.3) contains the proof of Theorem 1 and some corollaries. Section (3.2) gives the proof of Theorem 2, and (3.1) gives the calculation of vanishing cycles used above. In (3.4), Theorem 1 is applied to the study of the successive extensions that define a mixed sheaf. In particular, one proves that a mixed lisse sheaf admits a “weight filtration” by lisse subsheaves, and that the geometric monodromy of a pointwise pure lisse sheaf on a normal scheme is semisimple. Section (3.6) gives the local invariant-cycle theorem; (3.7) gives simplifications in the proofs of (1.8).

Section (4.1) contains the proof of the hard Lefschetz theorem. Sections (4.2) and (4.3) provide complements to SGA 7. In (4.3), the use of an ad hoc topos allows one to transpose the Morse-theoretic-looking arguments used by Lefschetz. In (4.4), one proves that the geometric monodromy group of the vanishing part of the cohomology of the hyperplane sections of a Lefschetz pencil (assumed sufficiently general in the wild characteristic-two case) is Zariski dense in an orthogonal or symplectic group, or is a Weyl group. The last case is studied in detail. In (4.5), one deduces the “gcd theorem” used in [7] to compare the ℓ -adic and crystalline cohomologies of projective smooth varieties.

In Section 5, the yoga of weights is applied to the $\mathbb{Q}_\ell$ -homotopy type. The definition of the latter makes essential use of the construction, by Miller and Grothendieck, of a graded differential (anti-)commutative algebra attached to a simplicial set and allowing one to compute its integral cohomology.

Finally, Section 6 develops the formalism of mixed sheaves. With proofs inspired by SGA 4 $\frac{1}{2}$ [Th. finitude], it is shown that their category is remarkably stable. One also defines a notion of pure complex, generalizing that of a lisse pointwise pure sheaf on a smooth scheme (6.2.4), (6.2.5). The direct image Rf_* by a proper morphism carries pure complexes to pure complexes. For them one generalizes the local invariant-cycle theorem and, under dimension hypotheses, the global invariant-cycle theorem and the hard Lefschetz theorem.

Notations and conventions

(0.1). Throughout this article, unless the contrary is explicitly stated, we fix a prime number ℓ , and we consider only noetherian separated algebraic spaces on which ℓ is invertible. We shall simply call them schemes.

The reader who dislikes algebraic spaces may replace, in (0.1), the words “algebraic space” by “scheme”.

(0.2). We fix an algebraic closure $\overline{\mathbb{Q}_\ell}$ of the field $\mathbb{Q}_\ell$ of ℓ -adic numbers. We denote by ι an isomorphism of fields from $\overline{\mathbb{Q}_\ell}$ to $\mathbb{C}$; see (1.2.11). Exception: in (1.10), the same letter ι will denote an isomorphism of fields from $\overline{\mathbb{Q}_\ell}$ to an algebraic closure $\overline{\mathbb{Q}_{\ell'}}$ of the field of ℓ' -adic numbers. The word sheaf will mean, according to the section, “sheaf for the étale topology”, “constructible $\overline{\mathbb{Q}_\ell}$ -sheaf”, or “Weil sheaf”. See (1.1.5), (1.3.2).

If $f : X \rightarrow Y$ is a morphism, and $\mathcal{F}$ is a sheaf on Y , we shall often write, when no ambiguity can arise,

$$H^*(X, \mathcal{F})$$

for $H^*(X, f^*\mathcal{F})$. The same convention applies to hypercohomology, with or without supports.

(0.3). A geometric point $\bar{x}$ of X is a morphism from the spectrum of an algebraically closed field - denoted $k(\bar{x})$ - to X . It is said to be localized at $x \in X$ if x is its image in X . It is called algebraic if $k(\bar{x})$ is an algebraic closure of $k(x)$.

Every geometric point defines an algebraic geometric point: replace $k(\bar{x})$ by the algebraic closure of $k(x)$ in $k(\bar{x})$. We shall sometimes use this construction to extend tacitly to geometric points terminology introduced for algebraic geometric points.

Every point x of X whose residue field is separably closed defines a geometric point $\bar{x}$: take the perfect closure of $k(x)$. We shall often identify x and $\bar{x}$.

If $f : X \rightarrow Y$ is a morphism, every geometric point $\bar{x}$ of X defines by composition a geometric point $f(\bar{x})$ of Y , sometimes denoted simply by $f\bar{x}$.

(0.4). Unless explicitly stated otherwise, locally means locally for the étale topology.

If x is a point of a scheme X , we write $X_{(x)}$ for the henselization of X at x , i.e. the spectrum of the henselization of the local ring of X at x . If $\bar{x}$ is a geometric point of X , we write $X_{(\bar{x})}$ for the strict localization, or strict henselization, of X at x . Its residue field is the separable closure of $k(x)$ in $k(\bar{x})$.

(0.5). A smooth curve over a field k is a smooth scheme purely of dimension one over k .

Let X be a scheme of finite type over a field k . If X is connected, its structural morphism $X \rightarrow \text{Spec}(k)$ admits a unique factorization

$$X \longrightarrow \text{Spec}(k') \longrightarrow \text{Spec}(k)$$

with k' a finite separable extension of k and X/k' geometrically connected. In general, each connected component of X admits such a factorization.

In particular, this gives a mechanical procedure for reducing properties of smooth curves over finite fields to those of absolutely irreducible smooth curves over finite fields. We shall often use it tacitly, passing from one formulation to the other.

(0.6). A trait is the spectrum of a discrete valuation ring V . It is henselian if V is henselian, and strictly local if it is henselian and its residue field is separably closed. The expression “a trait (S, η, s) ” means a trait S with closed point s and generic point η . The expression “a trait $(S, \bar{\eta}, s, \bar{s})$ ” means a trait (S, η, s) , together with a geometric point $\bar{s}$ localized at s , and with a generic geometric point $\bar{\eta}$ of the strict localization $S_{(\bar{s})}$.

(0.7). We shall always denote by p a prime number distinct from ℓ , by q a power of p , by $\mathbb{F}_q$ a finite field with q elements, and by F an algebraic closure of $\mathbb{F}_q$ (or of $\mathbb{F}_p$, if $\mathbb{F}_q$ has not been introduced).

The following conventions will often be in force. The fields $\mathbb{F}_q$ and F are fixed. Objects over $\mathbb{F}_q$ - schemes, or sheaves on $\mathbb{F}_q$ -schemes - are denoted with a subscript 0. Suppressing this subscript indicates extension of scalars from $\mathbb{F}_q$ to F . For instance, if $\mathcal{F}_0$ is a sheaf on the $\mathbb{F}_q$ -scheme X_0 , then $\mathcal{F}$ denotes its inverse image on $X = X_0 \otimes_{\mathbb{F}_q} F$.

Geometric points $\text{Spec}(K) \rightarrow X_0$ of X_0 will always be assumed to be defined by a geometric point of X ; that is, K is assumed to be endowed with an $\mathbb{F}_q$ -morphism $F \rightarrow K$.

(0.8). If X is a scheme of finite type over a field or over $\mathbb{Z}$, we denote by $|X|$ the set of its closed points. With the notation of (0.7), if X_0 is of finite type over $\mathbb{F}_q$, each point $x \in |X|$ defines a geometric point, again denoted x , of X_0 . We shall simply speak of the geometric point $x \in |X|$.

(0.9). In this article the point of view will be more Galoisian than in I; this will often lead us to write F where F^* appeared in I.

(0.10). A filtration F of an object V of an abelian category is called finite if there exist n and m such that $F^n V = V$ and $F^m V = 0$.

(0.11). We write $\varinjlim$ and $\varprojlim$ for inductive and projective limits. Quotation marks indicate a limit taken in a category of ind-objects or pro-objects; thus “ $\varinjlim X_i$ ” denotes the ind-object defined by the inductive system of the X_i (assumed filtered).

We write $:=$ for an equality whose right-hand side is the definition of the left-hand side.

1. Purity

(1.1) ℓ -adic sheaves

(1.1.1). We recall the definition of a constructible $\overline{\mathbb{Q}}_\ell$ -sheaf on a scheme X as in (0.1), following SGA 5, VI (1.4.2).

- a) Let A be a noetherian torsion ring. A sheaf of A -modules $\mathcal{F}$ is called constructible if there exists a finite partition of X into locally closed subsets X_i such that the restrictions $\mathcal{F}|_{X_i}$ are locally constant, with fibres finitely generated A -modules.
- b) Let R be a noetherian local ring of residual characteristic ℓ , and let $\mathfrak{m}$ be its maximal ideal. We assume that R is complete for the $\mathfrak{m}$ -adic topology. The category of constructible R -sheaves is the 2-projective limit of the categories of sheaves of R/I -modules, for I an open ideal of R .

More explicitly, a constructible R -sheaf $\mathcal{F}$ on X is a projective system of sheaves of R -modules, indexed by open ideals I of R , such that:

- (i) $\mathcal{F}_I$ is annihilated by I , and is constructible as a sheaf of R/I -modules;
- (ii) for $I \supset J$, the transition morphism from $\mathcal{F}_J$ to $\mathcal{F}_I$ induces an isomorphism

$$\mathcal{F}_J \otimes_{R/J} R/I \xrightarrow{\sim} \mathcal{F}_I.$$

The map $n \mapsto \mathfrak{m}^n$ from the positive integers to the set of open ideals of R is cofinal. Thus one may replace the projective limit over I by a projective limit over n , with R/I replaced by $R/\mathfrak{m}^n$.

The functor

$$\mathcal{F} \mapsto \varprojlim'' \mathcal{F}_I$$

from constructible R -sheaves to pro-sheaves of R -modules is fully faithful. We shall again call any pro-sheaf in its essential image a constructible R -sheaf. This leads one to abandon the notation $\mathcal{F}_I$ and to write instead

$$\mathcal{F} \otimes_R R/I.$$

Justification: for $\mathcal{F}$ as above, the projective system in J of the $\mathcal{F}_J \otimes_R R/I$ is essentially constant, with value $\mathcal{F}_I$.

We say that $\mathcal{F}$ is lisse (in another terminology, twisted constant) if the sheaves $\mathcal{F} \otimes_R R/I$ are locally constant.

- c) Let E be a finite extension of $\mathbb{Q}_\ell$, and let R be the integral closure of $\mathbb{Z}_\ell$ in E . The category of constructible E -sheaves is the quotient of the category of constructible R -sheaves by the thick subcategory of torsion sheaves. Equivalently:
 - (i) There is an essentially surjective functor

$$\mathcal{F} \mapsto \mathcal{F} \otimes_R E$$

from constructible R -sheaves to constructible E -sheaves.

(ii) One has

$$\mathrm{Hom}(\mathcal{F} \otimes_R E, \mathcal{G} \otimes_R E) = \mathrm{Hom}(\mathcal{F}, \mathcal{G}) \otimes_R E.$$

A constructible E -sheaf is lisse if it is locally of the form $\mathcal{F} \otimes_R E$, with $\mathcal{F}$ lisse.

d) For F a finite extension of E , there is a functor

$$\mathcal{F} \longmapsto \mathcal{F} \otimes_E F$$

from constructible E -sheaves to constructible F -sheaves, and

$$\mathrm{Hom}(\mathcal{F} \otimes_E F, \mathcal{G} \otimes_E F) = \mathrm{Hom}(\mathcal{F}, \mathcal{G}) \otimes_E F.$$

For an iterated extension, one has the canonical isomorphism

$$(\mathcal{F} \otimes_E F) \otimes_F K \simeq \mathcal{F} \otimes_E K. \quad (1.1.1.1)$$

The category of constructible $\overline{\mathbb{Q}}_\ell$ -sheaves is the $\overline{\mathbb{Q}}_\ell$ -linear 2-inductive limit of the categories of constructible E -sheaves, for finite extensions $E \subset \overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}_\ell$ becoming larger and larger. In other words:

(i) for $E \subset \overline{\mathbb{Q}}_\ell$ finite over $\mathbb{Q}_\ell$, there is an E -linear functor

$$\mathcal{F} \longmapsto \mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$$

from constructible E -sheaves to constructible $\overline{\mathbb{Q}}_\ell$ -sheaves, inducing isomorphisms on Hom after tensoring with $\overline{\mathbb{Q}}_\ell$;

(ii) there are canonical isomorphisms compatible with (1.1.1.1);

(iii) every constructible $\overline{\mathbb{Q}}_\ell$ -sheaf is of the form $\mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$ for suitable E and $\mathcal{F}$.

A constructible $\overline{\mathbb{Q}}_\ell$ -sheaf is lisse if it is locally of the form $\mathcal{F} \otimes_E \overline{\mathbb{Q}}_\ell$ with $\mathcal{F}$ lisse.

For these definitions to be usable, one needs the usual formalism for such sheaves. Such a formalism is partially constructed in SGA 5, V and VI. In those expositions, Jouanolou proves that constructible $\mathbb{Z}_\ell$ -sheaves on X form a noetherian abelian category, gives the relation between lisse constructible $\mathbb{Z}_\ell$ -sheaves and representations of the fundamental group, proves that constructible $\mathbb{Z}_\ell$ -sheaves are lisse on the strata of a stratification of X , defines tensor products and inverse images, and defines and studies the direct image functors with proper support $R^i f_!$ for $f : X \rightarrow Y$ a morphism of finite type between noetherian schemes.

Thanks to SGA 4 $\frac{1}{2}$ [finiteness], the theory also applies to the functors $R^i f_*$ and $R^i f_!$ when f is a morphism of finite type between schemes of finite type over an excellent regular scheme of dimension ≤ 1 . It likewise applies to the sheaves of vanishing cycles $R^i \Psi$ and $R^i \Phi$.

(1.1.2). In Section 6 we shall work essentially in the derived category of a category of ℓ -adic sheaves. This theory is not yet fully in place. We shall use the following framework.

Let R be the ring of integers of a finite extension E_λ of $\mathbb{Q}_\ell$, and let $\mathfrak{m}$ be its maximal ideal. If X is a scheme, define

$$D^-(X, R) := 2\text{-}\varprojlim_n D^-(X, R/\mathfrak{m}^n).$$

An object K of $D^-(X, R)$ is a projective system represented by complexes $K_n \in \mathrm{Ob} D^-(X, R/\mathfrak{m}^n)$ and by isomorphisms

$$K_{n+1} \overset{L}{\otimes}_{R/\mathfrak{m}^{n+1}} R/\mathfrak{m}^n \xrightarrow{\sim} K_n$$

in $D^-(X, R/\mathfrak{m}^n)$. The complex K_n will rather be denoted

$$K \otimes_R^L R/\mathfrak{m}^n.$$

We shall be especially interested in the subcategory $D_c^b(X, R)$ consisting of those $K \in \text{Ob } D^-(X, R)$ such that $K \otimes_R^L R/\mathfrak{m}$ is bounded with constructible cohomology. Then, for every n , $K \otimes_R^L R/\mathfrak{m}^n$ belongs to $D_{ctf}^b(X, R/\mathfrak{m}^n)$: bounded complexes of finite Tor-dimension whose cohomology sheaves are constructible.

The following remarks justify the use we shall make of $D_c^b(X, R)$.

- a) If K is in $D_c^b(X, R)$, the sheaves

$$\mathcal{H}^i K := \varprojlim_n \mathcal{H}^i(K \otimes_R^L R/\mathfrak{m}^n)$$

are constructible R -sheaves. The proof is reduced, by cutting up X , to the case where the cohomology sheaves of $K \otimes_R^L R/\mathfrak{m}^n$ are locally constant; then it is enough to examine their fibres at points, where one applies the argument of SGA 5, XV.

- b) Conversely, if $\mathcal{F}$ is a torsion-free constructible R -sheaf, the projective system of complexes concentrated in degree zero with term $\mathcal{F} \otimes_R R/\mathfrak{m}^n$ belongs to $D_c^b(X, R)$. We shall not need to consider the case in which $\mathcal{F}$ has torsion.
- c) Under suitable finiteness hypotheses, for instance when working with schemes of finite type over a regular scheme S of dimension ≤ 1 , the categories $D_{ctf}^b(X, R/\mathfrak{m}^n)$ are stable under the four operations Rf_* , f^* , $Rf_!$, $Rf^!$, and under the internal operations $\otimes^L$ and $R\mathcal{H}om$. Moreover these operations commute with reduction modulo $\mathfrak{m}^n$. They therefore extend trivially to $D_c^b(X, R)$. The same applies to the functors $R\Psi$ and $R\Phi$ of the theory of vanishing cycles.
- d) If, for all K, L in $D_c^b(X, R)$, the groups

$$\text{Hom}(K \otimes_R^L R/\mathfrak{m}^n, L \otimes_R^L R/\mathfrak{m}^n)$$

are finite, then $D_c^b(X, R)$ is a triangulated category, with distinguished triangles those whose reductions modulo $\mathfrak{m}^n$ are distinguished for all n .

In general, it is better to replace triangles by filtered complexes. One defines $DF_{ctf}^b(X, R/\mathfrak{m}^n)$ as the derived category of bounded filtered complexes, with finite filtration, whose associated graded object has finite Tor-dimension. Then one puts

$$DF_c^b(X, R) := 2\text{-}\varprojlim_n DF_{ctf}^b(X, R/\mathfrak{m}^n).$$

Whenever one would like to consider a triangle in $D_c^b(X, R)$, one can do better: construct an object of $DF_c^b(X, R)$ whose filtration has only two steps, the two components of the associated graded object being the vertices of the desired triangle and the underlying complex being the third vertex. By abuse of language we shall sometimes call such objects triangles and write them as short exact sequences.

- e) The definition of truncation functors $\tau_{\leq a}$ is delicate. If K is a complex of sheaves, $\tau_{\leq a} K$ is the complex

$$\cdots \rightarrow K^{a-1} \rightarrow \text{Ker}(d^a) \rightarrow 0 \rightarrow \cdots$$

This functor carries quasi-isomorphisms to quasi-isomorphisms and hence passes to the derived category, but it does not preserve finite Tor-dimension. The difficulty is bypassed

by a modified construction using the images of cohomology modulo sufficiently high powers of $\mathfrak{m}$. This construction uses in an essential way the fact that R is regular of dimension one.

Variante 1 ((1.1.3)). *Let $D_c^b(X, E_\lambda)$ be the category obtained from $D_c^b(X, R)$ by extension of scalars from R to E_λ : there is an essentially surjective functor*

$$D_c^b(X, R) \longrightarrow D_c^b(X, E_\lambda)$$

and

$$\mathrm{Hom}(K, L) \otimes_R E_\lambda \xrightarrow{\sim} \mathrm{Hom}(K \otimes_R E_\lambda, L \otimes_R E_\lambda).$$

One then puts

$$D_c^b(X, \overline{\mathbb{Q}}_\ell) := 2\text{-}\varinjlim_{E_\lambda \subset \overline{\mathbb{Q}}_\ell} D_c^b(X, E_\lambda).$$

(1.1.4). In the sequel we shall freely use, for $\mathbb{Z}_\ell$ -, $\mathbb{Q}_\ell$ -, or $\overline{\mathbb{Q}}_\ell$ -sheaves, the theorems known for constructible sheaves of $R/\mathfrak{m}^n$ -modules. The expositions SGA 5, V, VI, XV, and the preceding remarks justify our arguments; whenever a difficulty arises, it will be pointed out.

(1.1.5). In this section and the next, we shall simply say sheaf for constructible $\overline{\mathbb{Q}}_\ell$ -sheaf. This convention will be superseded by (1.3.2).

(1.1.6). An ℓ -adic representation of a profinite group π (or, more generally, of an extension of a finitely generated discrete group by a profinite group) on a $\overline{\mathbb{Q}}_\ell$ -vector space V is a homomorphism

$$\rho : \pi \rightarrow \mathrm{GL}(V)$$

such that there exist a finite extension E of $\mathbb{Q}_\ell$ in $\overline{\mathbb{Q}}_\ell$ and an E -structure V_E on V for which ρ factors through a continuous homomorphism from π to $\mathrm{GL}(V_E)$. Representations of π in an algebraic group over $\overline{\mathbb{Q}}_\ell$ are defined similarly.

For X connected and pointed by a geometric point $\bar{x}$, there is the usual relation between lisse sheaves and continuous representations of the fundamental group $\pi_1(X, \bar{x})$: the equivalence of categories sending a lisse sheaf to its fibre. We shall transport to sheaves the familiar terminology for representations. Whether or not X is connected, a lisse sheaf $\mathcal{F}$ on X is called simple or irreducible if it is nonzero and has no nontrivial lisse subsheaf; it is semisimple if it is a sum of simple sheaves. A Jordan–Holder series of $\mathcal{F}$ is a finite filtration of $\mathcal{F}$ by lisse subsheaves whose graded pieces are irreducible or zero. The constituents of $\mathcal{F}$ are the nonzero graded pieces for such a series, and the semisimplification of $\mathcal{F}$ is the direct sum of its constituents. Constituents and semisimplification are defined up to nonunique isomorphism.

(1.1.7). Let k be a finite field with q elements, and let $\bar{k}$ be an algebraic closure of k . The Frobenius substitution $\varphi \in \mathrm{Gal}(\bar{k}/k)$ is the automorphism $x \mapsto x^q$ of $\bar{k}$. The geometric Frobenius $F \in \mathrm{Gal}(\bar{k}/k)$ is the inverse of φ . The Weil group $W(\bar{k}/k)$ is the subgroup of $\mathrm{Gal}(\bar{k}/k)$ formed by the integral powers of F . It is isomorphic to $\mathbb{Z}$, and $\mathrm{Gal}(\bar{k}/k)$ is its profinite completion. We shall often identify $W(\bar{k}/k)$ with $\mathbb{Z}$, and $\mathrm{Gal}(\bar{k}/k)$ with $\widehat{\mathbb{Z}}$, by the isomorphism sending F to 1.

Specialize (1.1.6) to $X = \mathrm{Spec}(k)$. If $\mathcal{F}$ is a sheaf on $\mathrm{Spec}(k)$, its fibre $\mathcal{F}_{\bar{k}}$ at the geometric point $\mathrm{Spec}(\bar{k})$ is a $\overline{\mathbb{Q}}_\ell$ -vector space on which $\mathrm{Gal}(\bar{k}/k)$ acts by transport of structure. This action is continuous: the map $n \mapsto F^n$ extends by continuity to a homomorphism $\widehat{\mathbb{Z}} \rightarrow \mathrm{GL}(\mathcal{F}_{\bar{k}})$. Equivalently, the eigenvalues of F are ℓ -adic units. The functor $\mathcal{F} \mapsto \mathcal{F}_{\bar{k}}$ is an equivalence from the category of sheaves on $\mathrm{Spec}(k)$ to the category of $\overline{\mathbb{Q}}_\ell$ -vector spaces equipped with an automorphism F whose eigenvalues are ℓ -adic units.

(1.1.8). Let X be a scheme of finite type over $\mathbb{Z}$, $\mathcal{F}$ a sheaf on X , and $y : \text{Spec}(k) \rightarrow X$ a point of X with values in a finite field k . Let $\bar{k}$ be an algebraic closure of k . The inverse image of $\mathcal{F}$ on $\text{Spec}(k)$ falls under (1.1.7), so $\text{Gal}(\bar{k}/k)$ acts on $\mathcal{F}_{\bar{y}}$. By abuse of notation we write

$$\det(1 - F_y t, \mathcal{F})$$

for the corresponding determinant, and we speak similarly of the eigenvalues of F_y acting on $\mathcal{F}$.

If X is connected and pointed by a geometric point $\bar{x}$, one has a conjugacy class of maps

$$\text{Gal}(\bar{k}/k) = \pi_1(\text{Spec}(k), \text{Spec}(\bar{k})) \longrightarrow \pi_1(X, \bar{x}).$$

We write F_y for the image of geometric Frobenius by any of these maps. For $\mathcal{F}$ lisse, defined by a representation V of $\pi_1(X, \bar{x})$, one has

$$\det(1 - F_y t, \mathcal{F}) = \det(1 - F_y t, V),$$

which justifies the notation. In the special case where y is a closed point of X , this is simply the morphism of inclusion $\text{Spec}(k(y)) \rightarrow X$.

(1.1.9). When studying schemes of finite type over $\mathbb{F}_p$, it is often convenient to replace Galois groups and sheaves by the Weil groups and Weil sheaves defined below. The reader should note that the definition of Weil sheaves is relative to the choice of an algebraic closure F of $\mathbb{F}_p$.

Definition 1 ((1.1.10)). *With the notation of (0.7):*

- (i) A Weil sheaf $\mathcal{F}_0$ on X_0 consists of a sheaf $\mathcal{F}$ on X , together with an action of $W(F/\mathbb{F}_q)$ on $(X, \mathcal{F})$ inducing on $X = X_0 \otimes_{\mathbb{F}_q} F$ the action deduced from the action of $W(F/\mathbb{F}_q)$ on F .
- (ii) If $\bar{x}$ is a geometric point of X , the Weil group $W(X_0, \bar{x})$ is the inverse image of $W(F/\mathbb{F}_q)$ by the natural map

$$\pi_1(X_0, \bar{x}) \longrightarrow \pi_1(\text{Spec}(\mathbb{F}_q), \bar{x}) = \text{Gal}(F/\mathbb{F}_q).$$

Thus the topology on

$$W(X_0, \bar{x}) = \pi_1(X_0, \bar{x}) \times_{\text{Gal}(F/\mathbb{F}_q)} W(F/\mathbb{F}_q)$$

is the product topology; the subgroup

$$\pi_1(X, \bar{x}) = \text{Ker}(\pi_1(X_0, \bar{x}) \rightarrow \pi_1(\text{Spec}(\mathbb{F}_q), \bar{x}))$$

is open and closed. Replacing $\mathbb{F}_q$ by $\mathbb{F}_p$ does not change either the category of Weil sheaves or the Weil group.

(1.1.11). The inverse image on X of a sheaf on X_0 is endowed, by transport of structure, with an action of $\text{Gal}(F/\mathbb{F}_q)$. This identifies sheaves on X_0 with particular Weil sheaves: those for which the action of $W(F/\mathbb{F}_q)$ extends to a continuous action of $\text{Gal}(F/\mathbb{F}_q)$.

We shall admit for Weil sheaves the analogue of Grothendieck's results for ordinary sheaves. The generalization is easy, using (1.3.14).

(1.1.12). If $\tilde{X}$ is the universal covering of X_0 , pointed by $\bar{x}$, then there is an equivariant morphism, relative to the natural morphism $W(X_0, \bar{x}) \rightarrow W(F/\mathbb{F}_q)$,

$$(\tilde{X}, \text{ with action of } W(X_0, \bar{x})) \longrightarrow (X, \text{ with action of } W(F/\mathbb{F}_q)).$$

The inverse image on $\tilde{X}$ of a Weil sheaf is therefore endowed with an action of $W(X_0, \bar{x})$. If the sheaf is lisse, its fibre at $\bar{x}$ identifies with its global sections on $\tilde{X}$, giving a fibre functor

$$\{\text{lisse Weil sheaves}\} \longrightarrow \{\text{representations of } W(X_0, \bar{x})\}.$$

Here "representation" is defined as in (1.1.6). Standard arguments show that, if X_0 is connected, this functor is an equivalence of categories. As in (1.1.6), it allows us to transfer the language of representations to lisse Weil sheaves.

(1.1.13). Suppose X_0 is connected. For $y \in |X_0|$ (or more generally for y a point of X_0 with values in a finite field), the Frobenius elements F_y lie in $W(X_0, \bar{x})$; the image of F_y in $\text{Gal}(F/\mathbb{F}_q)$ is the integral power $q^{\deg y}$ of the Frobenius substitution. The element F_y of $W(X_0, \bar{x})$ is well-defined up to conjugacy.

We shall call degrees the arrows to the right in the standard diagram connecting $\pi_1(X, \bar{x})$, $W(X_0, \bar{x})$, and $W(F/\mathbb{F}_q) \simeq \mathbb{Z}$. One has $\deg F_y = \deg y$. The degree maps are surjective if X_0 is geometrically connected. In general, if the largest finite subfield of $\Gamma(X_0, \mathcal{O}_{X_0})$ has degree n over $\mathbb{F}_q$, their images are $n\mathbb{Z}$ and $n\widehat{\mathbb{Z}}$.

(1.1.14). Let $\mathcal{F}$ be a sheaf on X . If $F : X \rightarrow X$ is the Frobenius endomorphism deduced by scalar extension from the endomorphism $x \mapsto x^q$ of X_0 , and if $\text{id}_{X_0} \times F : X \rightarrow X$ is the action of geometric Frobenius, then, as in SGA 5, XV, there is a canonical isomorphism

$$(\text{id}_{X_0} \times F)^* \mathcal{F} \simeq F^* \mathcal{F}.$$

For a Weil sheaf $\mathcal{F}_0$ on X_0 , one deduces a Frobenius correspondence $F^* \mathcal{F} \rightarrow \mathcal{F}$, and the functor

$$\mathcal{F}_0 \longmapsto (\mathcal{F}, F^*)$$

is an equivalence from the category of Weil sheaves to the category of sheaves $\mathcal{F}$ on X endowed with an isomorphism $F^* \mathcal{F} \xrightarrow{\sim} \mathcal{F}$.

For constructible sheaves of $\mathbb{Z}/\ell^n$ -modules, or for constructible $\mathbb{Z}_\ell$ -sheaves, an analogous result holds. This is not so for $\mathbb{Q}_\ell$ - or $\overline{\mathbb{Q}}_\ell$ -sheaves: for example, take $\mathcal{F} = \overline{\mathbb{Q}}_\ell$ and let F^* be multiplication by an element of $\overline{\mathbb{Q}}_\ell^\times$ which is not an ℓ -adic unit.

The induced morphism

$$H^i(X, \mathcal{F}) \xrightarrow{F^*} H^i(X, F^* \mathcal{F}) \rightarrow H^i(X, \mathcal{F})$$

coincides with the action of $F \in W(F/\mathbb{F}_q)$ on cohomology by transport of structure. The same holds for cohomology with compact supports.

(1.1.15). We call $\pi_1(X_0, \bar{x})$ the arithmetic fundamental group, and $\pi_1(X, \bar{x})$ the geometric fundamental group. If X_0 is normal, these are profinite groups. If $\mathcal{F}_0$ is a lisse Weil sheaf on X_0 , its geometric monodromy group is the image of $\pi_1(X, \bar{x})$ in $\text{GL}(\mathcal{F}_{\bar{x}})$.

(1.2) Weights

Definition (1.2.1). Let q be a power of a prime number, and let $n \in \mathbb{Z}$. A number α is said to be *pure of weight n , relative to q* , if it is algebraic and all of its complex conjugates have absolute value $q^{n/2}$.

In what follows, “number” will mean “element of $\overline{\mathbb{Q}}_\ell$ ”.

Definition (1.2.2). Let X be a scheme of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be a sheaf on X .

- (i) We say that $\mathcal{F}$ is *pointwise pure* if there exists an integer n , the weight of $\mathcal{F}$, such that, for every $x \in |X|$, the eigenvalues of F_x are pure of weight n relative to $N(x)$.
- (ii) We say that $\mathcal{F}$ is *mixed* if it admits a finite filtration whose successive quotients are pointwise pure sheaves. The weights of those quotients which are nonzero are the pointwise weights of $\mathcal{F}$.

According to these definitions, the zero sheaf is pointwise pure and, for every n , has weight n . It is mixed, with empty set of weights.

Variant (1.2.3). If k is a finite field, sheaves on $\mathrm{Spec}(k)$ identify with ℓ -adic representations of $\mathrm{Gal}(\bar{k}/k)$. We transport the terminology of (1.2.2) to them.

Variant (1.2.4). The same terminology applies to Weil sheaves (1.1.10), and also to $\overline{\mathbb{Q}}_\ell$ -vector spaces endowed with an action of Frobenius, as far as the variant (1.2.3) is concerned.

Stabilities (1.2.5).

- (i) The category of pointwise pure sheaves of weight n is stable under passage to a quotient, to a subsheaf, by extension, by inverse image, and by direct image under a finite morphism.
- (ii) The tensor product of two pointwise pure sheaves of weights n and m is pointwise pure of weight $n + m$. The dual of a lisse pointwise pure sheaf of weight n is pointwise pure of weight $-n$.
- (iii) The category of mixed sheaves is stable under the operations of (i), and under tensor product.
- (iv) Finally, note that $\overline{\mathbb{Q}}_\ell(1)$ is pointwise pure of weight -2 .

(1.2.6). To show that $\alpha \in \overline{\mathbb{Q}}_\ell$ is pure of weight n , it suffices to verify that for every isomorphism ι from $\overline{\mathbb{Q}}_\ell$ to $\mathbb{C}$, the complex number $\iota\alpha$ has absolute value $q^{n/2}$. The number α is then automatically algebraic; otherwise α could be any transcendental number. Throughout what follows, our arguments will concern each isomorphism ι separately. This is why, although the important concepts are those of (1.2.2), the following terminology will be useful to us.

Let ι be an isomorphism of fields from $\overline{\mathbb{Q}}_\ell$ to $\mathbb{C}$. If q is a power of a prime number, and $\alpha \in \overline{\mathbb{Q}}_\ell$, put

$$w_q(\alpha) := 2 \log_q |\iota\alpha|; \quad (1.2.6.1)$$

this is the ι -weight of α , relative to q . For every real number β , a sheaf $\mathcal{F}$ is called *pointwise ι -pure of weight β* if, for every $x \in |X|$ and every eigenvalue α of F_x acting on $\mathcal{F}$, one has $w_{N(x)}(\alpha) = \beta$. We say that $\mathcal{F}$ is *ι -mixed* if it admits a finite filtration whose successive quotients are pointwise ι -pure. The weights of those of these quotients which are nonzero are the pointwise ι -weights of $\mathcal{F}$. One similarly transposes the variants (1.2.3) and (1.2.4). The analogue of (1.2.5) remains true. Note that, for k a finite field, a representation V of $W(\bar{k}/k)$ is automatically ι -mixed.

(1.2.7) Torsion. For $b \in \overline{\mathbb{Q}}_\ell^*$, we shall denote by $\overline{\mathbb{Q}}_\ell^{(b)}$ any Weil sheaf on $\mathrm{Spec}(\mathbb{F}_p)$ of rank one for which F acts by multiplication by b . For b an ℓ -adic unit, this is an ordinary sheaf. If a choice has been made of an algebraic closure $\bar{F}$ of $\mathbb{F}_p$, it will be normalized by the choice of an isomorphism between its geometric fibre at $\bar{F}$ and $\overline{\mathbb{Q}}_\ell$. For X a scheme of characteristic p (that is, over $\mathbb{F}_p$), $\mathcal{F}$ a sheaf on X , and b an ℓ -adic unit (respectively, X of finite type over $\mathbb{F}_p$, $\mathcal{F}$ a Weil sheaf, and $b \in \overline{\mathbb{Q}}_\ell^*$), we write $\mathcal{F}^{(b)}$ for the tensor product of $\mathcal{F}$ by the inverse image of $\overline{\mathbb{Q}}_\ell^{(b)}$; we say that $\mathcal{F}^{(b)}$ is deduced from $\mathcal{F}$ by torsion. The sheaf $\overline{\mathbb{Q}}_\ell^{(b)}$ is pointwise ι -pure of weight $w_p(b)$; the torsion $\mathcal{F} \mapsto \mathcal{F}^{(b)}$ therefore increases weights by $w_p(b)$.

Remark (1.2.8). In the definitions of pointwise pure sheaves and mixed sheaves, we have required the weights to be integers. In the definitions of pointwise ι -pure sheaves and ι -mixed sheaves, it is convenient to allow them to be arbitrary real numbers. The deep theorem (3.4.1) shows a posteriori that, for X of finite type over $\mathbb{F}_p$, this generality is largely illusory: every ι -mixed sheaf is a direct sum of sheaves deduced by torsion (1.2.7) from ι -mixed sheaves of integral weights, and every sheaf which is ι -mixed for every ι is a direct sum of sheaves deduced by torsion from mixed sheaves.

Every lisse sheaf is a successive extension of irreducible lisse sheaves, and we shall see in (1.3.6) that every lisse sheaf on X normal, connected, and of finite type over $\mathbb{F}_p$ is obtained by torsion from a sheaf whose determinant (the maximal exterior power) is defined by a finite-order character of the fundamental group of X . Therefore the following conjecture

Conjecture (1.2.9). On X of finite type over $\mathbb{F}_p$, every sheaf is ι -mixed, is a consequence of the more precise conjecture (1.2.10) below. One of our main tools, the theorem (1.5.1), is a partial result in the direction of (1.2.9).

Conjecture (1.2.10). Let X be normal, connected, and of finite type over $\mathbb{F}_p$, and let $\mathcal{F}$ be an irreducible lisse sheaf whose determinant is defined by a finite-order character of the fundamental group.

(i) $\mathcal{F}$ is pure of weight 0.

(ii) There exists a number field $E \subset \overline{\mathbb{Q}}_\ell$ such that, for $x \in |X|$, the polynomial

$$\det(1 - F_x t, \mathcal{F})$$

has coefficients in E .

(iii) For λ a non-archimedean place of E prime to p , the inverse roots in $\bar{E}_\lambda$ of the polynomial $\det(1 - F_x t, \mathcal{F})$, that is, the eigenvalues of Frobenius, are λ -adic units.

(iv) For λ dividing p , the valuation of the inverse roots verifies

$$|v(\alpha)/v(Nx)| \leq \frac{1}{2} \operatorname{rg} \mathcal{F}.$$

(v) For a suitable E (perhaps larger than in (ii)), and for each non-archimedean place λ prime to p , there exists an E_λ -sheaf compatible with $\mathcal{F}$, that is, with the same Frobenius eigenvalues.

(vi) For λ dividing p , one hopes for small crystalline companions.

The parts (i), (iii), and (iv) of the conjecture follow from the special case in which X is a curve. For $\mathcal{F}$ of rank 2 on a curve, Drinfeld's work reduces the conjecture, except for (vi) at present, to control of the constant of the functional equation of the L -function attached to $\mathcal{F}$; it is a question of $L(\mathcal{F}, \chi)$, with χ and its twists. In particular, if $\mathcal{F}$ belongs to an infinite compatible system of λ -adic representations, then (i) through (v) are verified.

Remark (1.2.11). I do not claim to believe in the existence of isomorphisms between $\overline{\mathbb{Q}}_\ell$ and $\mathbb{C}$; they are only a convenience of exposition. Whenever we prove that a number is pure, an easy fragment of the proof would suffice to establish that it is algebraic. For the rest of the arguments, it would then suffice to consider the complex embeddings of the subfield of $\overline{\mathbb{Q}}_\ell$ formed by the algebraic numbers, and this does not require the axiom of choice.

Definition (1.2.12). Let X be a scheme of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be a lisse sheaf on X . We say that $\mathcal{F}$ is *totally real* if, for every $x \in |X|$, the coefficients of the characteristic polynomial of F_x acting on $\mathcal{F}$ are totally real algebraic numbers.

The following variant will be useful to us in equal characteristic.

Variant (1.2.13). We say that $\mathcal{F}$ is ι -real if, for every $x \in |X|$, the polynomial

$$\iota \det(1 - F_x t, \mathcal{F})$$

has real coefficients.

Remark (1.2.14). Every lisse pointwise pure sheaf $\mathcal{F}$ is a direct factor of a totally real lisse pointwise pure sheaf, namely, if it is of weight n , the sheaf $\mathcal{F} \otimes \mathcal{F}^\vee(-n)$. Similarly, every lisse pointwise ι -pure sheaf of integral weight is a direct factor of a lisse pointwise ι -pure ι -real sheaf.

In these variants, one may take $\mathcal{F}$ to be a Weil sheaf. For Weil sheaves, every lisse pointwise ι -pure sheaf of arbitrary weight is a direct factor of a lisse pointwise ι -pure ι -real sheaf.

(1.3) Determinantal weights

The conventions of (0.7) are in force. Let X_0 be a normal geometrically connected scheme of finite type over $\mathbb{F}_q$, and let $\bar{x}$ be a geometric point of X_0 . In this section we give consequences of the following theorem.

Theorem (1.3.1). The image of the geometric fundamental group $\pi_1(X, \bar{x})$ in the quotient of $W(X_0, \bar{x})$ by the closure of its derived group is the product of a finite group of order prime to p by a pro- p group.

This theorem is well known, but I have not been able to find it in the literature. For our principal results, only the case in which X is a curve matters. We give below a proof in that case. The general case will be treated in (1.11.4).

Let $\bar{X}_0$ be the smooth projective completion of X_0 , and let $S = \bar{X}_0 - X_0$. Let k be the function field of X_0 , let k_v be its completion at a place v (identified with a closed point $v \in |\bar{X}_0|$), let r_v be the group of units of k_v , and let $k_{\mathbb{A}}^*$ be the idèle group of k . Class-field theory identifies $W(X_0, \bar{x})^{\text{ab}}$ with the idèle-class group

$$\left(\prod_{v \notin S} r_v \right) \backslash k_{\mathbb{A}}^* / k^*,$$

the projective limit of the divisor-class groups

$$C_m = \{\text{divisors on } \bar{X}_0\} / \{\text{those of the form } (\varphi), \varphi \equiv 1 \pmod{m}\}$$

for a “modulus” m which becomes larger and larger and is concentrated on S . The image of the geometric fundamental group is obtained by replacing $k_{\mathbb{A}}^*$ by the subgroup $k_{\mathbb{A}}^1$ of idèles of norm 1, that is, by replacing the C_m by the subgroups C_m^0 of divisor classes of degree 0. The kernel of the map

$$\left(\prod_{v \notin S} r_v \right) \backslash k_{\mathbb{A}}^1 / k^* \longrightarrow \left(\prod_v r_v \right) \backslash k_{\mathbb{A}}^1 / k^*$$

is a quotient of $\prod_{v \in S} r_v$, hence is the product of a pro- p group by a finite group; and the image group

$$\prod_v r_v \backslash k_{\mathbb{A}}^1 / k^*$$

of divisor classes of degree 0, for linear equivalence, is finite (Weil, BNT IV, Th. 7). Indeed the Picard variety $\text{Pic}^0(X)$ is of finite type, and this divisor-class group is the group of rational points over $\mathbb{F}_q$ of $\text{Pic}^0(X)$.

(1.3.2). In the remainder of this article, we shall simply call sheaves the Weil sheaves of (1.1.10). Constructible $\mathbb{Q}_\ell$ -sheaves will be called étale sheaves.

This convention will sometimes lead us to apply to $\bar{\mathbb{Q}}_\ell$ -sheaves results proved for $\mathbb{Q}_\ell$ -sheaves. The justification will always be immediate.

(1.3.3). Let $\mathcal{F}_0$ be a sheaf of rank 1 on X_0 . The group $W(X_0, \bar{x})$ acts on $\mathcal{F}_{\bar{x}}$ by homotheties, through a character

$$\chi : W(X_0, \bar{x}) \longrightarrow \overline{\mathbb{Q}}_\ell^*.$$

Conversely, for any such character χ , the equivalence (1.1.12) gives us a sheaf $\mathcal{L}_0(\chi)$ endowed with a trivialization $\mathcal{L}(\chi)_{\bar{x}} \simeq \overline{\mathbb{Q}}_\ell$.

The character χ factors through the largest abelian quotient of $W(X_0, \bar{x})$. By (1.3.1), the geometric monodromy group of $\mathcal{F}_0$, that is, the image under χ of $\pi_1(X, \bar{x})$, is therefore the product of a finite group by a pro- p group. This image is also a compact subgroup of E^* , for $E \subset \overline{\mathbb{Q}}_\ell$ a finite extension of $\mathbb{Q}_\ell$. It is therefore the product of a finite group by a pro- ℓ group and, since $p \neq \ell$, it is a finite group. In particular, some power χ^n ($n > 0$) of χ is trivial on $\pi_1(X, \bar{x})$, that is, is of the form

$$w \longmapsto b^{\deg(w)}.$$

If c is an n -th root of b , the character χ is the product of the character $w \mapsto c^{\deg(w)}$ by a character z of order n , that is, with values in the cyclic group of n -th roots of unity of $\overline{\mathbb{Q}}_\ell$. In particular, for $y \in |X_0|$, one has

$$|\iota\chi(F_y)| = |\iota c|^{\deg(y)};$$

the sheaf $\mathcal{F}_0$ is pointwise ι -pure, of weight the ι -weight of c relative to q (1.2.6.1). We have proved:

Proposition (1.3.4). Let $\mathcal{F}_0$ be a lisse sheaf of rank 1 on X_0 , and let $\chi : W(X_0, \bar{x}) \rightarrow \overline{\mathbb{Q}}_\ell^*$ be the corresponding character.

- (i) The geometric monodromy group of $\mathcal{F}_0$ is finite. More precisely, χ is the product of a character $w \mapsto c^{\deg(w)}$ by a character of finite order.
- (ii) $\mathcal{F}_0$ is pointwise ι -pure, of weight the weight of c relative to q .

Variant. If X_0 is a curve, the proof of (1.3.1) is complete - independent of (1.11) - and hence already gives (1.3.4). Here, following N. Katz, is how to deduce the general case of (1.3.4) from this particular case.

Let $E \subset \overline{\mathbb{Q}}_\ell$ be as above, and let M be the group of elements of finite order in E^* , that is, the roots of unity. It is a finite group. First one shows that, whatever $x, y \in |X_0|$ may be, one has

$$\chi(F_x)^{1/\deg(x)} = \chi(F_y)^{1/\deg(y)} \quad \text{in } E^* \otimes \mathbb{Q}$$

(written multiplicatively): by joining x and y by a curve, one reduces to the case in which X_0 is a curve and applies (1.3.4). For every zero-cycle of degree 0, $\sum n(x)x$, one therefore has $\chi(\prod F_x^{n(x)}) \in M$. It follows from Chebotarev's theorem that these products of Frobenius elements are dense in the geometric π_1 , which χ therefore sends into M : the restriction of χ to the geometric π_1 has finite order, and one concludes as above.

Definition (1.3.5). Let $\mathcal{F}_0$ be a lisse sheaf on X_0 . The determinantal weights (relative to ι) of $\mathcal{F}_0$ are the numbers

$$\frac{1}{d}(\iota\text{-weight of } \wedge^d \mathcal{F}'_0)$$

where $\mathcal{F}'_0$ runs through the constituents of $\mathcal{F}_0$, and d is its rank.

It is clear that a lisse sheaf pointwise ι -pure of weight β is also purely of determinantal weight β .

(1.3.6). If $\mathcal{F}_0$ is lisse and irreducible of rank n , and if $b \in \overline{\mathbb{Q}}_\ell^*$ is such that, for some $y \in |X_0|$,

$$\det(F_y) = b^{n \deg(y) [\mathbb{F}_q : \mathbb{F}_p]},$$

then the sheaf $\mathcal{F}_0^{(b^{-1})}$ of (1.2.7) is of determinantal weight 0 relative to every ι : its maximal exterior power is defined by a character of finite order.

(1.3.7). To study the behaviour of determinantal weights under tensor product and exterior power, it will be convenient to pass from the language of sheaves to that of representations of algebraic monodromy groups. The dictionary is inspired by Serre [11], II (1.3).

Let therefore $\mathcal{F}_0$ be a lisse sheaf on X_0 . Let G^0 be the algebraic subgroup of $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ which is the Zariski closure of the geometric monodromy group (1.1.15). By push-out from the exact sequence

$$0 \rightarrow \pi_1(X, \bar{x}) \rightarrow W(X_0, \bar{x}) \rightarrow \mathbb{Z} \rightarrow 0,$$

one defines an algebraic group G (not of finite type), extension of $\mathbb{Z}$ by G^0 :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi_1(X, \bar{x}) & \longrightarrow & W(X_0, \bar{x}) & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & G^0 & \longrightarrow & G & \longrightarrow & \mathbb{Z} \longrightarrow 0 \\ & & & & \downarrow & & \\ & & & & \mathrm{GL}(\mathcal{F}_{\bar{x}}) & & \end{array} \quad (1.3.7.1)$$

Here is the exact definition: if w is an element of $W(X_0, \bar{x})$ of degree 1, then $W(X_0, \bar{x})$ is the semidirect product of the group $w^{\mathbb{Z}}$ generated by w with $\pi_1(X, \bar{x})$. The image of w in $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ normalizes that of $\pi_1(X, \bar{x})$, that is, the geometric monodromy group. It therefore normalizes its Zariski closure G^0 , and G is the semidirect product of $\mathbb{Z}$ by G^0 relative to the action $\mathrm{int}(w)$ of $\mathbb{Z}$ on G^0 . The group G thus obtained fits into diagram (1.3.7.1), and this characterizes it up to a unique isomorphism.

Each linear representation of the algebraic group G defines an ℓ -adic representation of $W(X_0, \bar{x})$, hence a Weil sheaf on X_0 . The original sheaf $\mathcal{F}_0$ corresponds to the tautological representation (1.3.7.1) of G in $\mathrm{GL}(\mathcal{F}_{\bar{x}})$.

If $\mathcal{F}_0^{(i)}$ is a finite family of lisse sheaves, and one applies the preceding construction to the direct sum $\mathcal{F}_0$ of the $\mathcal{F}_0^{(i)}$, then in (1.3.7.1) one may replace $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ by the subgroup $\prod_i \mathrm{GL}(\mathcal{F}_{\bar{x}}^{(i)})$. Each sheaf $\mathcal{F}_0^{(i)}$ is determined by the representation of G in the corresponding factor $\mathrm{GL}(\mathcal{F}_{\bar{x}}^{(i)})$.

Theorem (1.3.8) (Grothendieck). Let $\mathcal{F}_0$ be a lisse sheaf on X_0 , let G and G^0 be as above, and let G^{00} be the neutral component of G^0 . Then the radical of the algebraic group G^{00} is unipotent.

We first prove:

Corollary (1.3.9). If $\mathcal{F}_0$ is semisimple, the Zariski closure G^0 of the geometric monodromy group of $\mathcal{F}_0$ is an extension of a finite group by a semisimple group.

If the representation of $W(X_0, \bar{x})$ on $\mathcal{F}_{0\bar{x}}$ is semisimple, the same is true of its restriction to the distinguished subgroup $\pi_1(X, \bar{x})$; the sum of the $\pi_1(X, \bar{x})$ -subspaces simple for $\pi_1(X, \bar{x})$ is W -stable, hence is a supplement to any other. We shall prove (1.3.9) assuming only that the representation of $\pi_1(X, \bar{x})$ is semisimple. Under this hypothesis, G^{00} is reductive, and it remains to prove that its largest central torus T_1 is trivial. It is therefore isogenous to the largest torus T quotient of G^{00} .

First suppose that $G^0 = G^{00}$ and that G is a product $G \simeq G^0 \times \mathbb{Z}$. The application $W(X_0, \bar{x}) \rightarrow G$ furnishes morphisms $W(X_0, \bar{x}) \rightarrow T$, such that the image of $\pi_1(X, \bar{x})$ is Zariski

dense. Since T is a product of multiplicative groups, it follows from (1.3.4) that this image is finite; hence $T = \{1\}$.

We reduce to this case in general by replacing X_0 by a finite cover and $\mathbb{F}_q$ by a finite extension, hence $W(X_0, \bar{x})$ and G by subgroups of finite index. The group $W(X_0, \bar{x})$ acts on T_1 while respecting the finite set $F \subset X(T_1)$ of characters whose kernels do not contain T_1 ; this set generates $X(T_1) \otimes \mathbb{Q}$. Hence G^0 , and therefore T_1 , acts faithfully; the group $W(X_0, \bar{x})$ therefore acts on T_1 through a finite quotient, and one is reduced to assuming this action trivial. The group of exterior automorphisms of G^{00} , finite on T_1 , is finite; one may suppose that $W(X_0, \bar{x})$ acts on G^{00} by inner automorphisms. Finally one is reduced to supposing that $G^0 = G^{00}$, and then $G \simeq G^{00} \times \mathbb{Z}$.

Prove (1.3.8). Let G be a Jordan–Hölder sequence for $\mathcal{F}_0$, let P be the subgroup of $\mathrm{GL}(\mathcal{F}_{\bar{x}})$ which respects the filtration G , let N be the subgroup of P which acts trivially on the successive quotients (unipotent radical), and let $L = P/N$. The analogue G_1^0 of G^0 for $\mathrm{Gr}_G(\mathcal{F}_0)$ is the image of G^0 in L : the algebraic group G^0 is an extension of G_1^0 by a subgroup of N , automatically unipotent, so that it suffices to prove (1.3.8) for $\mathrm{Gr}_G(\mathcal{F}_0)$, where (1.3.9) applies.

Lemma (1.3.10). Let G be a group scheme extension of $\mathbb{Z}$ by an algebraic group G^0 , whose connected component of the identity G^{00} is reductive. The following conditions are equivalent:

- (i) G admits a linear representation whose restriction to G^0 is faithful.
- (ii) G admits a linear representation whose restriction to G^0 has finite kernel.
- (iii) Let Z^{00} be the connected component of the centre of G^{00} . The restriction to Z^{00} of every inner automorphism of G is of finite order.
- (iv) The centre of G maps onto a finite-index subgroup of $\mathbb{Z}$.

The implication (i) $\Rightarrow$ (ii) is trivial. Prove that (ii) $\Rightarrow$ (iii): if ρ is a linear representation as in (ii), the weights of Z^{00} in this representation form a finite set of characters, rationally generating the character group of Z^{00} . This finite set is respected by the action of G on Z^{00} by inner automorphisms. A finite-index subgroup of G therefore acts trivially on this set, and hence on Z^{00} .

Prove that (iii) $\Rightarrow$ (iv). Let g be an element of G whose image in $\mathbb{Z}$ is 1, and let γ be the automorphism $x \mapsto gxg^{-1}$ of G^{00} . We know that the group of exterior automorphisms of a reductive group is the semidirect product of the automorphisms respecting a pinning by the adjoint group (inner automorphisms) (SGA 3 XXIV (1.3)), and that an automorphism preserving a pinning is of finite order as soon as its restriction to the connected centre is. Hypothesis (iii) therefore ensures that γ is the product of an automorphism of finite order preserving a pinning by an inner automorphism. Modifying g by an element of G^{00} , we may suppose γ of finite order. The restriction of $\mathrm{Int}(g)$ to G^0 is then also of finite order. If its order is n , then g^n is central (because it centralizes G^0 and g), and (iv) follows.

Finally, under hypothesis (iv), G admits a central subgroup Z , generated by an element g of nonzero degree in $\mathbb{Z}$; the group G/Z is algebraic and every faithful representation of G/Z gives a representation of G whose restriction to G^0 is faithful.

Corollary (1.3.11). Suppose $\mathcal{F}_0$ semisimple, and let Z be the centre of G . Then the kernel and cokernel of $\deg : Z \rightarrow \mathbb{Z}$ are finite.

The kernel $Z \cap G^0$ is contained in the centre of G^0 , which is finite by (1.3.9). Again by (1.3.9), condition (1.3.10)(iii) is trivially verified (one has $Z^{00} = \{e\}$), and the cokernel is finite by (1.3.10)(iv).

Corollary (1.3.12). Suppose $\mathcal{F}_0$ semisimple, and let g be a central element of degree $m \neq 0$ in G . For the sheaf defined by a representation V of G to be purely of ι -determinantal weight β ,

it is necessary and sufficient that all the eigenvalues α of g acting on V verify

$$|\iota\alpha| = q^{m\beta/2}.$$

One reduces to supposing V simple, in which case g is scalar. One then replaces V by its maximal exterior power, to reduce to the case in which V has rank 1. That case is left to the reader; see (1.3.4).

This interpretation of determinantal weight immediately gives the following proposition, which is the main result of this section.

Proposition (1.3.13).

- (i) Let $f : X'_0 \rightarrow X_0$ be a dominant morphism of normal connected schemes of finite type over $\mathbb{F}_q$. For a lisse sheaf on X_0 to be purely of ι -determinantal weight β , it is necessary and sufficient that its inverse image on X'_0 be so.
- (ii) If the lisse sheaves $\mathcal{F}_0$ and $\mathcal{G}_0$ on X_0 are purely of ι -determinantal weights β and γ , then $\mathcal{F}_0 \otimes \mathcal{G}_0$ is purely of ι -determinantal weight $\beta + \gamma$.
- (iii) Let $\mathcal{F}_0$ be a lisse sheaf on X_0 , and let $n(\beta)$ be the sum of the ranks of the constituents of $\mathcal{F}_0$ of ι -determinantal weight β . Then the determinantal weights of $\wedge^a \mathcal{F}_0$ are the sums

$$\sum a(\beta)\beta$$

with $\sum a(\beta) = a$ and $a(\beta) \leq n(\beta)$.

For (ii), take G defined by $\mathcal{F}_0$ and $\mathcal{G}_0$; see (1.3.7), last paragraph.

Proposition (1.3.14) (see (1.3.2)). For an irreducible lisse sheaf $\mathcal{F}_0$ of rank r on X_0 to be an étale sheaf, it is necessary and sufficient that $\wedge^r \mathcal{F}_0$ be one. Every irreducible lisse sheaf is therefore obtained by torsion (1.2.7) from an étale sheaf.

Let g be central in G as in (1.3.12). It acts on $\mathcal{F}_{0\bar{x}}$ by multiplication by a scalar u . The group G acts on $\wedge^r \mathcal{F}_{0\bar{x}}$ through a discrete quotient, which is also a quotient of $W(X_0, \bar{x})$; and for $\wedge^r \mathcal{F}_0$ to be an étale sheaf, it is necessary and sufficient that u be an ℓ -adic unit. Suppose this is so, and let E be a finite extension of $\mathbb{Q}_\ell$ such that the action of $W(X_0, \bar{x})$ factors through $\mathrm{GL}(r, E)$. It remains to show that the image of $W(X_0, \bar{x})$ in $\mathrm{GL}(r, E)$ is bounded; then the inverse image of an open subgroup will be open of finite index, and the representation will extend to the completion $\pi_1(X_0, \bar{x})$ of $W(X_0, \bar{x})$.

Let W_1 be the subgroup of finite index of $W(X_0, \bar{x})$ which is the inverse image of $G^{00} \times g^{\mathbb{Z}}$, let W_1^0 be the inverse image of G^{00} , and let ρ be the composite morphism

$$W_1 \longrightarrow G^{00} \times g^{\mathbb{Z}} \longrightarrow G^{00}.$$

Since the u^n ($n \in \mathbb{Z}$) form a bounded set, it suffices to show that $\rho(W_1) \subset G^{00}(E)$ is bounded. This subgroup normalizes $\rho(W_1^0)$, which is compact and Zariski dense. We conclude by the following lemma.

Lemma (1.3.15). In a semisimple group H over a local field E , the normalizer of a compact Zariski-dense subgroup $K \subset H(E)$ is compact.

We treat only the case in which E has characteristic 0. The logarithm map, first defined on a sufficiently small neighbourhood of e , is extended to the union of the compact subgroups of $H(E)$ by the formula

$$\log(x) = \frac{1}{n} \log(x^n).$$

The density of K , and the fact that H is semisimple, imply that $\log(K)$ generates $\text{Lie } H$ as a vector space over E . Let $\mathcal{O}_E$ be the valuation ring of E , and let L be the $\mathcal{O}_E$ -module generated by the compact set $\log K \subset \text{Lie } H$. It is stable under the normalizer of K . Since the adjoint representation has finite kernel, and since

$$\text{ad} : H(E) \longrightarrow \text{GL}(\text{Lie } H)$$

is proper, it follows that this normalizer is compact.

(1.4) Cohomology of curves and L -functions: reminders

This section gathers together some well-known results that we shall use constantly.

(1.4.1). Let X be a curve over an algebraically closed field k , and let $\mathcal{F}$ be a sheaf on X . The groups

$$H_c^0(X, \mathcal{F}), \quad H^0(X, \mathcal{F}), \quad H_c^2(X, \mathcal{F}), \quad H^2(X, \mathcal{F})$$

are elementary in nature.

- a) $H^0(X, \mathcal{F})$ is the group of global sections, and $H_c^0(X, \mathcal{F})$ is the subgroup of those with compact support; if X has no complete irreducible component, this means finite support. If X is connected, with base point x , and if $\mathcal{F}$ is lisse and defined by a representation V of $\pi_1(X, x)$, then

$$H^0(X, \mathcal{F}) = V^{\pi_1(X, x)}, \quad (1.4.1.1)$$

while

$$H_c^0(X, \mathcal{F}) = \begin{cases} V^{\pi_1(X, x)}, & \text{if } X \text{ is complete,} \\ 0, & \text{otherwise.} \end{cases} \quad (1.4.1.2)$$

- b) If U is the complement of a finite subset of X , then

$$H_c^2(U, \mathcal{F}) \simeq H_c^2(X, \mathcal{F}).$$

Taking U small enough so that U_{red} is smooth and $\mathcal{F}$ is lisse on U , this makes it possible to compute H_c^2 by Poincaré duality from the calculation of H^0 in a). If U_{red} is smooth and connected, with base point x , and if, on U , the sheaf $\mathcal{F}$ is defined by a representation V of $\pi_1(U, x)$, then

$$H_c^2(X, \mathcal{F}) = V_{\pi_1(U, x)}(-1). \quad (1.4.1.3)$$

(1.4.2). With the conventions of (0.7), let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a lisse sheaf on X_0 . Let $\mathcal{F}'_0$ (respectively $\mathcal{F}''_0$) be the largest subsheaf (respectively quotient sheaf) of $\mathcal{F}_0$ which becomes constant on X . It is the inverse image of a sheaf F'_0 (respectively F''_0) on $\text{Spec}(\mathbb{F}_q)$, and the discussion (1.4.1) gives

$$H^0(X, \mathcal{F}) = F', \quad (1.4.2.1)$$

$$H_c^0(X, \mathcal{F}) = F' \text{ or } 0 \text{ according as } X \text{ is complete or not,} \quad (1.4.2.2)$$

$$H_c^2(X, \mathcal{F}) = F''(-1). \quad (1.4.2.3)$$

The sheaves $\mathcal{F}'_0$ and F'_0 (respectively $\mathcal{F}''_0$ and F''_0) have the same determinantal weights, and these are among the determinantal weights of $\mathcal{F}_0$. From (1.4.2) we therefore get:

Corollary (1.4.3). If α is an eigenvalue of F^* on $H^0(X, \mathcal{F})$ or $H_c^0(X, \mathcal{F})$ (respectively on $H_c^2(X, \mathcal{F})$), then there exists a determinantal weight β of $\mathcal{F}_0$ such that

$$w_q(\alpha) = \beta \quad (\text{respectively } w_q(\alpha) = \beta + 2).$$

More trivially, one also has:

Corollary (1.4.4). If α is an eigenvalue of F^* on $H^0(X, \mathcal{F})$ or $H_c^0(X, \mathcal{F})$ (respectively on $H_c^2(X, \mathcal{F})$), then, for every $x \in |X_0|$, the number $\alpha^{\deg(x)}$ (respectively $(q^{-1}\alpha)^{\deg(x)}$) is an eigenvalue of F_x on $\mathcal{F}_0$.

(1.4.5). If X_0 is a scheme of finite type over $\mathbb{F}_q$, and if $\mathcal{F}_0$ is a sheaf on X_0 , then by Grothendieck one has an equality of formal t -adic series:

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \prod_i \det(1 - Ft, H_c^i(X, \mathcal{F}))^{(-1)^{i+1}}. \quad (1.4.5.1)$$

Applying ι to it, one obtains an equality between complex formal series, expressing that the left hand side is the Taylor expansion of the rational function on the right. If the product on the left converges for $|t| < R$, one obtains in that disc an equality between analytic functions. Here is a convergence criterion.

Proposition (1.4.6). Suppose that, for $x \in |X_0|$, the eigenvalues α of F_x on $\mathcal{F}$ satisfy

$$w_{N(x)}(\alpha) < \beta.$$

Then the product

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1}$$

converges for

$$|t| < q^{-\beta/2 - \dim X_0},$$

and hence has neither zero nor pole in that disc.

Set $d = \dim X_0$, and decompose X_0 into pieces each of which is quasi-finite over an affine space $\mathbb{A}^d$: one obtains an estimate

$$\#\{x \in |X_0| \mid \deg(x) = n\} < \text{Cte} \cdot q^{dn}.$$

This reduces the convergence of the product to that of the geometric series

$$\sum_n q^{dn} q^{n\beta/2} |t|^n.$$

(1.4.7). If $\dim X_0 \leq 1$, the H_c^i are zero for $i \notin \{0, 1, 2\}$. The formula deduced from (1.4.5.1) by applying ι therefore reduces to

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}))}{\det(1 - Ft, H_c^0(X, \mathcal{F})) \det(1 - Ft, H_c^2(X, \mathcal{F}))}. \quad (1.4.7.1)$$

If X_0 is a smooth affine curve and $\mathcal{F}_0$ is lisse, then H_c^0 is zero and the formula simplifies to

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}))}{\det(1 - Ft, H_c^2(X, \mathcal{F}))}. \quad (1.4.7.2)$$

In the uses we shall make of these formulae, the left hand side will be controlled by (1.4.6), and the poles on the right by (1.4.3).

(1.5) A criterion for purity

The conventions of (0.7) are in force. Let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$. The proof of the following theorem is parallel to that of Theorem (1.3.4), which it generalizes.

Theorem (1.5.1). The constituents of any lisse ι -real sheaf on X_0 are ι -pure.

Lemma (1.5.2). Let $\mathcal{F}_0$ be a lisse ι -real sheaf on X_0 , and let r be the largest of its determinantal weights, relative to ι (1.3.5). Then, for every $x \in |X_0|$, and every eigenvalue α of F_x on $\mathcal{F}_0$, one has

$$w_{N(x)}(\alpha) \leq r.$$

After removing from X_0 a point other than the one under consideration, one reduces to the case in which X_0 is affine. For every positive integer k , one then has the equality (1.4.7.2)

$$\prod_{x \in |X_0|} \det(1 - F_x t^{\deg(x)}, \mathcal{F}^{\otimes 2k})^{-1} = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}^{\otimes 2k}))}{\det(1 - Ft, H_c^2(X, \mathcal{F}^{\otimes 2k}))}. \quad (1.5.2.1)$$

In this formula:

a) The factors

$$\det(1 - F_x t^{\deg(x)}, \mathcal{F}^{\otimes 2k})^{-1}$$

on the left are formal series with constant term 1 and real non-negative coefficients. The proof is as in (1.3.3), (3.4), with “rational” replaced by “real”.

b) By (1.3.13)(ii), the determinantal weights of the constituents of $\mathcal{F}_0^{\otimes 2k}$ are at most $2kr$. By (1.4.3), the right hand side has no pole for

$$|t| < q^{-(2kr+2)/2}.$$

Because the factors on the left have non-negative coefficients, poles cannot be cancelled. In particular, if α is an eigenvalue of F_x on $\mathcal{F}_0$, then the contribution of α^{2k} gives a pole at absolute value

$$|\iota(\alpha)|^{-2k/\deg(x)}.$$

Consequently

$$|\iota(\alpha)|^{-2k/\deg(x)} \geq q^{-(2kr+2)/2},$$

or equivalently

$$w_{N(x)}(\alpha) \leq r + \frac{1}{k}.$$

Letting k tend to infinity gives the lemma.

(1.5.3) Proof of (1.5.1). Let $\mathcal{F}_0$ be a lisse ι -real sheaf on X_0 . For $\gamma \in \mathbb{R}$, let $\mathcal{F}_0(\gamma)$ be the sum of the constituents of $\mathcal{F}_0$ of determinantal weight γ , relative to ι , and let $n(\gamma)$ be the rank of $\mathcal{F}_0(\gamma)$. Let $x \in |X_0|$, and let α_i^γ ($1 \leq i \leq n(\gamma)$) be the eigenvalues of F_x on $\mathcal{F}_0(\gamma)$, counted with multiplicity. We have to prove that

$$w_{N(x)}(\alpha_i^\gamma) = \gamma.$$

By the definition of the determinantal weight, one has, for every γ ,

$$\sum_i w_{N(x)}(\alpha_i^\gamma) = n(\gamma)\gamma. \quad (1.5.3.1)$$

Suppose, as we may, that $\mathcal{F}_0(\beta) \neq 0$, and let N be the sum of the $n(\gamma)$ for $\gamma > \beta$. By (1.3.13)(iii), the determinantal weights of the $(N+1)$ -st exterior power of $\mathcal{F}_0$ are at most

$$\beta + \sum_{\gamma > \beta} n(\gamma)\gamma.$$

Since each product

$$\alpha_i^\beta \prod_{\gamma > \beta} \prod_{j=1}^{n(\gamma)} \alpha_j^\gamma$$

is an eigenvalue of F_x on $\wedge^{N+1} \mathcal{F}_0$, (1.5.2) gives

$$w_{N(x)}(\alpha_i^\beta) + \sum_{\gamma > \beta} \sum_j w_{N(x)}(\alpha_j^\gamma) \leq \beta + \sum_{\gamma > \beta} n(\gamma)\gamma.$$

Subtracting the equalities (1.5.3.1) for $\gamma > \beta$, one obtains

$$w_{N(x)}(\alpha_i^\beta) \leq \beta. \quad (1.5.3.2)$$

To deduce equality from this inequality, one may either pass to the dual sheaf, or observe that summing the inequalities (1.5.3.2) over i gives the equality (1.5.3.1) for β .

(1.6) Around Jacobson–Morosov

Proposition (1.6.1). Let N be a nilpotent endomorphism of an object V of an abelian category. There is then one and only one finite increasing filtration M of V such that

$$NM_i \subset M_{i-2}$$

and such that N^k induces an isomorphism

$$\mathrm{Gr}_k^M V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M V.$$

We prove existence; uniqueness is proved in the same way. We proceed by induction on an integer d such that $N^{d+1} = 0$. For $d = 0$ one takes the trivial filtration. Next put

$$M_d = V, \quad M_{d-1} = \mathrm{Ker} N^d, \quad M_{-d} = \mathrm{Im} N^d, \quad M_{-d-1} = 0,$$

so that $\mathrm{Gr}_d^M V$ and $\mathrm{Gr}_{-d}^M V$ are respectively the coimage and the image of N^d . On $\mathrm{Ker} N^d / \mathrm{Im} N^d$, the $(d-1)$ -st power of N is zero. The induction hypothesis therefore defines M on this quotient, and, for $d-1 \geq i \geq -d$, M_i on V is defined as the inverse image in $\mathrm{Ker} N^d$ of the corresponding subobject of $\mathrm{Ker} N^d / \mathrm{Im} N^d$.

Remark (1.6.2). The characterization (1.6.1) of M shows its compatibility with passage to the dual category.

Definition (1.6.3). The primitive part $P_i(V)$, or simply P_i , of $\mathrm{Gr}_i^M V$ is the kernel of the morphism induced by N from $\mathrm{Gr}_i^M V$ to $\mathrm{Gr}_{i-2}^M V$.

(1.6.4). If $i > 0$, then

$$N : \text{Gr}_i^M V \longrightarrow \text{Gr}_{i-2}^M V$$

is injective, since $N^{i-1}N$ is an isomorphism, and therefore $P_i = 0$. If $i \geq 0$, writing that

$$N \circ N^{i+1} : \text{Gr}_{i+2}^M V \longrightarrow \text{Gr}_i^M V \longrightarrow \text{Gr}_{i-2}^M V$$

is an isomorphism shows that $\text{Gr}_{-i}^M V$ is the direct sum of P_{-i} and of the image of $\text{Gr}_{i+2}^M V$ by N^{i+1} ; the morphism N induces an isomorphism from that image onto $\text{Gr}_{-i-2}^M V$. Repeating the construction, one obtains, by descending induction on $i \geq 0$, an isomorphism

$$\text{Gr}_i^M V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j}. \quad (1.6.4.1)$$

Through these isomorphisms, the morphism $N : \text{Gr}_i^M V \rightarrow \text{Gr}_{i-2}^M V$ is the evident inclusion for $i \geq 1$ and the evident projection for $i - 2 \leq -1$.

Schematically, the graded pieces are arranged as follows:

$$\begin{array}{rcll} \text{Gr}_3^M : & & P_3 & \cdots \\ \text{Gr}_2^M : & & P_2 & P_4 \cdots \\ \text{Gr}_1^M : & P_1 & P_3 & \cdots \\ \text{Gr}_0^M : P_0 & & P_2 & P_4 \cdots \\ \text{Gr}_{-1}^M : & P_1 & P_3 & \cdots \\ \text{Gr}_{-2}^M : & & P_2 & P_4 \cdots \\ \text{Gr}_{-3}^M : & & P_3 & \cdots \end{array}$$

with arrows induced by N running downward.

Lemma (1.6.5). The morphism

$$N : (V, M) \longrightarrow (V, M \text{ shifted by } 2)$$

is strictly compatible with the filtrations.

It is necessary to prove that $NM_{i+2} \supset \text{Im}(N) \cap M_i$. Distinguish two cases. If $i < 0$, the morphism $N : M_{i+2} \rightarrow M_i$ has surjective associated graded, hence is surjective, and $NM_{i+2} = M_i$, which proves the assertion a fortiori. If $i + 2 \geq 0$, the morphism

$$N : V/M_{i+2} \longrightarrow V/M_i$$

has injective associated graded, hence is injective, and $N^{-1}M_i \subset M_{i+2}$, again proving the assertion.

This permits one to exchange the constructions $\text{Ker } N$ and Gr^M .

Corollary (1.6.6). The inclusion of $\text{Ker } N$ in V induces isomorphisms

$$\text{Gr}_i^M(\text{Ker } N) \xrightarrow{\sim} P_i.$$

(1.6.7). Assume now that V is a finite-dimensional vector space over a field k . We first describe M when, in a basis e , the matrix of N is

$$\begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{pmatrix}. \quad (1.6.7.1)$$

Put $\dim V = d + 1$ and index the basis vectors by the integers going by steps of two from d to $-d$, so that $Ne_i = e_{i-2}$ for $i \neq -d$ and $Ne_{-d} = 0$. Then M_i is generated by the e_j with $j \leq i$.

In general, V is a sum of subspaces V_a stable under N , with N of the form (1.6.7.1) on each V_a , and the filtration M of V is the sum of the preceding filtrations of the V_a .

(1.6.8). If k has characteristic zero, one may interpret the filtration M in terms of the Jacobson–Morosov theorem: if

$$u : \mathrm{SL}(2) \longrightarrow \mathrm{GL}(V)$$

satisfies

$$du \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = N,$$

and if V_j is the subspace of V formed by those vectors v such that

$$u \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} (v) = \lambda^j v,$$

then M_i is the sum of the V_j for $j \leq i$. This is reduced to the case where V is an irreducible representation of $\mathrm{SL}(2)$, that is, a symmetric power of the evident representation, and one is then in the situation of (1.6.7).

This interpretation gives the behavior of M under tensor product and duality.

Proposition (1.6.9). Define the tensor product (V, N) of (V', N') and (V'', N'') by

$$V = V' \otimes V'', \quad N = N' \otimes 1 + 1 \otimes N'',$$

and the dual of (V', N') as $(V'^\vee, -{}^t N')$.

- (i) If k has characteristic zero, the filtration M of a tensor product is the tensor product of the filtrations M of the factors:

$$M_i(V' \otimes V'') = \sum_{i'+i''=i} M_{i'}(V') \otimes M_{i''}(V'').$$

- (ii) The filtration M of a dual is the dual of the filtration M of the original space:

$$M_i(V'^\vee) = M_{-i-1}(V')^\perp.$$

- (iii) Under these hypotheses, formation of Gr^M is therefore compatible with tensor product and passage to the dual.

(1.6.10). Assume henceforth that V is finite-dimensional over a field k of characteristic zero. On $\mathrm{Gr}^M(V)$ there is then a unique representation v of $\mathrm{SL}(2)$ such that

$$dv \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

is $N : \mathrm{Gr}_i^M V \rightarrow \mathrm{Gr}_{i-2}^M V$ and such that

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}$$

acts by multiplication by λ^j on $\mathrm{Gr}_j^M(V)$. Existence is seen by choosing u as in (1.6.8), identifying $\mathrm{Gr}^M(V)$ with V by means of the splitting V_j of the filtration, and transporting u to the graded object by this identification. Uniqueness follows from Bourbaki, Lie VIII, §11, no. 1, Lemma 1. Formation of v is compatible with tensor product and passage to the dual taken as in (1.6.9). This will allow us to determine the behavior of P_i under tensor product and duality.

(1.6.11). Choose once and for all, for each integer $d \geq 0$, an irreducible representation S_d of $\mathrm{SL}(2)$ of dimension $d + 1$, and a lowest-weight vector $e_{-d} \in S_d$ such that $e_{-d} \neq 0$ and

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} e_{-d} = \lambda^{-d} e_{-d}.$$

Put

$$P(d, d') = \{j \mid |d - d'| \leq j \leq d + d', j \equiv d + d' \pmod{2}\}, \quad (1.6.11.1)$$

and choose isomorphisms of representations

$$S_d \otimes S_{d'} \xrightarrow{\sim} \bigoplus_{j \in P(d, d')} S_j, \quad (1.6.11.2)$$

$$S_d^\vee \xrightarrow{\sim} S_d. \quad (1.6.11.3)$$

For V, N, M, P as above, there exists a unique isomorphism of $\mathrm{SL}(2)$ -representations

$$\varphi : \bigoplus_j S_j \otimes P_{-j} \xrightarrow{\sim} \mathrm{Gr}^M(V), \quad (1.6.11.4)$$

where $\mathrm{SL}(2)$ acts trivially on P_{-j} in the source and as in (1.6.10) on the target, such that the composite

$$P_{-j} \xrightarrow{e_{-j} \otimes -} S_j \otimes P_{-j} \longrightarrow \mathrm{Gr}_{-j}^M(V)$$

is the identity inclusion of P_{-j} . This yields isomorphisms

$$P_{-j} \simeq \mathrm{Hom}_{\mathrm{SL}(2)}(S_j, \mathrm{Gr}^M(V)). \quad (1.6.11.5)$$

(1.6.12). Let now V' and V'' be as in (1.6.10), let $V = V' \otimes V''$, and write P_j, P'_j, P''_j for $P_j(V), P_j(V')$, and $P_j(V'')$. The tensor product of the isomorphisms (1.6.11.4), followed by the choices (1.6.11.2), gives an isomorphism, depending only on the choices (1.6.11.2),

$$P_{-j} \xrightarrow{\sim} \bigoplus_{j \in P(j', j'')} P'_{-j'} \otimes P''_{-j''}. \quad (1.6.12.1)$$

For passage to the dual, one similarly obtains isomorphisms, depending only on the choices (1.6.11.3),

$$P_{-j}(V'^\vee) \xrightarrow{\sim} P_{-j}(V')^\vee. \quad (1.6.12.2)$$

Proposition (1.6.13). Let V be an object of an abelian category, endowed with a finite increasing filtration W , and let N be a nilpotent endomorphism of V respecting W . There is at most one finite increasing filtration M of V such that $NM_i \subset M_{i-2}$ and such that N^k induces isomorphisms

$$\mathrm{Gr}_k^M \mathrm{Gr}^W V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M \mathrm{Gr}^W V.$$

The proof is by induction on the length of W . This reduces one to the case where, for an integer a , one has $W_a = V$, and where the assertion is known for W_{a-1} , endowed with the induced filtration and the endomorphism induced by N . A simultaneous shift by the same amount on the filtrations W and M preserves the condition of (1.6.13), so that we may suppose $a = 0$.

Let M be a filtration satisfying the stated conditions. The filtration induced on W_{-1} satisfies the same conditions, hence is uniquely determined. The quotient filtration is the filtration (1.6.1) of $\mathrm{Gr}_0^W V$. Let c be such that $\mathrm{Gr}_i^M \mathrm{Gr}_0^W V$ is nonzero only for $-c \leq i \leq c$. For $i \geq 0$, the following identities hold:

- 1) for $i > c$, $M_{-i} = M_{-i}(W_{-1})$;
- 2) $M_{-i} = M_{-i}(W_{-1}) + N^i(M_i)$;
- 3) $M_i = \text{Ker}(N^{i+1} : V \rightarrow V/M_{-i-2})$.

These identities determine M uniquely, by descending induction on $|i|$, in terms of its restriction to W_{-1} and of N .

(1.6.14). In the applications, the abelian category is the category of vector spaces over a field k of characteristic zero, and instead of a single nilpotent endomorphism N of V one has a nilpotent action of a one-dimensional Lie algebra $\mathfrak{N}$. The preceding theory applies as soon as one chooses $N \neq 0$ in $\mathfrak{N}$. In (1.6.13), existence of M , and M itself when it exists, are independent of this choice. The isomorphism N^k has the canonical avatar

$$\text{Gr}_{i+k}^M \text{Gr}^W V(k) \xrightarrow{\sim} \text{Gr}_{i-k}^M \text{Gr}^W V, \quad (1.6.14.1)$$

where for a vector space Y one writes $Y(k) = Y \otimes \mathfrak{N}^{\otimes k}$.

In the situation of (1.6.1)–(1.6.12), the filtration M and the P_j do not depend on the choice of N , but the isomorphisms (1.6.4.1) and (1.6.12.1–2) do. If N is replaced by λN , each of their components, generically denoted $\alpha : X \rightarrow Y$, is multiplied by a convenient power λ^n of N . It therefore defines a map

$$\alpha \otimes N^{\otimes n} : X \longrightarrow Y(n)$$

independent of the choice of N . In this way the previous isomorphisms become canonical:

$$\text{Gr}_i^M(\text{Ker } N) \simeq P_i, \quad (1.6.14.2)$$

$$\text{Gr}_i^M V \simeq \bigoplus_{\substack{j \geq |i| \\ j \equiv i \pmod{2}}} P_{-j} \left(\frac{i+j}{2} \right), \quad (1.6.14.3)$$

$$P_{-j} \simeq \bigoplus_{j \in P(j', j'')} P'_{-j'} \otimes P''_{-j''} \left(\frac{j - j' - j''}{2} \right), \quad (1.6.14.4)$$

$$P_{-j}(V'^{\vee}) \simeq P_{-j}(V')^{\vee}(j). \quad (1.6.14.5)$$

To remember these formulas, one regards Gr_i and P_i as of weight i , and $\mathfrak{N}$ as of weight -2 ; both members are then always isobaric of the same weight. We shall also use a sheaf-theoretic variant of (1.6.14).

(1.7) Local monodromy

(1.7.1). Let R be a henselian discrete valuation ring, K its field of fractions, $k = R/\mathfrak{m}$ its residue field, $\overline{K}$ an algebraic closure of K , and $\overline{k}$ the residue field of the normalization of R in $\overline{K}$. The extension $\overline{k}/k$ is not algebraic in general. It is often convenient first to choose $\overline{k}$, and then $\overline{K}$, by taking an algebraic closure $\overline{k}$ of k , then an algebraic closure $\overline{K}$ of the field of fractions of the strict henselization of R .

The inertia group is defined as the kernel in the exact sequence

$$0 \longrightarrow I \longrightarrow \text{Gal}(\overline{K}/K) \longrightarrow \text{Gal}(\overline{k}/k) \longrightarrow 0. \quad (1.7.1.1)$$

It, or its image in a representation, is also called the local monodromy group. Recall that if k has exponential characteristic p , one has an exact sequence

$$0 \longrightarrow P \longrightarrow I \longrightarrow \widehat{\mathbb{Z}}_{(p')}(1) \longrightarrow 0,$$

where P is a pro- p group and

$$\widehat{\mathbb{Z}}_{(p')}(1) = \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$$

is the tame inertia group. Taking the ℓ -component gives

$$t_\ell : I \longrightarrow \mathbb{Z}_\ell(1)$$

whose kernel is a profinite group of order prime to ℓ .

(1.7.2). Let $\rho : I \rightarrow \mathrm{GL}(V)$ be an ℓ -adic representation of I . Grothendieck proved that the following hypothesis is often verified in practice:

(*) The restriction of ρ to a finite-index subgroup I_1 of I is unipotent.

If this is so, there is a unique representation $\bar{\rho}$ of $\mathbb{Q}_\ell(1)$, viewed as a one-dimensional Lie algebra in V , such that

$$\rho(\sigma) = \exp(\bar{\rho}(t_\ell(\sigma))) \quad (\sigma \in I_1 \subset I). \quad (1.7.2.1)$$

One calls $\bar{\rho}$ the logarithm of the unipotent part of the local monodromy. It may be identified with a nilpotent morphism

$$N : V(1) \longrightarrow V.$$

The defining formula is then written

$$\rho(\sigma) = \exp(Nt_\ell(\sigma)) \quad (\sigma \in I_1 \subset I). \quad (1.7.2.2)$$

The theory of (1.6.14) applies to this situation. The twist (k) of local systems is the usual Tate twist. The filtrations (1.6.1) and (1.6.13) are called, respectively, the local monodromy filtration of V and the local monodromy filtration of V relative to W .

(1.7.3). Assume the residue field finite with q elements. Define the Weil group $W(\bar{K}/K)$ as the subgroup of $\mathrm{Gal}(\bar{K}/K)$ inverse image of $W(\bar{k}/k)$. If V is an ℓ -adic representation of $W(\bar{K}/K)$, then by Grothendieck (SGA 7 I) condition (*) of (1.7.2) is automatically verified. The logarithm of the unipotent part of the monodromy

$$N : V(1) \longrightarrow V$$

is therefore defined. It commutes with the action of $W(\bar{K}/K)$.

Lemma (1.7.4). If F' and F'' are two liftings in $W(\bar{K}/K)$ of $F \in W(\bar{k}/k)$, then the eigenvalues of F' and F'' coincide up to multiplication by roots of unity.

To verify this, one may replace V by its semisimplification, and then suppose that the action of the inertia group factors through a finite quotient. In that case there is $n > 0$ such that F'^n and F''^n have the same image in $\mathrm{GL}(V)$, and the lemma follows.

Whatever lift one chooses, the ι -weights of the eigenvalues of F' or of F'' , relative to q , are therefore the same. Imitating (1.2), one then defines pure, mixed, ι -pure representations, and the weights of a representation.

Proposition-definition (1.7.5). Suppose that the ι -weights of V are integers. There exists then one and only one finite increasing filtration M of V , stable under $W(\bar{K}/K)$, such that $\mathrm{Gr}_i^M(V)$ is ι -pure of weight i . It is called the weight filtration. One has $NM_i(1) \subset M_{i-2}$.

Choose a Frobenius lift F' in $W(\bar{K}/K)$. Let V'_i be the sum of the generalized eigenspaces of F' in V corresponding to eigenvalues α such that $w_q(\alpha) = i$, and put $M'_i = \prod_{j \leq i} V'_j$. One proves that M' is independent of F' . This will make it stable under $W(\bar{K}/K)$ and gives $M = M'$. If

M'' is the filtration defined by another lift F'' , let I_1 be as in (1.7.2.1). Since I/I_1 is finite, there is $n > 0$ such that $F''^n \equiv F'''^n \pmod{I_1}$. For a suitable $\lambda \in \mathbb{Q}_\ell(-1)$, one then has in $\mathrm{GL}(V)$

$$F'''^n = \exp(\lambda N) F''^n,$$

hence

$$F'''^n = \exp(\mu N) F''^n \exp(\mu N)^{-1} \quad \text{with } \mu = \lambda/(1 - q^n).$$

Since V'_i is the sum of the generalized eigenspaces of F''^n corresponding to eigenvalues of weight i , the automorphism $\exp(\mu N)$ sends M' onto M'' . Since N respects M' , this proves the assertion.

Corollary (1.7.6). If V is ι -pure, the action of I factors through a finite quotient.

By twisting, one reduces to the case where V has integral weights. The filtration M is then reduced to one step. Since $NM_i \subset M_{i-2}$, one has $N = 0$, and the corollary follows.

Remark (1.7.7). The proof of (1.7.5) shows that any representation V admits a unique decomposition

$$V = \sum V^\alpha \quad (\alpha \in \overline{\mathbb{Q}_\ell}^*/\{\alpha \sim \alpha' \text{ if } \alpha/\alpha' \text{ is a root of unity}\}),$$

such that the eigenvalues of Frobenius liftings on V^α lie in the class α . Each V^α is obtained from a mixed representation by twisting.

Variant (1.7.8). If D is a smooth divisor in a regular scheme X , the preceding constructions have an analogue for sheaves on $X - D$ that are tamely ramified along D . More generally, let D be a normal-crossings divisor of X , the transverse union of smooth divisors D_i ($i \in I$). We work locally so that each D_i admits a global equation $t_i = 0$. Let E be the intersection of the D_i . For n invertible on X , put

$$X_n = X[t_i^{1/n}]_{i \in I},$$

the closed subscheme of the affine space over X , with coordinates T_i , defined by the equations $T_i^n = t_i$. The ramified covering $\pi : X_n \rightarrow X$ is etale over $X - D$ and totally ramified over E ; over E , π has the section $t_i^{1/n} = 0$. The restriction to E of a sheaf on X_n means its inverse image by this section. The group $(\mu_n)^I$ acts on X_n by multiplying $t_i^{1/n}$ by τ_i .

Let L be a set of prime numbers invertible on X , and let $\mathcal{F}$ be a sheaf of sets on $X - D$, locally constant, tamely ramified along D , and indeed L -ramified along D in the following sense. For d the generic point of an irreducible component of D , the henselization $X_{(d)}$ is a henselian trait; if $\bar{\eta}$ is a geometric generic point, require the inertia group to act on $\mathcal{F}_{\bar{\eta}}$ through an L -group. Abhyankar's lemma then gives an integer n , all of whose prime factors lie in L , such that the inverse image of $\mathcal{F}$ on $\pi^{-1}(X - D) \subset X_n$ is the restriction to $\pi^{-1}(X - D)$ of a locally constant sheaf on X_n . We call this the locally constant extension of $\mathcal{F}$ to X_n . We write $\mathcal{F}[E]$ for its restriction to E . It carries an action of μ_n^I deduced from the action on X_n . Putting

$$\mathbb{Z}_L(1) = \prod_{\ell \in L} \mathbb{Z}_\ell(1)$$

(a pro-sheaf on E), one obtains a functor

$$\mathcal{F} \longmapsto \mathcal{F}[E]$$

from locally constant sheaves of sets on $X - D$ that are L -ramified along D to locally constant sheaves of sets on E endowed with an action of $\mathbb{Z}_L(1)^I$. This functor passes to constructible $\overline{\mathbb{Q}_\ell}$ -sheaves by passage to the limit, with a parallel variant for Weil sheaves. In many cases,

in particular if X is of finite type over $\mathbb{Z}$, the action of $\mathbb{Z}_L(1)^I$ is quasi-unipotent: there are mutually commuting nilpotent endomorphisms

$$N_i \in \text{End}(\mathcal{F}[E])(-1)$$

such that, for σ in a suitable open subsheaf $n\mathbb{Z}_L(1)^I$ of $\mathbb{Z}_L(1)^I$, it acts by

$$\exp\left(\sum_i N_i \sigma_i\right).$$

We call N_i the logarithm of the unipotent part of the monodromy around D_i . In the case of a smooth divisor, one uses this endomorphism, as above, to define the local monodromy filtration of $\mathcal{F}[D]$, and, when a filtration W is present, the relative local monodromy filtration.

(1.7.9). Near an intersection of p of the D_i , the divisor D appears as the union of p smooth divisors, and the preceding constructions can be repeated. We use the following notation: for $J \subset I$, D_J is the intersection of the D_j ($j \in J$), X_J is the complement of the union of the D_i with $i \notin J$, and $D'_J = D_J \cap X_J$. The D_i with $i \notin J$ cut out on the regular scheme D_J a normal-crossings divisor with complement D'_J , while the D_j with $j \in J$ cut out on X_J a normal-crossings divisor with complement $X - D$; the intersection of their traces is D'_J .

Let $\mathcal{F}$ be a locally constant sheaf of sets on $X - D$, L -ramified along D . Applying (1.7.8) gives for each J a sheaf $\mathcal{F}[D'_J]$ on D'_J with an action of $\mathbb{Z}_L(1)^J$. Repeating the construction yields, for $K \subset I - J$, a locally constant sheaf $\mathcal{F}[D'_J][D'_{J \cup K}]$ on $D'_{J \cup K}$ with the compatible actions. Transitivity of inverse images gives an isomorphism

$$\mathcal{F}[D'_J][D'_{J \cup K}] \xrightarrow{\sim} \mathcal{F}[D'_{J \cup K}], \quad (1.7.9.1)$$

compatible with the action of $\mathbb{Z}_L(1)^{J \cup K}$. For $J \subset K \subset L$, the corresponding square is commutative. All this extends to constructible $\overline{\mathbb{Q}}_\ell$ -sheaves, and to Weil sheaves, by passage to the limit.

(1.7.10). The constructions (1.7.8) depend on the choice of the coordinates t_i . To make them canonical, one introduces the normal bundle T of E in X and the subbundles T_i tangent to the D_i . One replaces the construction of $\mathcal{F}[E]$ by that of a sheaf $\mathcal{F}[E]'$ on $T - \bigcup T_i$. A system of equations t_i defines a section of $T - \bigcup T_i$, and the sheaf $\mathcal{F}[E]$ defined in terms of the t_i is identified with the inverse image of $\mathcal{F}[E]'$ by this section. The construction is also compatible with base change by a smooth morphism $X' \rightarrow X$. The N_i and $\mathbb{Z}_L(1)^I$ act on $\mathcal{F}[E]'$, which plays the role of the restriction of $\mathcal{F}$ to a tubular neighborhood of E in X . It may also be regarded as an analogue of nilpotent orbits in the theory of variations of Hodge structure.

(1.7.11). Assume that X is henselian local, $X = \text{Spec}(R)$, and take for L the set of all primes invertible on X . We interpret (1.7.8) in Galois terms. Let k be the residue field of R , K its field of fractions, R_1 a strict henselization of R with residue field k_1 and fraction field K_1 , $\overline{K}$ an algebraic closure of K_1 , and $K_2 \subset \overline{K}$ the extension of K_1 obtained by adjoining the n -th roots of the t_i , for all n invertible in R . A locally constant sheaf of sets on $X - D$ is tamely ramified along D if and only if it becomes trivial over $\text{Spec}(K_2)$; this identifies the tame fundamental group of $X - D$ with $\text{Gal}(K_2/K)$. It is an extension

$$0 \longrightarrow \mathbb{Z}_L(1)^I \longrightarrow \pi_1^{\text{mod}}(X - D) \longrightarrow \text{Gal}(k_1/k) \longrightarrow 0.$$

The choice of the t_i and of their roots splits this extension. From a sheaf $\mathcal{F}$ on $X - D$, identified with a set F on which $\pi_1^{\text{mod}}(X - D)$ acts, one obtains an action of $\text{Gal}(k_1/k)$ on F , and a $\text{Gal}(k_1/k)$ -equivariant action of $\mathbb{Z}_L(1)^I$. This is the sheaf $\mathcal{F}[E]$ of (1.7.8).

(1.7.12). If k is finite, define $W^{\text{mod}}(X - D)$ as the inverse image of $W(k_1/k)$ in the tame fundamental group. Generalizing (1.7.3)–(1.7.7), with the same proof, one has:

- (1.7.12.1) The restriction to $\mathbb{Z}_L(1)^I$ of an ℓ -adic representation of $W^{\text{mod}}(X - D)$ is quasi-unipotent.
- (1.7.12.2) If F' and F'' are two liftings in W^{mod} of $F \in W(k_1/k)$, the eigenvalues of F' and F'' coincide up to multiplication by roots of unity. Thus the ι -weights are defined as in (1.7.4).
- (1.7.12.3) If the ι -weights of V are integers, there is on V one and only one weight filtration M , as in (1.7.5). The analogues of (1.7.6) and (1.7.7) are true.

(1.8) Local monodromy of pure sheaves

The conventions of (0.7) are in force. Let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$, and let $j : U_0 \hookrightarrow X_0$ be an open subset whose complement is the finite set S_0 .

Lemma (1.8.1). If $\mathcal{F}_0$ is a lisse sheaf on U_0 pointwise ι -pure of weight β , then, for every $x \in |S_0|$ and every eigenvalue α of F_x on $(j_*\mathcal{F}_0)_x$, one has

$$w_{N(x)}(\alpha) \leq \beta.$$

Removing, if necessary, a point of X_0 lying in U_0 , one reduces to the case where X_0 is affine. One then has the equality (1.4.7.2)

$$\prod_{x \in |U_0|} \det(1 - F_x t^{\deg x}, \mathcal{F}_0)^{-1} \prod_{x \in |S_0|} \det(1 - F_x t^{\deg x}, j_*\mathcal{F}_0)^{-1} = \frac{\det(1 - Ft, H_c^1(X, j_*\mathcal{F}))}{\det(1 - Ft, H^0(X, j_*\mathcal{F}))}. \quad (1.8.1.1)$$

In this formula:

- a) By (1.4.6), the first factor on the left converges for $|t| < q^{-(\beta+2)/2}$ and has neither zero nor pole in this region.
- b) By (1.4.3), the right-hand side has no pole for $|t| < q^{-(\beta+2)/2}$.
- c) Comparing a) and b), one finds that the remaining factors

$$\det(1 - F_x t^{\deg x}, j_*\mathcal{F}_0)^{-1} \quad (x \in |S_0|)$$

are without pole for $|t| < q^{-(\beta+2)/2}$.

If α is an eigenvalue of F_x on $(j_*\mathcal{F}_0)_x$, then $|\alpha| \leq N(x)^{(\beta+2)/2}$, that is

$$w_{N(x)}(\alpha) \leq \beta + 2. \quad (1.8.1.2)$$

To conclude, one applies (1.8.1.2) to tensor powers of $\mathcal{F}_0$. Since $(j_*\mathcal{F}_0)^{\otimes k}$ is a subsheaf of $j_*(\mathcal{F}_0^{\otimes k})$, the eigenvalues α^k satisfy $w_{N(x)}(\alpha^k) \leq k\beta + 2$, hence $w_{N(x)}(\alpha) \leq \beta + 2/k$. Letting $k \rightarrow \infty$ gives the lemma.

Remark (1.8.2). Compare the zeros of the two sides of (1.8.1.1). On the left, there are none for $|t| < q^{-(\beta+2)/2}$. On the right, the denominator does not vanish in this same region. Therefore the numerator cannot vanish there. Thus for every eigenvalue α of F on $H_c^1(X, j_*\mathcal{F})$, one has $w_q(\alpha) \leq \beta + 2$.

(1.8.3). Let $y \in S$, with image s in S_0 , let η be the generic point of the henselian trait with residue field $X_0(s)$, and let $\bar{\eta}$ be a geometric generic point of its strict henselization. The fibre $\mathcal{F}_{0\bar{\eta}}$ of $\mathcal{F}_0$ at $\bar{\eta}$ is a representation of the Weil group $W(\bar{\eta}/\eta)$. We write M for its local monodromy filtration (1.7.2).

Theorem (1.8.4). With the notation of (1.8.1) and (1.8.3), if $\mathcal{F}_0$ is pointwise ι -pure of weight β , the representation

$$\mathrm{Gr}_i^M(\mathcal{F}_{0\bar{\eta}})$$

of $W(\bar{\eta}/\eta)$ is ι -pure of weight $\beta + i$.

By twisting (1.2.7), we reduce to the case $\beta = 0$. Replace X_0 by a finite covering, and U_0 , $\mathcal{F}_0$, and s by their inverse images. This changes neither the weights, nor $\mathcal{F}_{0\bar{\eta}}$, nor the filtration M , and reduces us to the case where formula (1.7.2.1) for the representation of $W(\bar{\eta}/\eta)$ on $\mathcal{F}_{0\bar{\eta}}$ holds on all of I :

$$\rho(\sigma) = \exp(t_\ell(\sigma)N) \quad (\sigma \in I).$$

Under these hypotheses,

$$\mathrm{Ker} N = (\mathcal{F}_{0\bar{\eta}})^I,$$

and this vector space is the fibre of $j_*\mathcal{F}_0$ at $\bar{s}$. By (1.8.1), if α is an eigenvalue of F on $\mathrm{Ker} N$, then $|\iota(\alpha)| \leq 1$.

Let α be an eigenvalue of F on $P_{-j}(\mathcal{F}_{0\bar{\eta}})$. By (1.6.14.2), one first obtains $|\iota(\alpha)| \leq 1$. Apply this result to $\mathcal{F}_0 \otimes \mathcal{F}_0$. By (1.6.14.4),

$$P_{-j}(\mathcal{F}_{0\bar{\eta}}) \otimes P_{-j}(\mathcal{F}_{0\bar{\eta}})(-j)$$

is a direct factor of $P_0(\mathcal{F}_{0\bar{\eta}} \otimes \mathcal{F}_{0\bar{\eta}})$; hence $\alpha^2 q^j$ is an eigenvalue of F on that P_0 , and $|\iota(\alpha^2 q^j)| \leq 1$, i.e. $|\iota(\alpha)| \leq q^{-j/2}$. Applying this last result to the dual sheaf and using (1.6.14.5), one obtains also $|\iota(\alpha^{-1} q^{-j})| \leq q^{-j/2}$, i.e. $|\iota(\alpha)| \geq q^{-j/2}$. Thus the eigenvalues α of F on $P_{-j}(\mathcal{F}_{0\bar{\eta}})$ have absolute value $q^{-j/2}$, and one concludes by (1.6.14.3).

Corollary (1.8.5). Assume that $\mathcal{F}_0$ is a lisse sheaf on U_0 and that it admits a finite increasing filtration W by lisse subsheaves such that, for every i , $\mathrm{Gr}_i^W(\mathcal{F}_0)$ is pointwise ι -pure of weight i . Write again W for the induced filtration on $\mathcal{F}_{0\bar{\eta}}$. Then $\mathcal{F}_{0\bar{\eta}}$ is ι -mixed with integral weights; the local monodromy filtration of V relative to W exists, and it coincides with the weight filtration of $\mathcal{F}_{0\bar{\eta}}$ (1.7.5).

By (1.7.5) and (1.8.4) applied to $\mathrm{Gr}^W(\mathcal{F}_0)$, the characteristic properties (1.6.13) are satisfied by the weight filtration.

(1.8.6). Let D_0 be a smooth divisor in a scheme X_0 smooth over $\mathbb{F}_q$, let U_0 be the complementary open, and let $\mathcal{F}_0$ be a lisse sheaf on U_0 tamely ramified along D_0 . Suppose that $\mathcal{F}_0$ admits a finite increasing filtration W by lisse subsheaves such that $\mathrm{Gr}_i^W \mathcal{F}_0$ is pointwise ι -pure of weight i . Suppose that D_0 is defined by an equation t , and let $\mathcal{F}_0[D_0]$ be the sheaf on D_0 endowed with an action of $\mathbb{Z}_\ell(1)$ obtained from (1.7.8). Restricting $\mathcal{F}_0$ to curves transverse to D_0 , one deduces:

Corollary (1.8.7). With the notation of (1.8.6), the local monodromy filtration around D of $\mathcal{F}_0[D_0]$, relative to W , exists. For this filtration, denoted M , the sheaf $\mathrm{Gr}_i^M \mathcal{F}_0[D_0]$ is pointwise ι -pure of weight i .

Remark (1.8.8).

- 1) With the notation of (1.8.1), if $\mathcal{F}_0$ is lisse on U_0 , then $(j_*\mathcal{F}_0)_{\bar{s}} = (\mathcal{F}_{0\bar{\eta}})^I$. This group is deduced from $\mathrm{Ker} N$ by taking invariants by a finite group I/I_1 . Under the hypotheses of (1.8.4), $\mathrm{Ker} N \subset M_0$, and the induced filtration M on $(j_*\mathcal{F}_0)_{\bar{s}}$ has associated graded $\mathrm{Gr}_i^M(j_*\mathcal{F}_0)_{\bar{s}}$ ι -pure of weight $\beta + i$, and zero for $i > 0$.

- 2) With the notation of (1.8.6), if $\mathcal{F}_0$ is lisse on U_0 and tamely ramified along D_0 , the restriction of $j_*\mathcal{F}_0$ to D_0 is the subsheaf of invariants of a finite group in $\text{Ker } N \subset \mathcal{F}_0[D_0]$. If $\mathcal{F}_0$ is pointwise ι -pure of weight β , twisting (1.8.7) gives that, for the monodromy filtration M , $\text{Gr}_i^M(j_*\mathcal{F}_0|D_0)$ is pointwise ι -pure of weight $\beta + i$, and is zero for $i > 0$.
- 3) Under the hypotheses of (1.8.6), W and the monodromy define on $j_*\mathcal{F}_0|D_0$ a filtration M for which $\text{Gr}_i^M(j_*\mathcal{F}_0|D_0)$ is lisse and ι -pure of weight i .

Corollary (1.8.9). Let X_0 be a scheme of finite type over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be an open subset, and let $\mathcal{F}_0$ be an ι -mixed sheaf of pointwise weights $\leq p$ on U_0 . Then $j_*\mathcal{F}_0$ is again ι -mixed of pointwise weights $\leq p$. If the pointwise weights of $\mathcal{F}_0$ are integral, those of $j_*\mathcal{F}_0$ are integral as well.

We use the following devissages, justified by (1.2.5): devissage in $\mathcal{F}_0$ by exact sequences; replacement of the support by a locally closed subscheme; normalization by a finite surjective morphism; replacing the open by a dense smaller open to make infinity a divisor; and finally localization at the generic point at infinity. These reductions bring the proof to (1.8.8), 2).

Corollary (1.8.10). Let $\mathcal{F}_0$ be a lisse sheaf on X_0 of finite type over $\mathbb{F}_q$, and let $j : U_0 \hookrightarrow X_0$ be a dense open subset. If the restriction of $\mathcal{F}_0$ to U_0 is pointwise ι -pure, then $\mathcal{F}_0$ itself is pointwise ι -pure.

Let β be the weight of $\mathcal{F}_0|U_0$. Since $\mathcal{F}_0 \rightarrow j_*j^*\mathcal{F}_0$ and likewise for the dual, (1.8.9) shows that $\mathcal{F}_0$ is ι -mixed of weights $\leq \beta$, while its dual is ι -mixed of weights $\leq -\beta$. Thus the weights are also $\geq \beta$, and the assertion follows. Equivalently, one can reduce by curves and normalization to the case of a smooth curve and then use (1.8.1) directly.

Corollary (1.8.11). Every lisse ι -mixed sheaf $\mathcal{F}_0$ on a normal scheme X_0 of finite type over $\mathbb{F}_q$ is a successive extension of lisse pointwise ι -pure sheaves.

One may suppose X_0 connected and reduce to the case where the sheaf is irreducible. Since X_0 is normal, its restriction to any nonempty open subset is still irreducible; for a suitable open subset it is pointwise ι -pure by the definition of mixed sheaves, and (1.8.10) applies. Later, in (3.4.1), the normality hypothesis will be removed.

Corollary (1.8.12). Let $\mathcal{F}_0$ be a lisse ι -mixed sheaf on a connected scheme X_0 of finite type over $\mathbb{F}_q$. If $\mathcal{F}_0$ is ι -pure at one point of X_0 , then it is pointwise ι -pure.

For X_0 normal this follows from (1.8.11): the weights of $\mathcal{F}_0$ are read from its fibre at any point. The general case is obtained by applying this to the normalization and then descending the conclusion.

(1.8.13)–(1.8.14). The preceding discussion has the expected analogues for variations in Hodge theory. A variation of mixed Hodge structures on a smooth analytic space S consists of a local system $V_{\mathbb{Z}}$ of free finitely generated $\mathbb{Z}$ -modules, an increasing finite filtration W of $V_{\mathbb{Q}}$, and a decreasing finite filtration F of $V_{\mathbb{C}}$ by vector subbundles, holomorphically varying and satisfying Griffiths transversality, such that W and F define at each point of S a mixed Hodge structure. The variations occurring in algebraic geometry satisfy additional properties, for instance the polarizability of the variations $\text{Gr}^W(V)$.

Let (V, W, F) be such a variation on the punctured disk D^* , let $t \in D^*$, let V be the fibre at t , and let T be the monodromy. Suppose for simplicity that T is unipotent, and put $N = -\log T$.

Problem (1.8.15). Isolate a class of good variations of mixed Hodge structures on D^* such that, for a good unipotent variation (V, W, F) , there exists on V a finite increasing filtration M

with $NM_a \subset M_{a-2}$ and such that N^b induces isomorphisms

$$N^b : \mathrm{Gr}_{a+b}^M \mathrm{Gr}_a^W V \xrightarrow{\sim} \mathrm{Gr}_{a-b}^M \mathrm{Gr}_a^W V.$$

One would also like:

- a) the variation V to be asymptotic, in a suitable sense, to a nilpotent orbit: a good variation for which F at $t \exp(u)$ is the transform by $\exp(-uN)$ of the filtration deduced from F_t by horizontal transport from t to $t \exp(u)$;
- b) for a nilpotent orbit, (V, M, F) to be again a variation of mixed Hodge structures.

(1.9) Tame local monodromy of mixed sheaves

This subsection, which will not be used later, generalizes (1.8.7) to the case of normal-crossing divisors.

(1.9.1). Let X be a smooth scheme over $\mathbb{F}_q$, let D be a normal-crossing divisor on X , the transverse union of smooth divisors D_i ($i \in I$), and let $E = \bigcap_{i \in I} D_i$. We assume that each D_i is globally defined by an equation $t_i = 0$, in order to apply the construction of (1.7.8).

Let $\mathcal{F}$ be a lisse sheaf on $X - D$, tamely ramified along D . Suppose that it has a finite increasing filtration W by lisse subsheaves such that $\mathrm{Gr}_i^W \mathcal{F}$ is pointwise ι -pure of weight i . Let L be the set of primes distinct from p , and define $\mathcal{F}[E]$ as in (1.7.8). It is a lisse sheaf on E , equipped with an action of $\mathbb{Z}_L(1)^I$; this action is quasi-unipotent, hence gives nilpotent endomorphisms

$$N_i : \mathcal{F}[E](1) \longrightarrow \mathcal{F}[E].$$

Given a sheaf N , two finite increasing filtrations W' and W'' of N , and a finite family of endomorphisms

$$N_\alpha : N(1) \longrightarrow N \quad (\alpha \in A),$$

we shall consider the condition

$$L(W', W'', (N_\alpha)_{\alpha \in A}) :$$

the N_α stabilize W' , satisfy $N_\alpha W'_a(1) \subset W'_{a-2}$, and, for all rational numbers $c_\alpha > 0$, the endomorphism

$$N = \sum_{\alpha \in A} c_\alpha N_\alpha$$

induces isomorphisms

$$N^b : (\mathrm{Gr}_{a+b}^{W''} \mathrm{Gr}_a^{W'} N)(b) \xrightarrow{\sim} \mathrm{Gr}_{a-b}^{W''} \mathrm{Gr}_a^{W'} N.$$

By (1.6.13), if W' and the family N_α are given, there is at most one filtration W'' satisfying this condition. In its statement one may replace “positive rational number” by “positive integer,” since one can replace the c_α by nc_α and N by nN .

Theorem (1.9.2). With the hypotheses and notation of (1.9.1), there is on $\mathcal{F}[E]$ a unique system of finite increasing filtrations $W(J)$, $J \subset I$, by lisse subsheaves stable under $\mathbb{Z}_L(1)^I$, hence under the N_i , such that:

- a) $W(\emptyset)$ is the filtration induced from W by functoriality;
- b) for $J \subset K$, the condition

$$L(W(J), W(K), (N_k)_{k \in K-J})$$

is satisfied.

Moreover, $\mathrm{Gr}_a^{W(I)} \mathcal{F}[E]$ is pointwise ι -pure of weight a .

The uniqueness of the $W(J)$ follows at once from (1.6.13); it is enough to have the conditions for $\emptyset \subset K$. For existence, one first treats the following special case.

Lemma (1.9.3). Let $(c_k)_{k \in I}$ be a system of positive integers. The sheaf $\mathcal{F}[E]$ admits a unique finite increasing filtration $W(I)$ by lisse subsheaves such that

$$N = \sum_{k \in I} c_k N_k$$

satisfies the condition $L(W, W(I), N)$. Moreover $\mathrm{Gr}_a^{W(I)} \mathcal{F}[E]$ is pointwise ι -pure of weight a .

By uniqueness, it is enough to prove the existence of the required filtration at one geometric point of each connected component of E . After a finite extension of the base field, one may suppose the point rational. Complete the equations t_i to a system of local coordinates (t_i, u_j) centered at that point. Restrict $\mathcal{F}$ to the curve defined parametrically by $u_j = 0$, $t_i = t^{c_i}$. On the strict localization of this curve, the geometric generic fibre identifies with $\mathcal{F}[E]$, and the logarithm of monodromy is precisely $N = \sum c_i N_i$. The assertion follows from (1.8.5).

(1.9.4). We deduce the theorem from this special case. The filtration $W(I)$ of (1.9.3) is independent of the choice of the c_k , since it is characterized independently of them as a weight filtration. For $J \subset I$, apply (1.9.3) to the corresponding partial intersection. This gives a weight filtration $W(J)$. Reapplying the construction to the residual divisor components gives, for $J \subset K$, the condition

$$L(W(J), W(K), (N_k)_{k \in K-J}).$$

The compatibilities with the identifications of (1.7.9) then give the required system on $E = D_I$.

Variant (1.9.5). Let S be a scheme of finite type over $\mathbb{Z}$, $f : X \rightarrow S$ a smooth morphism, $D \subset X$ a relative normal-crossing divisor, and $\mathcal{F}$ a lisse sheaf on $X - D$, tamely ramified along D . A variant of (1.9.2) holds in this setting; it is verified fibre by fibre by reduction to (1.9.2). The case of a normal-crossing divisor in a regular scheme of finite type over $\mathbb{Z}$ appears inaccessible, already for $S = \mathrm{Spec}(\mathbb{Z}[1/\ell])$.

Remark (1.9.6). As explained in [2], one expects the theory of variations of Hodge structures, always assumed polarizable, often to run parallel to that of lisse pointwise pure sheaves. Cattani and Kaplan have proved an analogue of (1.9.2), restricted to the pure case, for variations of Hodge structures on $(D^*)^I$. In their context, the coefficients c_α in the condition $L(W', W'', (N_\alpha))$ may even be taken to be real > 0 , not merely rational > 0 .

(1.10) Non-archimedean absolute values

The conventions of (0.7) are in force. Let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be an open subset whose complement is the finite set S_0 , and let $\mathcal{F}_0$ be a lisse sheaf on U_0 . In (1.8) the trace formula and local monodromy were used to study the archimedean absolute values of the Frobenius eigenvalues at the points of S_0 . Here we study absolute values which need not be archimedean.

Choose either an isomorphism $\iota : \overline{\mathbb{Q}}_\ell \simeq \mathbb{C}$, or a prime ℓ' , an algebraic closure $\overline{\mathbb{Q}}_{\ell'}$, and an isomorphism $\overline{\mathbb{Q}}_\ell \simeq \overline{\mathbb{Q}}_{\ell'}$. In the latter case the natural absolute value is normalized by $|\ell'| = 1/\ell'$.

Lemma (1.10.1). Let $b > 0$. If, for every $x \in |U_0|$ and every eigenvalue α of F_x on $\mathcal{F}_0$, one has

$$|\alpha| \leq b^{\deg x},$$

then, for every $x \in |S_0|$ and every eigenvalue α of F_x on $j_* \mathcal{F}_0$, one still has

$$|\alpha| \leq b^{\deg x}.$$

One repeats the proof of (1.8.1), treating first the non-archimedean case and indicating the archimedean modifications. In the trace formula, the first factor converges for $|t| < 1/b$, while the right hand side has no pole in this disk. Hence the remaining factors

$$\det(1 - F_x t^{\deg x}, j_* \mathcal{F}_0) \quad (x \in |S_0|)$$

have no pole there, which proves the lemma.

(1.10.2). Let $s \in |S_0|$, and let $\bar{s}, \eta, \bar{\eta}$ be as in (1.8.3). We call “eigenvalues of F_s on $\mathcal{F}_0$ ” the eigenvalues of a lift of $F_s \in W(\bar{s}/s)$ to $W(\bar{\eta}/\eta)$ acting on the fibre. This terminology is used only when the choice of the lift is immaterial.

Theorem (1.10.3). In the non-archimedean case, let $0 < b < c < \infty$. If, for every point $x \in |U_0|$ and every eigenvalue α of F_x on $\mathcal{F}_0$,

$$b^{\deg x} \leq |\alpha| \leq c^{\deg x},$$

then the same inequalities hold for every $s \in |S_0|$ and every eigenvalue of F_s on $\mathcal{F}_0$.

After replacing U_0 by a finite cover, local monodromy may be assumed unipotent. The action of inertia is then written

$$\sigma \longmapsto \exp(t_\ell(\sigma)N).$$

Lemma (1.10.1) gives the upper bound on the eigenvalues on $\text{Ker } N$, and (1.6.14.3) propagates it to the whole representation. The lower bound is obtained by applying the upper bound to the dual sheaf.

(1.10.4)–(1.10.8). Applying (1.10.3) to exterior powers gives relations between Newton polygons attached to the Frobenius endomorphisms at points of U_0 and those at the points of S_0 . We record only the form needed later. For $\ell' \neq p$, Frobenius eigenvalues are in practice ℓ' -adic units, and one obtains:

Corollary (1.10.5). If, for every $x \in |U_0|$, the eigenvalues α of F_x on $\mathcal{F}_0$ are ℓ' -adic units, then for every $s \in |S_0|$ the eigenvalues of F_s on $\mathcal{F}_0$ are still ℓ' -adic units.

For $\ell' = p$ it is useful to use valuations. If N is a power of p , let v_N be the valuation of $\mathbb{Q}_p$ normalized by $v_N(N) = 1$. The preceding argument, applied to exterior powers and determinants, gives corresponding inequalities for the valuations $v_N(\alpha)$.

Corollary (1.10.9). Suppose that for almost all $x \in |U_0|$ the eigenvalues α of F_x on $\mathcal{F}_0$ satisfy

$$\gamma \leq v_N(\alpha) \leq \delta.$$

If n is the integer part of $\delta - \gamma$, then the logarithm N of the unipotent part of the local monodromy at a point $s \in |S_0|$ satisfies

$$N^{n+1} = 0.$$

Indeed, if m is minimal with $N^{m+1} = 0$, then F_s has eigenvalues α and $q^m \alpha$, and one applies (1.10.3).

(1.11) Specialization of monodromy

Proposition (1.11.1). Let S be an irreducible scheme, let $f : X \rightarrow S$ be a smooth morphism, purely of relative dimension 1, with geometrically connected fibres, let g be a section of f , and let $\mathcal{F}$ be a lisse $\mathbb{Z}_\ell$ -sheaf on X . For each geometric point $\bar{s}$ of S , let $M_{\bar{s}}$ be the image of

$$\pi_1(X_{\bar{s}}, g(\bar{s}))$$

in

$$\mathrm{Aut}((\mathcal{F}/\ell^n \mathcal{F})_{g(\bar{s})}).$$

Then there is an open subset $U \subset S$ such that, for every n and every geometric point $\bar{s}$ of U , $M_{\bar{s}}$ is the fibre at $\bar{s}$ of a lisse subsheaf M^n of

$$\mathrm{Aut}(g^* \mathcal{F}/\ell^n \mathcal{F})$$

on U .

One considers the finite étale cover

$$I_n = \mathrm{Isom}(\mathcal{F}/\ell^n \mathcal{F}, f^* g^* \mathcal{F}/\ell^n \mathcal{F})$$

of X , with the identity section above g . After shrinking S , one constructs simultaneously the subsheaves M^n and a tower of finite étale covers. The construction is compatible in n , and specialization of the fundamental group gives the desired local constancy.

Remarks (1.11.2)–(1.11.5). For the groups $\pi_1(X_{\bar{s}}, g(\bar{s}))$ to form a local system on S , it is enough that X admit a relative compactification

$$X \hookrightarrow \overline{X}$$

where $\overline{X}$ is proper and smooth over S , of relative dimension 1, and X is the complement of a finite étale subscheme T of S ; it is enough for such a compactification to exist after a finite radicial surjective base change.

With these notations, if $X_n \rightarrow X$ is one of the finite étale covers constructed above, one also has compatible compactifications

$$X_n \hookrightarrow \overline{X}_n$$

whose complements T_n are finite over S . For a geometric point $\bar{s}$, a point $t \in T_{\bar{s}}$, and inertia I_t , the image of I_t in $M_{\bar{s}}$ specializes compatibly when $(\bar{s}, t)$ specializes. This follows after reducing to the strictly local case and using tame coverings of the compactified curves.

The proposition remains valid for any morphism of finite type f : localization on S and Mayer–Vietoris reduce to the case where f is flat with normal fibres; restriction to an open then reduces to the smooth quasi-projective case.

2. The Hadamard–de la Vallée-Poussin method

(2.1) The method

(2.1.1). Let Γ be a group isomorphic to $\mathbb{Z}$ or to $\mathbb{R}$, let $\omega_1 : \Gamma \rightarrow \mathbb{R}_{>0}$ be a non-trivial quasi-character, let G be a locally compact group extension of Γ by a compact group G^0 , let Σ be an infinite countable set, and let $(x_v)_{v \in \Sigma}$ be a family of conjugacy classes in G . We impose hypotheses (A) and (B).

(A1) If $\Gamma \simeq \mathbb{R}$, the extension G is trivial. This hypothesis is in fact automatic.

(A2) If $\Gamma \simeq \mathbb{Z}$, the centre of G maps to a subgroup of finite index in Γ .

For $s \in \mathbb{C}$, put $\omega_s = \omega_1^s$. If $\Gamma \simeq \mathbb{Z}$, we write $q > 1$ and $\deg : \Gamma \simeq \mathbb{Z}$ so that

$$\omega_1(\gamma) = q^{-\deg(\gamma)}.$$

Let

$$N_v = \omega_{-1}(x_v).$$

The set of connected components of G is discrete, and we give G the corresponding Riemann-surface structure. The hypotheses (B) are:

$$N_v > 1$$

and

$$\prod_{v \in \Sigma} (1 - N_v^{-s})^{-1}$$

converges absolutely for $\Re(s) > 1$. For $\Gamma \simeq \mathbb{Z}$, this is the same as requiring $\deg(v) > 0$ and

$$\#\{v \mid \deg(v) = n\} = O(q^{n(1+\epsilon)})$$

for every $\epsilon > 0$.

For a finite-dimensional continuous complex representation v of G , the product

$$L(v) = \prod_{v \in \Sigma} \det(1 - v(x_v))^{-1}$$

converges absolutely in the region $\Re(v) > 1$, and one writes $L(v, s) = L(v\omega_s)$.

Examples (2.1.2)–(2.1.3). If $\Gamma \simeq \mathbb{R}$, the situation is described by a compact group G^0 , conjugacy classes $x_v^0 \subset G^0$, and real numbers $N_v > 1$; this is the framework considered by Serre. If X_0 is a smooth absolutely irreducible curve over $\mathbb{F}_q$, with function field K , one takes the Weil group extension

$$1 \longrightarrow \pi_1(X, \bar{x}) \longrightarrow W(X_0, \bar{x}) \longrightarrow \mathbb{Z} \longrightarrow 0,$$

sets $\Sigma = |X_0|$, and takes x_v to be geometric Frobenius.

(2.1.4). For $\Re(s) > 1$, the logarithmic derivative is

$$-\frac{L'}{L}(v, s) = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-ns} \operatorname{Tr}(v(x_v^n)).$$

Equivalently, for a unitary virtual representation v and real $\sigma > 1$,

$$-\frac{L'}{L}(\omega_\sigma v) = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-n\sigma} \operatorname{Tr} v(x_v^n).$$

If μ is a measure on G , or on the set of conjugacy classes of G , its Fourier transform is

$$\hat{\mu}(v) = \int \operatorname{Tr} v(g) d\mu.$$

For positive μ of finite total mass one has

$$\hat{\mu}(v \otimes \bar{v}) \geq 0.$$

The last displayed logarithmic derivative is the Fourier transform of the positive measure

$$\mu_\sigma = \sum_{v \in \Sigma} \sum_{n > 0} \log N_v N_v^{-n\sigma} \delta_{x_v^n}.$$

Therefore, for every unitary virtual representation v ,

$$-\frac{L'}{L}(\omega_\sigma(v \otimes \bar{v})) \geq 0 \quad (\sigma > 1).$$

Let $r(v)$ denote the order of the pole, the negative of the order of zero, of $L(v)$ at ω_1 . If r extends additively to the Grothendieck group of unitary representations and satisfies the positivity inequalities obtained above, then the Hadamard–de la Vallée-Poussin argument implies that $r(v) = 0$ for $v \neq 1$, except possibly for a one-dimensional character of order 2.

Variant and proof (2.1.5)–(2.1.8). It is enough to prove the assertion after replacing G by the compact group obtained from its unitary representations. In the compact case, let $[p : \tau]$ denote the multiplicity of an irreducible representation τ in p . For any finite set T of irreducible representations and any $\varepsilon > 0$, there is a representation p such that

$$\dim \tau [p \otimes \bar{p} : \tau] < (1 + \varepsilon)[p \otimes \bar{p} : 1] \quad (\tau \in T).$$

This follows from density of the algebra of central characters in the algebra of continuous central functions. Using the positivity of $r(p \otimes \bar{p})$, letting $\varepsilon \rightarrow 0$ and increasing T , one obtains the desired vanishing.

(2.1.9)–(2.1.12). For the Riemann zeta function, $G^0 = 1$, Σ is the set of primes, and $N_p = p$. The function

$$-\frac{\zeta'}{\zeta}(\sigma + it)$$

is the Fourier transform of the positive measure

$$\sum_{p,n} \log p p^{-n\sigma} \delta_{n \log p},$$

and hence is positive definite. The classical inequality

$$3A_\sigma(0) + 4\Re A_\sigma(t) + 2\Re A_\sigma(2t) \geq 0$$

is the form used by Mertens to derive the theorem of Hadamard and de la Vallée-Poussin that $\zeta(s)$ has no zero on $\Re(s) = 1$.

Assume now that $\Gamma \simeq \mathbb{Z}$, that $L(1)$ is meromorphic for $\Re(s) > 1$ with a simple pole at ω_1 , no other pole, and no zero in that domain. For a central element $z \in G$ of positive degree d , translate the pieces G_{i+nd} to G_i . The measures

$$\mu = \sum_{v,n>0} \deg(v) q^{-n \deg(v)} \delta_{x_v^n}$$

and the Haar comparison measure have Fourier–Laplace transforms whose difference is holomorphic on $\Re(s) \geq 0$. Consequently the translated restrictions of μ converge vaguely to the corresponding translated Haar measure. This gives the equidistribution theorem (2.1.12).

(2.2) Purity and compactness

(2.2.1)–(2.2.4). Let $G_{\mathbb{C}}$ be an algebraic group extension

$$1 \longrightarrow G_{\mathbb{C}}^0 \longrightarrow G_{\mathbb{C}} \longrightarrow \mathbb{Z} \longrightarrow 0$$

with $G_{\mathbb{C}}^0$ reductive, and choose a real form $G_{\mathbb{R}}$ as in (2.2.1). The exact sequence

$$1 \longrightarrow G_{\mathbb{R}}^0 \longrightarrow G_{\mathbb{R}} \longrightarrow \mathbb{Z} \longrightarrow 0$$

has compact connected part. Two elements of $G_{\mathbb{R}}$ conjugate in $G_{\mathbb{C}}$ are already conjugate in $G_{\mathbb{R}}$. An element $g \in G_{\mathbb{C}}$ has a unique Jordan decomposition

$$g = g_s g_u = g_u g_s,$$

with g_s semisimple and g_u unipotent.

Let X_0 be a smooth connected scheme over $\mathbb{F}_q$, let $\mathcal{F}_0$ be a lisse $\mathbb{Q}_{\ell}$ -sheaf, and let G be the Zariski closure of the image of $W(X_0, \bar{x})$ in $\mathrm{GL}(\mathcal{F}_{\bar{x}})$, with the grading by degree. Assume that G^0 is reductive and that some representation with finite kernel on G^0 gives an ℓ -mixed sheaf. Then every linear representation of G gives an ℓ -mixed lisse sheaf.

Example (2.2.5). If $\mathcal{F}_0$ is lisse and ℓ -mixed on X_0 , and the restriction of the corresponding representation to $\pi_1(X, \bar{x})$ is semisimple, then the preceding hypotheses hold.

Proposition (2.2.6). Let $v \in |X_0|$, and let F_v be the image of a geometric Frobenius in G . The semisimple part $(F_v)_s$ is conjugate to an element of $G_{\mathbb{R}}$, uniquely up to $G_{\mathbb{R}}$ -conjugacy.

The uniqueness follows from the real-conjugacy lemma. The existence follows by applying the weight bounds to a faithful representation and using the compatibility of semisimple parts with representations.

(2.2.7)–(2.2.8). For X_0 a smooth connected scheme over $\mathbb{F}_q$, set

$$\Sigma = |X_0|, \quad x_v = (F_v)_s \in G_{\mathbb{R}}/\text{conjugacy}, \quad N_v = q^{\deg v}.$$

These data satisfy the hypotheses (A) and (B') of (2.1). If $\dim X_0 = 1$, they satisfy (B) as well. If $\dim X_0 = N$, the same statements hold after replacing the character by $q^{-N \deg}$; in what follows the character $q^{-\deg}$ is retained and the statements of (2.1) are adjusted accordingly.

The functors extension of scalars by ι and restriction to $G_{\mathbb{R}}$ give equivalences between linear representations of G , linear representations of $G_{\mathbb{C}}$, and continuous complex representations of $G_{\mathbb{R}}$. For a representation τ of $G_{\mathbb{R}}$, let $\mathcal{F}_0(\tau_{\ell})$ be the lisse sheaf obtained from the corresponding ℓ -adic representation. If τ is irreducible, this sheaf is pointwise pure of weight $2N(\tau)$. Moreover the L -function $L(\tau, s)$ is expressed by the formal zeta function

$$L(\tau, s) = Z(\mathcal{F}_0(\tau_{\ell}), t), \quad t = q^{-s}.$$

Proposition (2.2.9). If $\dim X_0 = 1$, the function $L(\tau)$ satisfies the non-vanishing condition (C).

Grothendieck's formula gives

$$Z(\mathcal{F}_0(\tau_{\ell}), t) = \frac{\det(1 - Ft, H_c^1(X, \mathcal{F}(\tau_{\ell})))}{\det(1 - Ft, H_c^0(X, \mathcal{F}(\tau_{\ell}))) \det(1 - Ft, H_c^2(X, \mathcal{F}(\tau_{\ell})))}.$$

If τ is unitary and is not a power of the quasi-character, then the restriction of τ to G^0 is non-trivial and the denominator is 1. The assertion follows from (2.2.8). For the trivial character, the assertion is the classical assertion for the zeta function of X_0 . A possible zero at a quadratic twist is excluded by passing to the corresponding double cover.

Corollary (2.2.10). Let X_0 be a smooth curve over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a lisse pointwise ι -pure sheaf of weight β on X_0 . If α is an eigenvalue of F on $H_c^1(X, \mathcal{F})$, then

$$w_q(\alpha) < \beta + 2.$$

This improves the trivial estimate coming from (1.8.2), and will be improved further in (3.2).

The assertion is invariant under finite extension of the base field. One reduces to the absolutely irreducible case, and then by exact sequences to the case where $\mathcal{F}_0$ is irreducible. The hypotheses of (2.2.5) are then satisfied. After twisting, one may suppose $\beta = 0$. The corresponding representation is unitary on $G_{\mathbb{R}}$, and (2.2.9), together with the cohomological expression of L , yields the stated strict inequality.

3. The fundamental theorem

(3.1) A calculation of vanishing cycles

The results of this paragraph will be used to compute, modulo integers, the weights of certain groups of vanishing cycles.

(3.1.1). Let S be a smooth projective surface over an algebraically closed field k , let D be a normal-crossings divisor on S , let $V = S - D$, let $j : V \hookrightarrow S$ be the inclusion, and let $\mathcal{F}$ be a sheaf on V , tamely ramified along D . We shall study the cohomology groups $H_c^*(V, \mathcal{F})$ by the method of Lefschetz pencils.

As in I.5, embed S in a projective space P and sweep it out by a pencil of hyperplanes $(H_t)_{t \in A^*}$. The pencil has axis A , and $\tilde{S}$ denotes the space of pairs (x, t) with $x \in H_t$. Let $\tilde{V}$ be the inverse image of V in $\tilde{S}$. The two projections give a diagram

$$\begin{array}{ccc} \tilde{V} & \xrightarrow{\tilde{j}} & \tilde{S} \xrightarrow{\pi} S \\ f \downarrow & & \downarrow f \\ A^* & & A^* \end{array}$$

(the repeated letter f should cause no confusion). The fibers of $f : \tilde{S} \rightarrow A^*$ are the hyperplane sections S_t .

We impose the following general-position hypotheses.

- (A) The axis A is transverse to S and disjoint from D ; hence $\tilde{S}$ is smooth, obtained from S by blowing up the finite set $S \cap A$, and none of these points lies on D .
- (B) The only singularities of f are ordinary quadratic points, and none of them lies on D .
- (C) On the normalization D' of D , the only singularities of f are ordinary quadratic points; none of them lies above a singular point of D .
- (D) Each exceptional fiber contains only one exceptional point.

An exceptional point is one of the following: a critical point of f on S , a critical point of f on D' , or a singular point of D . Thus exceptional fibers are of one of three types: a curve with an ordinary double point, a tangency of D with S_t , or two branches of D meeting on S_t .

There is a canonical isomorphism

$$H_c^*(\tilde{V}, \pi^* \mathcal{F}) \simeq H_c^*(V, \mathcal{F}) \oplus H^*(S \cap A, \mathcal{F})(-1)[-2]. \quad (3.1.1.3)$$

For what follows it would suffice to know the injectivity

$$\pi^* : H_c^*(V, \mathcal{F}) \hookrightarrow H_c^*(\tilde{V}, \pi^* \mathcal{F}), \quad (3.1.1.4)$$

whose Poincare-dual transpose has a retraction. We use the Leray spectral sequence

$$E_2^{pq} = H^p(A^*, R^q f_! \pi^* \mathcal{F}) \Rightarrow H_c^{p+q}(\tilde{V}, \pi^* \mathcal{F}). \quad (3.1.1.5)$$

The sheaves $R^q f_! \pi^* \mathcal{F} = R^q f_!(\tilde{j}_! \pi^* \mathcal{F})$ are lisse except at the exceptional values of f .

Let t be an exceptional value, let $A_{(t)}^*$ be the henselization of A^* at t , and let $\bar{\eta}$ be a geometric generic point. Applying the theory of vanishing cycles to $\tilde{j}_! \pi^* \mathcal{F}$ over $A_{(t)}^*$, the sheaves of vanishing cycles Φ^q are concentrated at the exceptional point x of S_t , and one has a long exact sequence

$$\cdots \rightarrow (R^q f_! \pi^* \mathcal{F})_t \rightarrow (R^q f_! \pi^* \mathcal{F})_{\bar{\eta}} \rightarrow \Phi^q \rightarrow (R^{q+1} f_! \pi^* \mathcal{F})_t \rightarrow \cdots \quad (3.1.1.6)$$

(3.1.2). Assume in addition that the local monodromy of $\mathcal{F}$ around D is unipotent. If $d \in D$, let $S_{(d)}$ be the strict localization of S at d , and $V_{(d)}$ the inverse image of V . The hypothesis gives a finite filtration F of $\mathcal{F}|_{V_{(d)}}$ by lisse subsheaves whose graded pieces extend as constant sheaves over $S_{(d)}$. We denote by $\text{Gr}_i^F(\mathcal{F})_d$ the fiber at d of this constant prolongation.

For a two-element set B , write $\mathbb{Z}^B/\mathbb{Z}$ for the quotient by the diagonal copy of $\mathbb{Z}$; equivalently it is the free rank-one group endowed with two opposite identifications with $\mathbb{Z}$, indexed by B .

(3.1.3) Case (a). Since $\tilde{j}_! \pi^* \mathcal{F}$ is lisse at the ordinary double point x , the vanishing cycles are those of the constant sheaf, tensored by the fiber of $\mathcal{F}$. Picard-Lefschetz theory in relative dimension one gives

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.3.1)$$

and, if B is the two-element set of branches of S_t at x ,

$$\Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \mathcal{F}_x(-1) \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.3.2)$$

It may be described by the Kummer covering $T^\ell = uv^{-1}$ when u and v are local equations of the two branches.

(3.1.4) Case (b). First suppose $\mathcal{F} = \mathbb{Q}_\ell$. From the exact sequence

$$0 \rightarrow j_! \mathbb{Q}_\ell \rightarrow \mathbb{Q}_\ell \rightarrow i_* \mathbb{Q}_{\ell D} \rightarrow 0 \quad (3.1.4.1)$$

and the vanishing-cycle long exact sequence one obtains

$$\Phi^q(j_! \mathbb{Q}_\ell) \simeq \Phi^{q-1}(\mathbb{Q}_{\ell D}). \quad (3.1.4.2)$$

If B is the set of two points of the henselized normalization $D'_{(x)}$ above the generic point, then

$$\Phi^q(j_! \mathbb{Q}_\ell) = 0 \quad (q \neq 1), \quad (3.1.4.3)$$

and the non-zero group is the corresponding one-dimensional branch quotient. In general, by devissage with the filtration F of (3.1.2),

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.4.5)$$

and

$$\text{Gr}_i^F \Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \text{Gr}_i^F(\mathcal{F})_x \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.4.6)$$

(3.1.5) Case (c). Let B be the set of the two branches of D through x . With $i : D' \rightarrow S$ the normalization, locally near x one has

$$0 \rightarrow j_! \mathbb{Q}_\ell \rightarrow \mathbb{Q}_\ell \rightarrow i_* \mathbb{Q}_\ell \rightarrow \mathbb{Q}_{\ell x} \otimes \mathbb{Q}_\ell^B/\mathbb{Q}_\ell \rightarrow 0. \quad (3.1.5.1)$$

Since f and fi are smooth at the relevant points, the only contribution to vanishing cycles is again in degree one. Thus

$$\Phi^q(\tilde{j}_! \pi^* \mathcal{F}) = 0 \quad (q \neq 1), \quad (3.1.5.3)$$

and, for F as in (3.1.2),

$$\text{Gr}_i^F \Phi^1(\tilde{j}_! \pi^* \mathcal{F}) \simeq \text{Gr}_i^F(\mathcal{F})_x \otimes (\mathbb{Z}^B/\mathbb{Z}) \otimes \mathbb{Q}_\ell. \quad (3.1.5.4)$$

(3.2) Dimension one

The conventions of (0.7) remain in force.

Proposition 1 (3.2.1). *Let X_0 be a smooth absolutely irreducible curve over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a lisse sheaf pointwise ι -pure and ι -real on X_0 . Then the polynomials*

$$\iota \det(1 - Ft, H_c^1(X, \mathcal{F}))$$

have real coefficients.

In Grothendieck's formula, the local factors on the left are real polynomials by hypothesis and the rational function on the right is real. If β is the weight of $\mathcal{F}_0$, the zeros of the factor attached to H_c^2 are precisely the poles of absolute value $q^{-(\beta+2)/2}$, hence are stable under complex conjugation. Poincaré duality deals with the proper case and the denominator in the cohomological expression is real; therefore the numerator is real.

Remark 1 (3.2.2). The arguments of (0.5) remove the absolute irreducibility hypothesis.

Theorem 3 (3.2.3). *Let X_0 be a smooth projective curve over $\mathbb{F}_q$, let $j : U_0 \hookrightarrow X_0$ be a dense open subset, and let $\mathcal{F}_0$ be a lisse sheaf on U_0 pointwise ι -pure of weight β . Then the eigenvalues α of F on $H^i(X, j_*\mathcal{F})$ satisfy*

$$w_q(\alpha) = \beta + i.$$

For $i = 1$, $H^1(X, j_*\mathcal{F})$ is the image of $H_c^1(U, \mathcal{F})$ in $H^1(U, \mathcal{F})$. The proof starts from the following induction statement.

Lemma 1 (3.2.4(k)). *Let U_0 be a smooth curve over $\mathbb{F}_q$ and let $\mathcal{F}_0$ be lisse, pointwise ι -pure of weight zero on U_0 . Then the eigenvalues α of F on $H_c^1(U, \mathcal{F})$ satisfy*

$$w_q(\alpha) < 1 + 2^{-k}.$$

The assertion for sheaves on a compactification $j : U_0 \hookrightarrow X_0$ follows from the long exact sequence attached to

$$0 \rightarrow j_!j^*\mathcal{G}_0 \rightarrow \mathcal{G}_0 \rightarrow \mathcal{G}_0/j_!j^*\mathcal{G}_0 \rightarrow 0.$$

Assume (3.2.4)(k), and prove (3.2.4)(k + 1). After finite base extension, choose a compactification X_0 and suppose first that the local monodromy along $X - U$ is tame and unipotent. Put

$$S_0 = X_0 \times X_0, \text{quad} V_0 = U_0 \times U_0,$$

let $D_0 = S_0 - V_0$, and take a Lefschetz pencil satisfying the general-position hypotheses of (3.1). Let

$$\mathcal{G}_0 = \mathcal{F}_0 \boxtimes \mathcal{F}_0 = p_1^*\mathcal{F}_0 \otimes p_2^*\mathcal{F}_0.$$

Lemma 2 (3.2.7). *The lisse sheaf $w^*R^1f_!\mathcal{G}_0$ on the non-exceptional locus W_0 is a direct factor of an ι -real sheaf.*

Indeed $\mathcal{G}_0$ is pointwise ι -pure and ι -real, and the fiber of $R^1f_!\mathcal{G}_0$ at a point is the H_c^1 of a curve with the restricted coefficient sheaf; apply (3.2.1).

Lemma 3 (3.2.8). *The sheaf $R^1f_!\mathcal{G}_0$ has no section supported on the exceptional set T .*

Lemma 4 (3.2.9). *If $\text{Gr}_i^G R^1f_!\mathcal{G}_0$ does not become constant on A^* , its restriction to W_0 is pointwise ι -pure of integral weight.*

The proof uses (1.8.4) at the boundary points and the calculation of vanishing cycles in (3.1). For each exceptional t , the five-term exact sequence

$$0 \rightarrow (R^1 f_! \mathcal{G}_0)_t \rightarrow (R^1 f_! \mathcal{G}_0)_{\bar{\eta}} \rightarrow \Phi^1 \rightarrow (R^2 f_! \mathcal{G}_0)_t \rightarrow 0 \quad (3.2.9.1)$$

shows both (3.2.8) and the claimed integrality of the weights of the nonconstant graded pieces.

Lemma 5 (3.2.10). *The constituents of $w^* R^1 f_! \mathcal{G}_0$ are pointwise pure of weights < 2 .*

This follows from (2.2.10) by testing the weights at one point.

Lemma 6 (3.2.11). *The eigenvalues α of F on*

$$H^1(A^*, R^1 f_! \mathcal{G})$$

satisfy $w_q(\alpha) < 2 + 2^{-k}$.

The constant graded pieces contribute no H^1 , and the nonconstant ones are handled by (3.2.5)(k).

Lemma 7 (3.2.12). *The eigenvalues on $E_2^{0,2}$ and on $E_2^{2,0}$ of the Leray spectral sequence are of weights < 2 .*

This is a direct consequence of the fiber formula and two applications of (1.4.4).

From the Leray spectral sequence (3.1.1.5), the eigenvalues on $H_c^2(\tilde{V}, \pi^* \mathcal{G})$ have weight $< 2 + 2^{-k}$. Since

$$H_c^1(U, \mathcal{F}) \otimes H_c^1(U, \mathcal{F}) \subset H_c^2(V, \mathcal{G}) \subset H_c^2(\tilde{V}, \pi^* \mathcal{G}),$$

Kunneth's formula gives the induction step. The tame-unipotent hypothesis is removed by a finite cover; pullback injects cohomology. Duality then gives Theorem (3.2.3).

(3.3) The general case

Theorem 4 (3.3.1). *Let $f : X \rightarrow Y$ be a morphism of schemes of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be a sheaf on X . If $\mathcal{F}$ is mixed of weights $\leq n$, then, for each i , $R^i f_! \mathcal{F}$ is mixed of weights $\leq n + i$.*

The proof uses successive devissages. Exact sequences reduce the assertion in the coefficient sheaf; an open-closed decomposition reduces it in X ; base change to an open part and its complement reduces it in Y ; Leray spectral sequences give transitivity. Universal homeomorphisms do not affect étale cohomology. Finally, morphisms of relative dimension zero preserve pointwise purity and have no higher direct images. These reductions reduce to the case where f has relative dimension one, $\mathcal{F}$ is lisse and pointwise pure, and X is the complement in a smooth projective relative curve of a divisor finite étale over Y , with tame ramification along the divisor.

On each geometric fiber, Theorem (3.2.3) gives purity for $R^i f_! j_* \mathcal{F}$. The local-monodromy theorem (1.8.4) gives the weights of the graded pieces of the local monodromy filtration on $i^* j_* \mathcal{F}$. The finite part is handled by the relative-dimension-zero case, and the exact sequence

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* i^* j_* \mathcal{F} \rightarrow 0$$

finishes the proof.

(3.3.2). Lower bounds for weights are obtained from the p -adic estimates of SGA 7 XXI, 5. A sheaf is called integral if all Frobenius eigenvalues are algebraic integers. The weights of a mixed integral sheaf are necessarily non-negative, since the norm of a pure algebraic integer of weight n has absolute value $q^{dn/2}$.

Corollary 1 (3.3.3). *Let $f : X \rightarrow Y$ be of finite type over $\mathbb{Z}$, and let $\mathcal{F}$ be an integral sheaf on X . If $\mathcal{F}$ is mixed of weights $\leq n$, then $R^i f_* \mathcal{F}$ is mixed of weights between 0 and $n + i$. If i is larger than the maximum dimension d of the fibers, the weights lie between $2(i - d)$ and $n + i$.*

For $Y = \text{Spec } \mathbb{F}_q$ this becomes:

Corollary 2 (3.3.4). *Let X_0 be of finite type over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be mixed of weights $\leq n$ on X_0 . Then $H_c^i(X, \mathcal{F})$ is mixed of weights $\leq n + i$. If $\mathcal{F}_0$ is integral, the weights are non-negative; if moreover $i > d = \dim X$, they are $\geq 2(i - d)$.*

Corollary 3 (3.3.5). *If X_0 is smooth of finite type over $\mathbb{F}_q$ and $\mathcal{F}_0$ is lisse mixed of weights $\geq n$, then $H^i(X, \mathcal{F})$ is mixed of weights $\geq n + i$.*

This is Poincare duality applied to the dual sheaf. Hence:

Corollary 4 (3.3.6). *If X_0 is smooth of finite type over $\mathbb{F}_q$ and $\mathcal{F}_0$ is lisse pure of weight n , then the image of*

$$H_c^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})$$

is pure of weight $n + i$.

(3.3.7)–(3.3.9). If an algebraic number α is pure of weight n relative to q , normalize p -adic valuations by $v(q) = 1$ and attach to α the pair $(v(\alpha), v(q^n \alpha^{-1}))$. For integral eigenvalues both coordinates are non-negative and sum to n . For the constant sheaf, the preceding bounds localize these pairs in the corresponding triangles in the (r, s) -plane. In particular, for X_0 proper and smooth one obtains again the main theorem of I, with projective replaced by proper:

$$\det(1 - Ft, H^i(X, \mathbb{Q}_\ell)) \in \mathbb{Z}[t]$$

is independent of $\ell \neq p$, and all complex roots have absolute value $q^{i/2}$.

(3.3.10)–(3.3.11). The theorem remains valid with “mixed” replaced by “ ι -mixed” and real-valued weights. The only weights in $R^i f_* \mathcal{F}$ are those $\leq \beta + i$ congruent modulo $\mathbb{Z}$ to a weight already occurring in $\mathcal{F}$. The smoothness assumption is used only to invoke Poincare duality, and may therefore be replaced by the corresponding rational-homology-manifold condition

$$Ra^! \mathbb{Q}_\ell = \mathbb{Q}_\ell(n)[2n]$$

for the structural morphism $a : X_0 \rightarrow \text{Spec } \mathbb{F}_q$.

(3.4) Application: the structure of mixed sheaves

Theorem 5 (3.4.1). *Let $\mathcal{F}_0$ be an ι -mixed sheaf on a scheme X_0 of finite type over $\mathbb{F}_q$.*

(i) There is a unique functorial decomposition

$$\mathcal{F}_0 = \bigoplus_{b \in \mathbb{R}/\mathbb{Z}} \mathcal{F}_0(b)$$

according to weights modulo $\mathbb{Z}$.

(ii) If the pointwise weights are integers and $\mathcal{F}_0$ is lisse, then $\mathcal{F}_0$ has a unique finite increasing weight filtration W by lisse subsheaves such that $\mathrm{Gr}_i^W \mathcal{F}_0$ is pointwise pure of weight i ; morphisms are strictly compatible with these filtrations.

(iii) If $\mathcal{F}_0$ is lisse and pointwise pure, and if X_0 is normal, then $\mathcal{F}$ on X is semisimple.

Lemma 8 (3.4.2). *For lisse sheaves $\mathcal{F}_0$ and $\mathcal{G}_0$ on X_0 there is an exact sequence*

$$0 \rightarrow H^0(X, \mathrm{Hom}(\mathcal{F}, \mathcal{G}))_F \rightarrow \mathrm{Ext}^1(\mathcal{F}_0, \mathcal{G}_0) \rightarrow H^1(X, \mathrm{Hom}(\mathcal{F}, \mathcal{G}))^F.$$

Here superscripts and subscripts denote invariants and coinvariants for the action of the Weil group.

Lemma 9 (3.4.3). *If $\mathcal{F}_0$ and $\mathcal{G}_0$ are lisse on smooth X_0 and pointwise pure of weights β and γ , then a geometrically non-trivial extension of $\mathcal{F}_0$ by $\mathcal{G}_0$ can exist only if $\beta \equiv \gamma \pmod{\mathbb{Z}}$ and $\beta > \gamma$.*

The same proof gives the corresponding vanishing criterion for Ext^1 with $\beta \geq \gamma$.

The proof of semisimplicity takes the largest semisimple lisse subsheaf of the underlying geometric sheaf. It is Frobenius-stable, hence comes from X_0 . The preceding extension criterion forces the quotient extension to be geometrically trivial, contradicting maximality unless the quotient is zero.

The uniqueness in (i) and (ii) follows by looking at generalized eigenspaces of the Frobenius at closed points. Existence is first proved when X_0 is smooth and the sheaf is an iterated extension of lisse pointwise pure sheaves; the extension criterion gives the splitting by weights modulo $\mathbb{Z}$ and the weight filtration. The general case follows by induction on $\dim X_0$ using an open-closed decomposition and specialization.

Variant 2 (3.4.9). *Every lisse mixed sheaf has a filtration by pointwise weights by lisse subsheaves, and every pure sheaf is geometrically semisimple.*

Potentially pure sheaves. A sheaf on a scheme X of finite type over an algebraically closed field is said to have a property potentially if it is obtained by base change from a family over a finite-type base whose closed fibers have the property. If $f : Y \rightarrow X$ is proper and smooth, the sheaves $R^i f_* \mathbb{Q}_\ell$ are potentially pointwise pure. Specialization then gives:

Corollary 5 (3.4.12). *If X is normal of finite type over an algebraically closed field and $\mathcal{F}$ is lisse and potentially pointwise ι -pure, then $\mathcal{F}$ is semisimple.*

Corollary 6 (3.4.13). *For S normal connected and $f : X \rightarrow S$ proper and smooth, the sheaves $R^i f_* \mathbb{Q}_\ell$ are semisimple.*

(3.5) Application: equidistribution theorems

Return to the notation and hypotheses of (2.2). Thus X_0 is normal geometrically connected of dimension N , and for every irreducible representation τ of $G_{\mathbb{R}}$ the corresponding sheaf is pointwise pure of weight $2N(\tau)$. By Grothendieck's cohomological interpretation of L -functions, together with (3.3.4) and (3.3.10), $L(\tau)$ extends meromorphically to G ; in the half-plane relevant to N it is holomorphic and invertible except for the simple pole at the trivial character. Therefore the equidistribution results (2.1.10)–(2.1.12) apply.

For $n > 0$, let $X_{(n)}$ be the base change of X_0 to $\mathbb{F}_{q^n}$. A rational point $x \in X_0(\mathbb{F}_{q^n})$ gives a geometric Frobenius element F_x of degree n . If a closed point x_0 has degree $d \mid n$, then the d rational points above it contribute conjugates of $F_{x_0}^{n/d}$. Thus, on conjugacy classes,

$$\sum_{x_0 \in |X_0|} \deg(x_0) [F_{x_0}^{n/\deg(x_0)}] = \sum_{x \in X_0(\mathbb{F}_{q^n})} [F_x]. \quad (3.5.2.1)$$

After identifying components of $G_{\mathbb{R}}$ by a positive-degree central element z , Theorem (2.1.12) gives:

Theorem 6 (3.5.3). *For every component index i , the measures obtained from the Frobenius classes of points of $X_0(\mathbb{F}_{q^{nd+i}})$, normalized by the expected factor $q^{-N(nd+i)}$, converge vaguely to the translated Haar measure on the corresponding component of $G_{\mathbb{R}}$.*

Example: elliptic curves and Sato–Tate. Let $f : E_0 \rightarrow X_0$ be a family of elliptic curves and put $\mathcal{F}_0 = R^1 f_* \mathbb{Q}_{\ell}$. If the modular invariant j is nonconstant, the geometric monodromy group is all of $\mathrm{SL}(V)$: after imposing level $n \geq 3$ prime to p , the associated modular cover maps dominantly to the moduli scheme of elliptic curves with level n structure, and the known topological monodromy contains a finite-index subgroup of $\mathrm{SL}(H^1(E_x, \mathbb{Z}_{\ell}))$.

Then $G_{\mathbb{R}}$ is $\mathrm{SU}(2) \times \mathbb{Z}$, acting on $\iota V \simeq \mathbb{C}^2$ by $(g, n) \mapsto q^{n/2} g$. Identify conjugacy classes in $\mathrm{SU}(2)$ with $[0, \pi]$ by eigenvalues $e^{\pm i\theta}$. Haar measure pushes forward to

$$\frac{2}{\pi} \sin^2 \theta \, d\theta.$$

For $x \in X_0(\mathbb{F}_{q^n})$ define the Frobenius angle $\theta(x)$ by

$$\#E_x(\mathbb{F}_{q^n}) = 1 - 2q^{n/2} \cos \theta(x) + q^n$$

(up to the sign convention of geometric Frobenius). Hence:

Proposition 2 (3.5.7). *If j is nonconstant, then as $n \rightarrow \infty$ the normalized counting measures of the angles $\theta(x)$, $x \in X_0(\mathbb{F}_{q^n})$, converge vaguely to the Sato–Tate measure $\frac{2}{\pi} \sin^2 \theta \, d\theta$.*

(3.6) Application: the local invariant-cycle theorem

Let k be algebraically closed, let S be the spectrum of the henselization of $k[T]$ at (T) , with closed point s , generic point η , and geometric generic point $\bar{\eta}$. For a proper morphism $f : X \rightarrow S$ there is a specialization map

$$\mathrm{sp}^* : H^*(X_s) \longrightarrow H^*(X_{\bar{\eta}}),$$

whose image lies in the inertia invariants.

Theorem 7 (3.6.1). *Assume that X is essentially smooth over k and that X_{η} is smooth over η . Then, in $\mathbb{Q}_{\ell}$ -cohomology,*

$$\mathrm{sp}^* : H^*(X_s) \longrightarrow H^*(X_{\bar{\eta}})^{\mathrm{Gal}(\bar{\eta}/\eta)}$$

is surjective.

The proof spreads the situation to a family over a smooth curve over a finite field, so that Frobenius weights are available. The Hochschild–Serre spectral sequence for the inertia group gives exact sequences of Wang type

$$0 \rightarrow H^{i-1}(X_{\bar{\eta}})_I(-1) \rightarrow H^i(X_{\eta}) \rightarrow H^i(X_{\bar{\eta}})^I \rightarrow 0. \quad (3.6.1.5)$$

Together with the long exact sequence for cohomology with support and Poincaré duality this yields a cross of exact sequences. The terms above and below have incompatible weights:

Lemma 10 (3.6.2). *$H^i(X_{\bar{\eta}})^I$ is mixed of weights $\leq i$.*

Lemma 11 (3.6.3). *$H^{2N-i-1}(X_{\bar{\eta}})_I(-N)$ is mixed of weights $> i$.*

Applying the exact functor W_i to the cross gives the surjectivity.

In the complex-analytic analogue, for a proper map $X \rightarrow D$ smooth over the punctured disk, the same result follows by replacing Frobenius weights with the weight filtration of mixed Hodge theory; this is the form related to Steenbrink’s mixed Hodge structure on vanishing cohomology.

(3.7) Return to (1.8)

The results of (3.3) and (1.8) simplify the proofs of (1.8.2) and (1.8.4). For a family of elliptic curves $f : E_0 \rightarrow S_0$ over a smooth S_0 , the image

$${}^k W = \operatorname{Im} \left(H_c^1(S, \operatorname{Sym}^k(R^1 f_* \mathbb{Q}_\ell)) \rightarrow H^1(S, \operatorname{Sym}^k(R^1 f_* \mathbb{Q}_\ell)) \right)$$

is pure of weight $k + 1$. This replaces the earlier reduction of the corresponding modular-form estimate to the usual Weil conjecture and gives a cleaner input to the Ramanujan–Petersson application.

Finally, let $\psi : \mathbb{F}_p \rightarrow \mathbb{C}^*$ be a nontrivial additive character. If $Q \in \mathbb{F}_q[X_1, \dots, X_n]$ has degree d prime to p and its homogeneous part Q_d defines a smooth projective hypersurface, then

$$\left| \sum_{x_1, \dots, x_n \in \mathbb{F}_q} \psi \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p} Q(x_1, \dots, x_n) \right| \leq (d-1)^n q^{n/2}. \quad (3.7.2.1)$$

The sharper statement is

$$H_c^i(\mathbb{A}^n, \mathcal{F}(\psi Q)) = 0 \quad (i \neq n), \quad \dim H_c^n(\mathbb{A}^n, \mathcal{F}(\psi Q)) = (d-1)^n, \quad (3.7.2.3)$$

and $H_c^n(\mathbb{A}^n, \mathcal{F}(\psi Q))$ is pure of weight n . One proves this by considering the universal polynomial over the open parameter scheme S_0 where the degree- d part is smooth. The sheaves $R^i f_* \mathcal{F}(\psi Q_S)$ are lisse by local acyclicity; since S_0 is connected and the sheaves are mixed, it suffices by (1.8.12) to check a single polynomial, namely $Q = \sum X_i^d$. Kunneth then reduces to the one-variable case $Q = X^d$, already treated in (1.8.11).

4. Lefschetz pencils

In this chapter, except in number (4.5), we work over an algebraically closed field k of characteristic p . Let ℓ be a prime number distinct from p . The cohomology groups considered have coefficients in $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, or $\mathbb{Z}/m$, where m is a power of ℓ .

(4.1) The hard Lefschetz theorem

In this number we suppose chosen, over k , an isomorphism between $\mathbb{Z}_\ell$ and $\mathbb{Z}_\ell(1)$. This spares us from writing Tate twists.

Theorem 8 (4.1.1). *Let X be a projective smooth variety over k , let $\mathcal{L}$ be an ample invertible sheaf on X , and let $\xi = c_1(\mathcal{L}) \in H^2(X, \mathbb{Q}_\ell)$. Suppose that X is pure of dimension n . Then, for each $i > 0$, cup-product by ξ^i ,*

$$\xi^i : H^{n-i}(X, \mathbb{Q}_\ell) \longrightarrow H^{n+i}(X, \mathbb{Q}_\ell),$$

is an isomorphism.

We prove the theorem by induction on n . The case $n = 0$ is trivial; suppose $n \geq 1$. Since $c_1(\mathcal{L}^{\otimes N}) = N c_1(\mathcal{L})$, the assertion for $\mathcal{L}$ is equivalent to the assertion for any positive tensor power. We may therefore assume $\mathcal{L}$ very ample. Let Y be a smooth hyperplane section of X for the projective embedding defined by $\mathcal{L}$. We first reduce the induction step to the following lemma.

Lemma 12 (4.1.2). *The intersection form on $H^{n-1}(Y)$ is nondegenerate on the image of $H^{n-1}(X)$.*

For $i = 0$ the assertion is trivial. For $i \geq 1$, cup-product by $\xi = c_1(Y)$ is the composite of the restriction morphism $H^*(X) \rightarrow H^*(Y)$ and its transpose under Poincare duality, the Gysin morphism $H^*(Y) \rightarrow H^{*+2}(X)$. Thus ξ^i factors as

$$H^{n-i}(X) \longrightarrow H^{(n-1)-(i-1)}(Y) \xrightarrow{\xi^{i-1}} H^{(n-1)+(i-1)}(Y) \longrightarrow H^{n+i}(X).$$

For $i > 1$ the two extreme arrows are isomorphisms by weak Lefschetz, and the middle arrow is an isomorphism by induction. For $i = 1$, weak Lefschetz gives only the injection $H^{n-1}(X) \hookrightarrow H^{n-1}(Y)$, whose transpose $H^{n-1}(Y) \rightarrow H^{n+1}(X)$ is surjective. The composite is an isomorphism precisely when Lemma (4.1.2) holds.

After replacing $\mathcal{L}$ by a tensor power, assume that the embedding admits a Lefschetz pencil of hyperplane sections. Let $Y = Y_t$ be a smooth member of such a pencil $(Y_t)_{t \in \mathbb{P}^1}$. We prove (4.1.2) by interpreting the image of $H^{n-1}(X)$ in $H^{n-1}(Y)$ in terms of monodromy.

Let S be the set of parameters t for which Y_t is singular. For $t \in \mathbb{P}^1 - S$, the groups $H^i(Y_t)$ form the fibers of a lisse $\mathbb{Q}_\ell$ -sheaf on $\mathbb{P}^1 - S$, and $\pi_1(\mathbb{P}^1 - S, t_0)$ acts on $H^i(Y)$.

Proposition 3 (4.1.3). *The image of $H^{n-1}(X)$ in $H^{n-1}(Y)$ is the subspace $H^{n-1}(Y)^{\pi_1(\mathbb{P}^1 - S, t_0)}$ of monodromy invariants.*

This is proved in SGA 7, XVIII, (5.6.10), under the hypothesis that Y satisfies hard Lefschetz, which is our induction hypothesis. Over $\mathbb{C}$, in integral cohomology, a direct proof is given in SGA 7, XIX, (4), by combining the corresponding statement with the Picard–Lefschetz formula. In (4.3) below we transpose this direct argument to $\mathbb{Z}_\ell$ -cohomology over an arbitrary base field.

We also know by (3.4.13) that $H^{n-1}(Y)$ is a completely reducible representation of $\pi_1(\mathbb{P}^1 - S, t_0)$. It remains only to apply the following algebraic lemma.

Lemma 13 (4.1.4). *Let V be a completely reducible linear representation of a group π , equipped with a π -invariant nondegenerate bilinear form Φ . Then the restriction of Φ to V^π is nondegenerate.*

By complete reducibility, write $V = V^\pi \oplus W$ for a subrepresentation W having no trivial quotient. Then W is orthogonal to V^π : indeed Φ induces a morphism $W \rightarrow (V^\pi)^*$, hence a trivial quotient if it is nonzero. It follows that $\Phi|_{V^\pi}$ is nondegenerate.

Corollary 7 (4.1.5). *The odd Betti numbers of a nonsingular projective variety are even.*

It is enough to assume the variety connected and pure of dimension n . If $\xi = c_1(\mathcal{L})$ with $\mathcal{L}$ ample, then the form $\text{Tr}(\xi^i xy)$ on $H^{n-i}(X)$ is nondegenerate by hard Lefschetz and Poincare duality. If $n - i$ is odd, this form is alternating; hence $b_{n-i} = b_{n+i}$ is even. This result should hold without the projectivity hypothesis, but Deligne does not know how to prove it in characteristic different from zero.

(4.1.6) Weak Lefschetz. Let X and Y be as above, except that X is pure of dimension $n + 1$ and Y is pure of dimension n . The key input is the cohomological-dimension theorem for affine varieties: in $\mathbb{Z}/m$ -cohomology,

$$H^i(X - Y) = 0 \quad (i > n + 1).$$

Since $X - Y$ is smooth, Poincare duality also gives

$$H_c^i(X - Y) = 0 \quad (i < n + 1).$$

The long exact sequence of relative cohomology then gives

$$H^i(X) \xrightarrow{\sim} H^i(Y) \text{quad}(i < n), \quad H^n(X) \hookrightarrow H^n(Y).$$

The transpose Gysin morphisms satisfy

$$H^i(Y) \xrightarrow{\sim} H^{i+2}(X) \quad (i > n), \quad H^n(Y) \rightarrow H^{n+2}(X).$$

Passing to the projective limit gives the same conclusions in $\mathbb{Z}_\ell$ - and $\mathbb{Q}_\ell$ -cohomology. In $\mathbb{Z}_\ell$ -cohomology the universal-coefficient formula further gives that $H^{n+1}(X - Y, \mathbb{Z}_\ell)$ is torsion-free, and hence $H^n(Y, \mathbb{Z}_\ell)/H^n(X, \mathbb{Z}_\ell)$ is torsion-free.

(4.2) Reminders on Lefschetz pencils

In (1.3) and (1.4) the essential results of the theory of Lefschetz pencils were summarized, excluding the case where $p = 2$ and the hyperplane sections have even dimension. The aim here is to repair that omission in part and, in doing so, to complete SGA 7, XVIII: in the wild case, the theory holds for general Lefschetz pencils, not only for generic ones.

(4.2.1) Local theory. Let $f : X \rightarrow S$ be as in (1.4.2), of relative dimension n . Write $n = 2n'$ or $2n' + 1$. Let $\delta \in H^n(X_{\bar{\eta}})(n')$ be the vanishing cycle, determined up to sign. In the exceptional case n even and $p = 2$, omitted in (1.4.3), the inertia group has several characters of order two. For one of them, depending on the ordinary quadratic singularity of f , the local monodromy is again given by the Picard–Lefschetz formula of loc. cit. The statements hold in $\mathbb{Z}/m$ -, $\mathbb{Z}_\ell$ -, and $\mathbb{Q}_\ell$ -cohomology.

Corollary 8 (4.2.2). *The image of $H^n(X_s)$ in $H^n(X_{\bar{\eta}})$ is the orthogonal complement of δ . In $\mathbb{Z}_\ell$ - or $\mathbb{Q}_\ell$ -cohomology it is also the subspace of invariants under inertia.*

The second assertion follows from the Picard–Lefschetz formula: the term $(x, \delta)\delta$ vanishes if and only if $(x, \delta) = 0$. This is clear when δ is not torsion; if δ is torsion, the pairing (x, δ) is identically zero.

(4.2.3) Global theory. Retain the notation of (1.5.6). Thus $X \subset P$ is a connected nonsingular projective variety of dimension $n + 1$, $(X_t)_{t \in D}$ is a Lefschetz pencil of hyperplane sections, S is the set of exceptional parameters, and $j : U \hookrightarrow D$ is the complementary open. Identify D with a line in the dual projective space $P^\vee$, and let $X^\vee \subset P^\vee$ be the image of X by the correspondence “the hyperplane H is tangent to X at x .” A Lefschetz pencil is called transverse if D is transverse to $X^\vee$.

A path between two geometric points of a scheme is an isomorphism between the corresponding fiber functors on étale covers. For any path c on U , from u to a geometric generic point of the henselization $D_{(t)}$, one obtains an identification

$$H^n(X_{\bar{\eta}}, \mathbb{Z}_\ell) \simeq H^n(X_{\bar{u}}, \mathbb{Z}_\ell),$$

and hence a vanishing cycle $\pm\delta_c$ in $H^n(X_{\bar{u}}, \mathbb{Z}_\ell)(n')$.

Definition 2 (4.2.4). (i) *The set of vanishing cycles is the set of all cycles $\pm\delta_c$ obtained in this way, with c and $t \in S$ varying.*

(ii) *The vanishing part of $H^n(X_{\bar{u}})$ is the subspace generated by the $\lambda\delta$, where $\lambda \in \mathbb{Z}_\ell(-n')$ and δ is a vanishing cycle.*

Since the projective line D is simply connected, the group $\pi_1(U, u)$ is topologically generated by conjugates of local inertia groups I_t , $t \in S$. Thus the monodromy group is topologically generated by the images of these conjugates. For n odd these images are transvections $x \mapsto x + \lambda(x, \delta)\delta$; for n even they are reflections $x \mapsto x - (-1)^{n'}(x, \delta)\delta$.

Proposition 4 (4.2.6). *For a generic Lefschetz pencil the vanishing cycles, taken up to sign, are all conjugate under the monodromy group.*

If $X^\vee$ is not a hypersurface, the assertion is trivial. Otherwise $X^\vee$ is irreducible. Let η be the generic point of the Grassmannian of lines in $P^\vee$, and let D_η be the corresponding projective line. On the geometric generic line the critical locus is the intersection with $X^\vee$, which is connected over the generic point of $X^\vee$. The geometric fundamental-group comparison and Bertini's theorem then show that the monodromy action is transitive on the set of critical points, whence the conjugacy of vanishing cycles.

Corollary 9 (4.2.7). *For $p = 2$ and n even, assume that the Lefschetz pencil is sufficiently general. Then:*

- (i) *if n is odd, the homomorphisms from $\mathbb{Z}_\ell(1)$ to the monodromy group, given by the Picard–Lefschetz formula $x \mapsto x + \lambda(x, \delta)\delta$, are conjugate to one another;*
- (ii) *if n is even, the reflections $x \mapsto x - (-1)^{n'}(x, \delta)\delta$, for δ a vanishing cycle, are conjugate to one another.*

For a generic pencil this follows from (4.2.6) and the local Picard–Lefschetz formula; the general sufficiently general case follows by specialization (1.11).

Corollary 10 (4.2.8). *Under the hypotheses of (4.2.7), the vanishing cycles, taken up to sign, are all conjugate in $H^n(X_{\bar{u}}, \mathbb{Q}_\ell)$.*

Modulo torsion, $\pm\delta$ is determined by the corresponding Picard–Lefschetz transformation.

(4.3) Complement to SGA 7, XIX, (4)

Choose once and for all an isomorphism over k between $\mathbb{Z}_\ell$ and $\mathbb{Z}_\ell(1)$, so that Tate twists are suppressed. In this number Deligne gives a direct proof of (4.1.3), generalizing SGA 7, XIX, (4), to an arbitrary base field and to $\mathbb{Z}_\ell$ -cohomology.

Let P be a projective space over k , let $X \subset P$ be projective, smooth, and pure of dimension $n + 1$, let $A \subset P$ be a codimension-two linear subspace meeting X transversally, and suppose that the hyperplanes through A sweep out on X a Lefschetz pencil. Let $\tilde{X}$ be the blow-up of X along $A \cap X$, with exceptional divisor $\tilde{A}$, and let

$$f : \tilde{X} \longrightarrow \mathbb{P}^1$$

be the morphism whose fibers are the members Y_t of the pencil. Let S be the set of critical values, choose $0 \in \mathbb{P}^1 - S$, and put $Y = Y_0$.

The geometric situation is represented by

$$\begin{array}{ccccc} \tilde{A} & \hookrightarrow & \tilde{X} & \longrightarrow & X \\ \downarrow & & \downarrow \hat{f} & & \\ 0 & \hookrightarrow & \mathbb{P}^1 & \xlongequal{\quad} & \mathbb{P}^1. \end{array} \quad (4.3.1.1)$$

Let $\text{Ev}(Y)$ be the vanishing part of $H^n(Y)$ and let $\text{Ev}(Y)^\perp$ denote its orthogonal complement for $(x, y) = \text{Tr}(xy)$.

Lemma 14 (4.3.2). *In $\mathbb{Z}/m$ -cohomology,*

$$H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m) \simeq \text{Ev}(Y)^\perp \subset H^n(Y).$$

In $\mathbb{Z}_\ell$ - or $\mathbb{Q}_\ell$ -cohomology this group is also the subspace of $\pi_1(\mathbb{P}^1 - S, 0)$ -invariants.

The sheaf $R^n f_* \mathbb{Z}/m$ has no section with punctual support, and its H^0 injects into the fiber $H^n(Y)$. Its image is the fiber at 0 of the largest constant subsheaf. The local theory identifies this largest locally constant subsheaf with $\text{Ev}(Y)^\perp$, and since $\mathbb{P}^1$ is simply connected it is constant.

Proposition 5 (4.3.3). *The subspace $\mathrm{Ev}(Y)^\perp$ is the image of $H^n(X)$ in $H^n(Y)$.*

In $\mathbb{Z}/m$ -cohomology this may be written as the exact cross

$$\begin{array}{ccccccc} & & \mathrm{Ev}(Y)^* & & & & \\ & & \downarrow & & & & \\ 0 & \longrightarrow & \mathrm{Ev}(Y) & \longrightarrow & H^n(Y) & \longrightarrow & H^{n+2}(X) \longrightarrow 0 \\ & & \downarrow & & & & \\ & & H^n(X) & & & & \end{array} \quad (4.3.3.1)$$

where the composite $H^n(X) \rightarrow H^n(Y) \rightarrow H^{n+2}(X)$ is cup-product with the class of Y . In $\mathbb{Z}_\ell$ -cohomology, since the duals $\mathrm{Ev}(Y)^*$ modulo ℓ^j need not form a projective system, one instead passes to the limit in the exact sequence

$$0 \rightarrow H^n(X) \rightarrow H^n(Y) \rightarrow (\mathbb{Z}/\ell^j)^B,$$

where the vanishing cycles indexed by B generate the vanishing part. This gives the analogous cross

$$\begin{array}{ccccccc} & & \mathrm{Ev}(Y)^* & & & & \\ & & \downarrow & & & & \\ 0 & \longrightarrow & \mathrm{Ev}(Y) & \longrightarrow & H^n(Y) & \longrightarrow & H^{n+2}(X) \longrightarrow 0 \\ & & \downarrow & & & & \\ & & H^n(X) & & & & \end{array} \quad (4.3.3.2)$$

in $\mathbb{Z}_\ell$ -cohomology.

(4.3.4) Relative cohomology. Let $u : T' \rightarrow T''$ be a morphism of topoi. Triples $(\mathcal{F}', \mathcal{F}'', \varphi : u^* \mathcal{F}'' \rightarrow \mathcal{F}')$ form a new topos T . If $\mathcal{F} = (\mathcal{F}', \mathcal{F}'', \varphi)$ is an abelian sheaf of T , set

$$H^0(T'' \text{ mod } T', \mathcal{F}) = \ker(H^0(T'', \mathcal{F}'') \rightarrow H^0(T', \mathcal{F}')).$$

The right-derived functors give the relative cohomology of T'' modulo T' . Equivalently, for $K \in D^+(T)$, there is a triangle

$$R\Gamma(T'' \text{ mod } T', K) \rightarrow R\Gamma(T'', K'') \rightarrow R\Gamma(T', K') \rightarrow,$$

and hence a long exact sequence of cohomology.

(4.3.5) Sheaves on $\mathbb{P}^1$. Let X be an irreducible complete variety over k , let x, y be two closed points, and let $K \in D^+(X)$. A path from a geometric generic point of the henselization at x to one at y gives a morphism

$$H^*(X - \{x\}, K) = H^*(X \text{ mod } \{x\}, K) \rightarrow H^*(X_{(y)} \text{ mod } \bar{y}, K). \quad (4.3.5.1)$$

Proposition 6 (4.3.6). *Let $t \in \mathbb{P}^1$, and let $K \in D^b(\mathbb{P}^1)$ have constructible cohomology sheaves, with finite support in negative degrees. Let S be a finite subset of $\mathbb{P}^1$ such that the cohomology sheaves of K are locally constant outside S . Then the product of the morphisms (4.3.5.1), for $X = \mathbb{P}^1$, $x = t$, and $y \in S$, is injective:*

$$H^1(\mathbb{P}^1 \text{ mod } t, K) \rightarrow \prod_{s \in S} H^1(\mathbb{P}^1_{(s)} \text{ mod } \bar{s}, K).$$

The proof first removes a subcomplex supported in negative degrees and reduces to a sheaf $\mathcal{H}$ without punctual sections, locally constant outside S . Then $H^1(\mathbb{P}^1 \text{ mod } t, \mathcal{H})$ classifies $\mathcal{H}$ -torsors on $\mathbb{P}^1$ trivialized at t . Such torsors are determined by their local restrictions because $\mathbb{P}^1$ is simply connected and the images of the local monodromy groups topologically generate $\pi_1(\mathbb{P}^1 - S)$.

(4.3.7) Plan of the proof of (4.3.3). The injection of $H^n(X)$ into $H^n(Y)$ factors through $H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m) \subset H^n(Y)$. It remains to show that, for a class α in this H^0 , the obstruction $d\alpha \in H^{n+1}(X - Y)$ to extending α from Y to X vanishes. Since $X - Y = \tilde{X} - Y - \tilde{A}$, and if φ is the projection of $\tilde{X} - \tilde{A}$ to $\mathbb{P}^1$, one has

$$H^{n+1}(X - Y) = H^1(\mathbb{P}^1 \bmod 0, R\varphi_* \mathbb{Z}/m[n]).$$

The weak Lefschetz theorem shows that the lower cohomology sheaves have finite support. Proposition (4.3.6) then reduces the obstruction to local obstructions, which are precisely the pairings (α, δ_s) with vanishing cycles.

(4.3.8) Proof of (4.3.3). Let $\alpha \in H^n(Y)$ and let $\tilde{\alpha}$ be its inverse image on the inverse image of Y in $\tilde{X}$. The obstruction to extending α to X is the same as the obstruction to extending $\tilde{\alpha}$ to $\tilde{X}$. In the derived-category form of the Leray spectral sequence, take a flasque resolution $(\mathbb{Z}/m)^*$ of $\mathbb{Z}/m$ on $\tilde{X}$, put $K = f_*(\mathbb{Z}/m)^*$ and $L = f_*((\mathbb{Z}/m)^*|_{\tilde{A}})$. The exact sequence of simple complexes associated with

$$\Gamma(\mathbb{P}^1, K) \rightarrow s(K_0 \rightarrow L_0) \rightarrow L_0$$

identifies the obstruction as an element of

$$H^1(\mathbb{P}^1 \bmod 0, s(K \rightarrow L)[n]),$$

where $s(K \rightarrow L)$ represents $R\varphi_* \mathbb{Z}/m$. Applying (4.3.6), the obstruction is tested after localization at each critical value. These local obstructions vanish exactly for classes coming from $H^0(\mathbb{P}^1, R^n f_* \mathbb{Z}/m)$, and (4.3.3) follows.

Corollary 11 (4.3.9). *In $\mathbb{Q}_\ell$ -cohomology one has an orthogonal direct-sum decomposition, for the intersection form,*

$$H^n(Y) = H^n(X) \oplus \text{Ev}(Y).$$

This reformulation of hard Lefschetz follows from (4.3.3) and (4.1.2).

(4.3.10) In $\mathbb{Z}_\ell$ -cohomology, the cross (4.3.3.2) shows that the intersection of the vanishing part of $H^n(Y)$ with the fixed part $H^n(X) = H^n(Y)^{\pi_1}$ is

$$\ker(\xi : H^n(X) \rightarrow H^{n+2}(X)).$$

In $\mathbb{Q}_\ell$ -cohomology this kernel is zero by hard Lefschetz. Thus one recovers Lefschetz's fundamental lemma: a cycle which is both invariant and vanishing is zero. In $\mathbb{Z}_\ell$ -cohomology this may fail; after replacing the embedding by a multiple, this kernel may coincide with the whole torsion subgroup of $H^n(X)$.

(4.4) Monodromy of Lefschetz pencils

Retain the notation of (4.3.1). Work in $\mathbb{Q}_\ell$ -cohomology, and let G be the group of automorphisms of $\text{Ev}(Y)$ preserving the nondegenerate intersection form. The group $\pi_1(\mathbb{P}^1 - S, 0)$ acts on $\text{Ev}(Y)$ preserving this form; let M be its image in G .

Theorem 9 (4.4.1). *Assume X irreducible. If $p = 2$ and n is even, assume in addition that the pencil is sufficiently general. Then either:*

- (a) M is open in G ;
- (b) n is even and M is finite.

Since G is a semisimple algebraic group over $\mathbb{Q}_\ell$, it is enough to prove Zariski density of M in the nonfinite case. After extending scalars to $\mathbb{C}$, this is reduced to two elementary algebraic lemmas applied to the Zariski closure of M and to the conjugacy class of Picard–Lefschetz transformations.

Lemma 15 (4.4.2a). *Let V be a finite-dimensional complex vector space with a nondegenerate alternating form. Let $M \subset \mathrm{Sp}(V)$ be an algebraic subgroup and let R be an M -orbit in V generating V . Suppose that M is the smallest algebraic subgroup containing all transvection one-parameter subgroups*

$$U_\delta = \{x \mapsto x + \lambda(x, \delta)\delta \mid \lambda \in \mathbb{C}\}, \quad \delta \in R.$$

Then $M = \mathrm{Sp}(V)$.

Lemma 16 (4.4.3a). *If $\delta', \delta'' \in R$ are not orthogonal, then every nonzero linear combination of δ' and δ'' belongs to R .*

The subgroup generated by $U_{\delta'}$ and $U_{\delta''}$ is the group SL_2 of their span; its nonzero vectors form a single orbit. The proof of (4.4.2a) follows by taking a maximal subspace W for which $R \cap W$ is dense, using (4.4.3a) to show no vector outside W can be nonorthogonal to W , and concluding from nondegeneracy and generation that $W = V$ and hence $M = \mathrm{Sp}(V)$.

Lemma 17 (4.4.2b). *Let V be a finite-dimensional complex vector space with a nondegenerate symmetric form. Let $M \subset O(V)$ be an algebraic subgroup and let R be an M -orbit generating V . Suppose $(\delta, \delta) = 2$ for all $\delta \in R$, and suppose that M is the smallest algebraic subgroup of $O(V)$ containing the reflections*

$$s_\delta : x \mapsto x - (x, \delta)\delta, \quad \delta \in R.$$

Then M is finite or $M = O(V)$.

Lemma 18 (4.4.3b). *Let $\delta', \delta'' \in R$. If (δ', δ'') is not of the form $2 \cos \theta$ with θ a rational multiple of 2π , then the vectors δ in the span of δ', δ'' satisfying $(\delta, \delta) = 2$ belong to R . The same conclusion holds if $(\delta', \delta'') = \pm 2$.*

The first case uses the fact that the product $s_{\delta'} s_{\delta''}$ is a rotation of infinite order in the two-plane, whose algebraic envelope is the connected orthogonal group of that plane. The second case uses the transvection obtained when the plane has a one-dimensional radical. The proof of (4.4.2b) then proceeds as in the alternating case, except that if all maximal subspaces have dimension one, the scalar products in R take only finitely many values; hence R is finite and so is M .

Deligne then studies the finite case.

Proposition 7 (4.4.5). *If M is finite, then the analogous group in ℓ' -adic cohomology is finite for every $\ell' \neq p$. As a quotient of $\pi_1(\mathbb{P}^1 - S, 0)$ it is independent of ℓ , and the character of its representation on $\mathrm{Ev}(Y, \mathbb{Q}_{\ell'})$ has integer values independent of ℓ' .*

The proof removes the trivial factors from the virtual representation

$$\sum_i (-1)^i H^i(Y)$$

and reduces by specialization to the case where the pencil is defined over a finite field. For $x \in \mathbb{P}_0^1 - S_0$ rational over $\mathbb{F}_{q^m}$, the Lefschetz trace formula identifies the trace of Frobenius with the number of $\mathbb{F}_{q^m}$ -rational points of the fiber. The required independence follows from Chebotarev.

Lemma 19 (4.4.6). *Let U_0 be normal and geometrically connected over $\mathbb{F}_q$, with base point u . Let G be the quotient of $W(U_0, u)$ by a subgroup of finite index in $\pi_1(U, u)$, and let V be a virtual ℓ -adic representation of G . Let V' be a virtual ℓ' -adic representation of $W(U_0, u)$. If the characters of V and V' agree and are rational on powers of Frobenius elements, then V' is inflated from a virtual representation of G , and the two characters are equal and rational.*

The proof uses Chebotarev in the form that every element of sufficiently large degree in G is a Frobenius element. A character of G is determined by its restriction to elements of sufficiently large degree, via a Vandermonde independence argument.

Lemma 20 (4.4.7). *If two virtual ℓ -adic representations of $W(U_0, u)$ have equal characters on powers of Frobenius elements, then their characters are equal.*

One passes to the Zariski closure G attached to the sum of the representations, cuts according to the ℓ -adic valuations of eigenvalues of a central element of positive degree, twists to reduce to representations of $\pi_1(U_0, u)$, and applies Chebotarev density.

(4.4.8) Assume now the hypotheses of (4.4.1), with $n = 2n'$ even, M finite, and $\text{Ev}(Y) \neq 0$. The intersection form on $\text{Ev}(Y)$ corresponds, after the twist (n') , to a $\mathbb{Q}_\ell$ -valued form; vanishing cycles $\delta \in \text{Ev}(Y)(n')$ satisfy

$$(\delta, \delta) = (-1)^{n'} 2.$$

Theorem 10 (4.4.9). *Under the hypotheses of (4.4.8):*

- (a) *the $\mathbb{Z}$ -module L generated by the vanishing cycles is such that $L \otimes \mathbb{Q}_\ell = \text{Ev}(Y)(n')$, and the form is integral on L ;*
- (b) *the vanishing cycles $\pm\delta$ form in L a root system R of type A , D , or E ; the group M is the corresponding Weyl group.*

The reflections s_δ are characterized intrinsically as the elements conjugate to nontrivial elements of local inertia groups. For two such reflections,

$$\text{Tr}(s_\delta s_{\delta'}) = (\dim \text{Ev}(Y) - 2) + (\delta, \delta')^2.$$

Thus $\text{Tr}(s_\delta s_{\delta'}) - \dim \text{Ev}(Y) + 2$ has a square root in $\mathbb{Q}_\ell$ for every $\ell \neq p$; hence it is the square of an integer. This gives $(\delta, \delta') \in \mathbb{Z}$, proves integrality, and identifies R as an irreducible simply-laced root system.

(4.5) Application: the gcd theorem

The conventions (0.7) are in force. The following theorem is used by Katz and Messing to compare ℓ -adic and crystalline cohomology.

Theorem 11 (4.5.1). *Let X_0 be a smooth projective absolutely irreducible variety of dimension $n + 1$ over $\mathbb{F}_q$, and let $(X_t)_{t \in \mathbb{P}^1}$ be a Lefschetz pencil of hyperplane sections of X , defined over $\mathbb{F}_q$. If $p = 2$ and n is even, assume the pencil sufficiently general. Let $S \subset \mathbb{P}^1$ be the set of t for which X_t is singular. Then*

$$\det(1 - Ft, H^n(X))$$

is the least common multiple of the polynomials $f(T) = \prod_i (1 - \alpha_i T)$ having the following property: for every $t \in \mathbb{P}^1 - S$ defined over $\mathbb{F}_{q^k}$, if F is Frobenius on X_t relative to its natural $\mathbb{F}_{q^k}$ -structure, then

$$\prod_i (1 - \alpha_i^k T) \quad \text{divides} \quad \det(1 - FT, H^n(X_t)).$$

Using the orthogonal decomposition

$$H^n(X_t) = H^n(X) \oplus \text{Ev}(X_t),$$

one sees that $\det(1 - FT, H^n(X))$ has the required property. To prove minimality Deligne proves the following lemma.

Lemma 21 (4.5.2). *Let*

$$P(T) = \prod_i (1 - \gamma_i T), \quad Q(T) = \prod_j (1 - \beta_j T),$$

with all γ_i, β_j nonzero. If for every $t \in \mathbb{P}^1 - S$ defined over $\mathbb{F}_{q^k}$ one has

$$P^{(k)}(T) \mid Q^{(k)}(T) \det(1 - FT, \text{Ev}(X_t)),$$

where $P^{(k)}(T) = \prod_i (1 - \gamma_i^k T)$ and similarly for Q , then P divides Q .

It is enough to prove the lemma when $P = 1 - \gamma T$. Suppose by contradiction that γ is distinct from the β_j . For each j , let k_j generate the ideal of integers k for which $\gamma^k = \beta_j^k$, and let N be the product of the nonzero k_j . The hypothesis then gives, for $k \equiv 1 \pmod{N}$,

$$1 - \gamma^k T \mid \det(1 - FT, \text{Ev}(X_t)). \quad (4.5.3)$$

Let u be a geometric point of $\mathbb{P}^1 - S$, and let G be the image of the arithmetic fundamental group in $\text{GL}(E) \times \mathbb{Z}$, where $E = \text{Ev}(X_u)$ and the projection to $\mathbb{Z}$ is the degree. For $t \in (\mathbb{P}^1 - S)_0(\mathbb{F}_{q^k})$ the Frobenius element F_t has degree k , and

$$\det(1 - FT, \text{Ev}(X_t)) = \det(1 - p(F_t)T).$$

By Chebotarev, (4.5.3) implies:

$$\gamma^{\deg(g)} \text{ is an eigenvalue of } p(g) \quad (g \in G, \deg(g) \equiv 1 \pmod{N}). \quad (4.5.4)$$

It remains to show that this is impossible even for elements of degree one.

Lemma 22 (4.5.4). *If $g \in \text{GL}(E)$ normalizes the geometric monodromy group M , then the elements gm for $m \in M$ have no common eigenvalue.*

One may replace M by its Zariski closure. This closure is either a symplectic group, an orthogonal group, or a Weyl group of type A , D , or E . In the symplectic and orthogonal cases, after multiplying by a scalar and by an element of the group, the assertion reduces to the fact that the elements of the symplectic or orthogonal group have no common eigenvalue; already 1 and -1 have none in common. In the Weyl case, one reduces to an automorphism of the Dynkin diagram. If the diagram automorphism is trivial, take a Coxeter transformation. If it is opposition, replace by its negative. The remaining cases are D_{2k} with an order-two diagram automorphism and D_4 with an order-three automorphism; in the first, two explicit signed permutation matrices have disjoint spectra, and in the second, using the Hurwitz quaternion realization of D_4 , g and $-g$ have no common eigenvalue. This proves the theorem.

5. Application to the $\mathbb{Q}_\ell$ -homotopy type

The aim of this section is to justify some of the results announced in [3]. The problem is to attach to each algebraic variety X over an algebraically closed field k a differential graded $\mathbb{Q}_\ell$ -algebra $A(X)$, well-defined up to quasi-isomorphism and functorial in X , which is the $\mathbb{Q}_\ell$ -homotopy type of X . In particular, one wants

$$H^*A(X) = H^*(X, \mathbb{Q}_\ell).$$

One also wants that, if $k = \mathbb{C}$, the minimal model of $A(X)$ be obtained from the Sullivan minimal model of the rational homotopy type of the topological space $X(\mathbb{C})$ by extending scalars from $\mathbb{Q}$ to $\mathbb{Q}_\ell$.

The method which I had in mind for constructing $A(X)$ when I wrote [3] has evaporated. The method used here depends essentially on a construction, due independently to Grothendieck and to Miller [8], described in (5.1).

(5.1) The $\mathbb{Z}\{t\}$ -de Rham complex, after Grothendieck and Miller

(5.1.1) The algebra of polynomials with divided powers over $\mathbb{Z}$ is the subring

$$\mathbb{Z}\{t\} \subset \mathbb{Q}[t]$$

with $\mathbb{Z}$ -basis

$$\frac{t^n}{n!} \quad (n \geq 0).$$

The t -grading on $\mathbb{Z}\{t\}$ is the one for which $t^n/n!$ has degree n .

(5.1.2) We shall abbreviate “differential graded” as DG. DG algebras are always assumed commutative in the sense of the Koszul sign rule. The standard example is the DG algebra of exterior differential forms on a variety.

(5.1.3) Let A be a commutative ring and set

$$A\{t\} = A \otimes_{\mathbb{Z}} \mathbb{Z}\{t\}.$$

For $A = \mathbb{Q}$, this is just the polynomial algebra $\mathbb{Q}[t]$. A t -graded DG $A\{t\}$ -algebra is a DG $A\{t\}$ -algebra whose homogeneous components are themselves given a grading, the t -grading, making it a graded $A\{t\}$ -module and such that the product and the differential d have t -degree 0.

(5.1.4) The construction of the $\mathbb{Z}\{t\}$ -de Rham complex, defined below, is a contravariant functor

$$X \longmapsto \Omega(X)$$

from simplicial sets to t -graded DG $\mathbb{Z}\{t\}$ -algebras. We write $\Omega^p(X)$ for the part of exterior degree p . For a bihomogeneous element,

$$0 \leq \text{exterior degree} \leq t\text{-degree.} \quad (5.1.4.1)$$

There is moreover a functorial morphism of bigraded $\mathbb{Z}\{t\}$ -algebras

$$H^*\Omega(X) \longrightarrow \bigoplus_{p \leq n} H^p(X, \mathbb{Z}) \cdot \frac{t^n}{n!}, \quad (5.1.4.2)$$

which identifies the cohomology of $\Omega(X)$ with the direct sum on the right.

(5.1.5) More generally, for every commutative ring A —the cases needed below are $A = \mathbb{Z}$, $A = \mathbb{Z}/\ell^n$, and $A = \mathbb{Q}$ —there is a functor

$$X \longmapsto \Omega(X, A),$$

the $A\{t\}$ -de Rham complex, satisfying the same formalism as (5.1.4), with $\mathbb{Z}$ replaced by A . It is covariant in A , and the analogue of (5.1.4.2) is functorial in A .

For $A = \mathbb{Q}_\ell$, there is a morphism from $\Omega(X, \mathbb{Q})$ to the t -graded DG $\mathbb{Q}[t]$ -algebra obtained from the de Rham–Sullivan complex $\Omega_{\text{DRS}}(X)$ by extension of scalars to $\mathbb{Q}_\ell[t]$. This morphism is compatible with (5.1.4.2). It identifies the part of t -degree n of $\Omega(X, \mathbb{Q})$ with

$$\frac{t^n}{n!} \otimes \{\text{forms in } \Omega_{\text{DRS}}(X) \text{ of degree } \leq n \text{ in the barycentric coordinates of each simplex}\}.$$

If B is a quotient of A , the natural morphism

$$\Omega^p(X, A) \otimes_A B \longrightarrow \Omega^p(X, B)$$

is an isomorphism except possibly in the component of t -degree p .

(5.1.6) Some preliminaries are needed before the idea of the construction and then the construction itself are given.

(5.1.6.1) If a is a finite set, write

$$\mathbb{Z}\{(x_i)_{i \in a}\}$$

for the subring of $\mathbb{Q}[(x_i)_{i \in a}]$ with $\mathbb{Z}$ -basis the monomials

$$\prod_{i \in a} \frac{x_i^{r_i}}{r_i!}.$$

Let $\Omega'(a)$ be the corresponding de Rham complex: it is the subcomplex of the de Rham complex of $\mathbb{Q}[(x_i)_{i \in a}]$ which is freely generated over $\mathbb{Z}\{(x_i)_{i \in a}\}$ by the products of dx_i 's. Put

$$t = \sum_{i \in a} x_i,$$

and let $\Omega(a)$ be the quotient of $\Omega'(a)$ by the ideal generated by dt . This is a t -graded DG $\mathbb{Z}\{t\}$ -algebra if x_i and dx_i are assigned t -degree 1. It is called the t -graded DG $\mathbb{Z}\{t\}$ -algebra of relative differential forms, relative to t , polynomial and integral with divided powers, on $\mathbb{R}^a$.

(5.1.6.2) For any ring A , set

$$\Omega(a, A) = \Omega(a) \otimes_{\mathbb{Z}} A.$$

(5.1.6.3) An intrinsic description is as follows.

a) If L is a free A -module, consider

$$\Omega(L) = \Gamma^*(L) \otimes \bigwedge^*(L),$$

where Γ^* denotes divided powers. The exterior degree is that of the exterior power, and the t -degree is the sum of the exterior and divided-power degrees. The differential, compatible with divided powers, is characterized by sending a generator in the divided-power factor to the corresponding generator in the exterior factor.

b) If M is a free quotient of L , consider the corresponding quotient

$$\Omega(L, M)$$

of $\Omega(L)$.

c) Then

$$\Omega(a, A) \simeq \Omega(A^a, A^a/A\Delta),$$

where $A\Delta$ is the diagonal copy of A in A^a .

(5.1.6.4) The t -graded DG algebra $\Omega(a)$ is a contravariant functor of a . This is clear from (5.1.6.3), since the pair

$$(\mathbb{Z}^a, \mathbb{Z}^a/\mathbb{Z}\Delta)$$

is contravariant in a . More concretely, to a map $\varphi : v \rightarrow a$ one associates φ^* , sending x_i to $\sum_{\varphi(j)=i} x_j$. For example, if $v \subset a$ and φ is the inclusion, φ^* is obtained by putting $x_i = 0$ for $i \notin v$.

(5.1.7) Here is the idea of the construction in the parallel setting of Godement's "simplicial schemes." Start with a set S , and a collection $\mathcal{S}$ of finite nonempty subsets of S , called simplexes. Suppose that every one-element subset of S is a simplex and that every nonempty subset of a simplex is a simplex. For $a \in \mathcal{S}$, let $|a|$ be the usual simplex in $\mathbb{R}^{(S)}$ spanned by the basis vectors e_i , $i \in a$, and let X be the union of the $|a|$'s. Let $\tilde{X}$ be the cone on X , the union of the cones with vertex 0 spanned by the $|a|$'s.

It is more convenient to consider the homotopy-equivalent spaces X' , the union of the affine spaces a' spanned by the $|a|$'s, and $\tilde{X}'$, the union of the vector spaces $\tilde{a}'$ spanned by the $|a|$'s. Let

$$t : \mathbb{R}^{(S)} \longrightarrow \mathbb{R}$$

be the function "sum of coordinates." The space X' is the fiber over $t = 1$ of the restriction $\tilde{X}' \rightarrow \mathbb{R}$ of t .

For each a , let $\Omega(a)$ be the t -graded DG $\mathbb{Z}\{t\}$ -algebra of relative forms, relative to t , polynomial and integral with divided powers, on $\tilde{a}'$. By definition, a p -form in $\Omega^p(X)$ is a family

$$\omega = (\omega_a)_{a \in \mathcal{S}}, \quad \omega_a \in \Omega^p(a),$$

such that, if $v \subset a$, then ω_v is the restriction of ω_a .

(5.1.8) Here is the formal definition. Restrict the functor $a \mapsto \Omega(a)$ of (5.1.6.4) to the finite ordered sets

$$\Delta_n = [0, n]$$

and the increasing maps between them. This gives a simplicial t -graded DG $\mathbb{Z}\{t\}$ -algebra Ω_Δ . Put

$$\Omega(X, A) = \text{Hom}(X, \Omega_\Delta \otimes A),$$

where this means the direct sum, over the exterior and t -degrees, of the groups of morphisms of simplicial sets from X to the corresponding bihomogeneous component of $\Omega_\Delta \otimes A$.

(5.1.9) For the proof of the fundamental properties of this construction, I refer to [8]. Let us only indicate that:

- a) the morphism of (5.1.5) to the de Rham–Sullivan complex is obtained by sending x_i to the barycentric coordinate x_i times t ;
- b) the key point is the calculation of the homology groups—or homotopy groups—of the simplicial group involved: each H_i is isomorphic to $\mathbb{Z}$ and is placed in exterior degree and t -degree i . This calculation follows easily from (5.1.6.3) if one recalls that the simplicial group $a \mapsto A^a$ is homotopic to 0: its normalized complex is reduced to $A^{-1} \rightarrow A$ in homological degrees 0 and 1. One has

$$\Gamma^*(A^a) \simeq \Gamma^*(0) = A,$$

and $\Lambda(A^a/A\Delta)$ is computed by Quillen’s decalage theorems: its normalized object is the symmetric algebra of the A -module A , with Sym^n placed in homological degree n .

(5.2) The $\mathbb{Q}_\ell$ -homotopy type

(5.2.1) Let X be a filtered projective system of simplicial sets X^α . Put

$$H^i(X, \mathbb{Z}/\ell^n) = \varinjlim_\alpha H^i(X^\alpha, \mathbb{Z}/\ell^n). \quad (5.2.1.1)$$

The relation (5.1.4.2) still holds between these groups and the t -graded DG algebra over $\mathbb{Z}/\ell^n\{t\}$:

$$\Omega(X, \mathbb{Z}/\ell^n) = \varinjlim_\alpha \Omega(X^\alpha, \mathbb{Z}/\ell^n), \quad (5.2.1.2)$$

by exactness of filtered inductive limits.

Assume now:

$$(F) \quad \text{the groups } H^i(X, \mathbb{Z}/\ell) \text{ are finite.}$$

The arguments of SGA 5, V, (5.3.1) apply and show that each

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n) \quad (5.2.1.3)$$

is a finitely generated $\mathbb{Z}_\ell$ -module and that one has the usual exact sequences

$$0 \rightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \rightarrow H^i(X, \mathbb{Z}/\ell^n) \rightarrow \text{Tor}_{\mathbb{Z}_\ell}(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \rightarrow 0.$$

Put

$$\Omega(X, \mathbb{Z}_\ell) = \varprojlim_n \Omega(X, \mathbb{Z}/\ell^n), \quad (5.2.1.4)$$

a t -graded DG $\mathbb{Z}_\ell\{t\}$ -algebra. The groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, hence their projective system satisfies the Mittag–Leffler condition. In each bidegree (i, p) , except for $i = p$, the transition morphisms in $\Omega^p(X, \mathbb{Z}/\ell^n)$ are surjective by (5.1.5), and the Mittag–Leffler condition is again satisfied. For $i = p$, one has an exact sequence

$$\Omega^{p-1}(X, \mathbb{Z}/\ell^n) \xrightarrow{d} \Omega^p(X, \mathbb{Z}/\ell^n) \longrightarrow H^p(X, \mathbb{Z}/\ell^n) \longrightarrow 0,$$

and hence again Mittag–Leffler, since cohomology satisfies it.

Thus passage to the projective limit in (5.2.1.4) is exact and gives a morphism

$$H^*\Omega(X, \mathbb{Z}_\ell) \longrightarrow H^*(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell\{t\}, \quad (5.2.1.5)$$

which identifies the cohomology of $\Omega(X, \mathbb{Z}_\ell)$ with the direct sum of the

$$H^p(X, \mathbb{Z}_\ell) \cdot \frac{t^n}{n!} \quad (p \leq n).$$

It is essential in this construction that the inductive limit over α has preceded the projective limit over n .

For any t -graded module M over $\mathbb{Q}[t]$, for instance over $\mathbb{Q}_\ell[t]$, write

$$\varinjlim_t M$$

for the inductive limit of the system of homogeneous components M_k , where the transition morphism from M_k to M_{k+1} is multiplication by t . Put

$$H^*(X, \mathbb{Q}_\ell) = H^*(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \quad (5.2.1.6)$$

and

$$A(X) = \varinjlim_t (\mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \Omega(X, \mathbb{Z}_\ell)). \quad (5.2.1.7)$$

It follows from (5.2.1.5), by passage to the limit, that

$$H^*A(X) = H^*(X, \mathbb{Q}_\ell). \quad (5.2.1.8)$$

(5.2.2) Let X be an algebraic variety over an algebraically closed field k . It can be defined over a countable algebraically closed subfield $k_1 \subset k$. Let X_1/k_1 be such that X is obtained from X_1 by extension of scalars. Since k_1 is countable, one can find a countable projective system, indexed by the integers, of étale and surjective morphisms

$$X_\alpha \longrightarrow X_1$$

such that, for every étale and surjective morphism $U \rightarrow X_1$, there are an α and a factorization of $X_\alpha \rightarrow X_1$ through U .

For each α and each finite set a , let

$$(X_\alpha/X_1)^a$$

be the fiber product over X_1 of the family, indexed by a , of copies of X_α . When a runs through the ordered finite sets $[0, n]$, these fiber products form a simplicial scheme. We take the associated projective system of simplicial sets and set

$$A(X) = A(\{(X_\alpha/X_1)^\bullet\}_\alpha). \quad (5.2.2.2)$$

It would have been more natural to take, instead of these fiber products, a projective cofinal system of étale hypercoverings of X_1 ; according to Artin, this is essentially equivalent.

By the definition of étale cohomology, if one had used hypercoverings, and by Artin with the definition just chosen, one has

$$H^*(\{(X_\alpha/X_1)^\bullet\}_\alpha, \mathbb{Z}/\ell^n) = H^*(X_1, \mathbb{Z}/\ell^n).$$

On the other hand,

$$H^*(X_1, \mathbb{Z}/\ell^n) \simeq H^*(X, \mathbb{Z}/\ell^n).$$

The finiteness condition (F) is satisfied by SGA 4 $\frac{1}{2}$, Théorème de finitude, (1.1). By definition of ℓ -adic cohomology, one therefore obtains an isomorphism

$$H^*A(X) \simeq H^*(X, \mathbb{Q}_\ell).$$

(5.2.3) Suppose X is defined over a countable subfield $k_0 \subset k$, and take k_1 to be the algebraic closure of k_0 in k . One may then choose the projective system of the X_α 's defined over k_0 . The group $\text{Gal}(k_1/k_0)$ then acts on the resulting system, hence, by transport of structure, on $A(X)$. This action is compatible with the action of Galois on the ℓ -adic cohomology of X .

One should keep in mind that $A(X)$ is enormous—this was the price to pay for its depending canonically on X . Consequently, one can hardly speak of continuity for the Galois action. One can only say that Galois acts continuously on each $\Omega(X, \mathbb{Z}/\ell^n)$, given the discrete topology, for every n .

(5.2.4) Now take $k = \mathbb{C}$, and let X_α be a projective system as in (5.2.2). One can then find a projective system of spaces X^α over $X(\mathbb{C})$ and maps

$$a_\alpha : X^\alpha \longrightarrow X_\alpha(\mathbb{C})$$

such that each X^α is the disjoint union of the open sets of an open covering of $X(\mathbb{C})$, and every finite intersection of these open sets is either empty or contractible.

Let $X(\mathbb{C})$ denote the projective system of nerves

$$N^\alpha = \pi_0((X^\alpha/X(\mathbb{C}))^\bullet)$$

of the coverings X^α . Each of these nerves has the homotopy type of $X(\mathbb{C})$. The projective system of the a_α 's gives a morphism from $X(\mathbb{C})$ to the étale system X .

Construction (5.2.5) We construct isomorphisms between the minimal models of the DG $\mathbb{Q}_\ell$ -algebra $A(X)$ and the tensor product with $\mathbb{Q}_\ell$ of Sullivan's minimal models $\mathcal{M}(X(\mathbb{C}))$ of the rational homotopy type of the topological space $X(\mathbb{C})$.

By SGA 4, XVI, (4.1), the morphism

$$\Omega(X, \mathbb{Z}/\ell^n) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}/\ell^n)$$

is a quasi-isomorphism: in cohomology it induces the isomorphisms deduced from the comparison isomorphisms

$$H^i(X, \mathbb{Z}/\ell^n) \simeq H^i(X(\mathbb{C}), \mathbb{Z}/\ell^n).$$

There are also quasi-isomorphisms

$$\Omega(N^\alpha, \mathbb{Z}/\ell^n) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}/\ell^n).$$

Passing to limits, one obtains quasi-isomorphisms

$$\Omega(X, \mathbb{Z}_\ell) \longrightarrow \Omega(X(\mathbb{C}), \mathbb{Z}_\ell) \longleftarrow \Omega(N^\alpha, \mathbb{Z}_\ell).$$

There are moreover quasi-isomorphisms

$$\Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Z}_\ell \longrightarrow \Omega(N^\alpha, \mathbb{Z}_\ell)$$

and

$$\varinjlim_t \Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Q} \longrightarrow \Omega_{\text{DRS}}(N^\alpha) \longrightarrow \Omega_{\text{DRS}}(X(\mathbb{C})).$$

Thus there is a chain of quasi-isomorphisms

$$A(X) \simeq A(X(\mathbb{C})) \longleftarrow (\varinjlim_t \Omega(N^\alpha, \mathbb{Z}) \otimes \mathbb{Q}) \otimes \mathbb{Q}_\ell \longrightarrow \Omega_{\text{DRS}}(N^\alpha) \otimes \mathbb{Q}_\ell \longrightarrow \Omega_{\text{DRS}}(X(\mathbb{C})) \otimes \mathbb{Q}_\ell.$$

Passing to minimal models replaces these quasi-isomorphisms by isomorphisms.

(5.3) Gradings by the weight

(5.3.1) Let A be a DG algebra over a field K of characteristic 0. For us K will be $\mathbb{Q}$ or $\mathbb{Q}_\ell$. Assume $H^0(A) = K$, and that the $H^i(A)$ are finite-dimensional. For the definition of the minimal model $\mathcal{M} \rightarrow A$ of A , I refer to [6]. It follows easily from the inductive construction of $\mathcal{M}$ that there is an increasing exhaustive filtration of $\mathcal{M}$ by subalgebras of finite type $\mathcal{M}_n$, $n \geq 0$, such that:

(*) every automorphism of $\mathcal{M}$ is homotopic to an automorphism preserving each $\mathcal{M}_n$.

If an automorphism σ of $\mathcal{M}$ preserves this filtration, it makes sense to speak of its eigenvalues and of the grading of $\mathcal{M}$ by generalized eigenspaces. After extension of scalars to an algebraic closure $\bar{K}$, the algebra $\mathcal{M} \otimes_K \bar{K}$ is the direct sum of

$$\mathcal{M}(\alpha) = \bigcup_m \ker(\sigma - \alpha)^m, \quad \alpha \in \bar{K}^*,$$

and these subspaces form a grading of type $\bar{K}^*$.

Again by the inductive construction of $\mathcal{M}$, every eigenvalue of σ on $\mathcal{M} \otimes \bar{K}$ is a product of eigenvalues of σ on the $H^i(A) \otimes \bar{K}$, and even on the spherical quotients of the $H^i(A) \otimes \bar{K}$. More precisely, an eigenvalue of σ on $\mathcal{M}^n \otimes \bar{K}$ is a product

$$\prod_i \alpha_i$$

of eigenvalues α_i of σ on $H^{n(i)}(A)$, with $\sum_i n(i) > n$.

Let Γ be the subgroup of $\bar{K}^*$ generated by the eigenvalues of σ . If

$$v : \Gamma \longrightarrow \mathbb{Z}$$

is a homomorphism, the sums

$$\mathcal{M}(n) = \bigoplus_{v(\alpha)=n} \mathcal{M}(\alpha)$$

form a $\mathbb{Z}$ -grading of $\mathcal{M} \otimes \bar{K}$. If, for every eigenvalue α of σ on $H^*(A)$, one has $v(\alpha) \geq 0$, this grading is in nonnegative degrees. If, for every eigenvalue α of σ on $H^i(A)$, one has $v(\alpha) \geq i$, then $\mathcal{M}^n \otimes \bar{K}$ is entirely in degrees $> n$. If Γ is contained in a group Γ' , stable under $\text{Gal}(\bar{K}/K)$, and v extends to a Galois-invariant homomorphism

$$v' : \Gamma' \longrightarrow \mathbb{Z},$$

then this grading of $\mathcal{M} \otimes \bar{K}$ is defined over K .

(5.3.2) The case of interest is the following. Let $K = \mathbb{Q}_\ell$. For a suitable integer $q > 1$, a power of a prime, one may take Γ' to be the group of algebraic numbers α all of whose complex conjugates have the same absolute value, this absolute value being of the form $q^{n(\alpha)/2}$ for an integer $n(\alpha)$. Put

$$v'(\alpha) = n(\alpha).$$

The ℓ -adic gradings by weight mentioned in [3] are obtained in this way, taking σ to be a Frobenius. Here are some examples.

(5.3.3) Let X be a complex algebraic variety. It can be defined over a subfield $k_0 \subset \mathbb{C}$ of finite type over $\mathbb{Q}$, i.e. it is obtained by extension of scalars from some X_0/k_0 . This permits $\text{Gal}(\bar{k}_0/k_0)$ to act on $A(X)$, defined as in (5.2). Let $\mathcal{M}$ be the Sullivan minimal model of the rational homotopy type of the topological space $X(\mathbb{C})$. By (5.2.5), $\mathcal{M} \otimes \mathbb{Q}_\ell$ is a minimal model of $A(X)$. More precisely, there is a natural homotopy class of quasi-isomorphisms

$$\varphi : \mathcal{M} \otimes \mathbb{Q}_\ell \longrightarrow A(X),$$

which induce in cohomology the natural isomorphisms

$$H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_\ell \simeq H^*(X, \mathbb{Q}_\ell).$$

For every automorphism s of $A(X)$, one therefore has an automorphism $\tilde{s}$ of $\mathcal{M} \otimes \mathbb{Q}_\ell$, well-determined up to homotopy, such that $s\varphi$ is homotopic to $\varphi\tilde{s}$. It may be chosen to preserve a filtration as in (5.3.1). This applies in particular to $s \in \text{Gal}(\bar{k}_0/k_0)$. If s is chosen to be a Frobenius as in [3], the hypotheses of (5.3.2) are satisfied by $\tilde{s}$, and the corresponding grading of $\mathcal{M} \otimes \mathbb{Q}_\ell$ induces on cohomology a grading which splits the tensor product with $\mathbb{Q}_\ell$ of the weight filtration of Hodge theory on $H^*(X(\mathbb{C}), \mathbb{Q})$. A Sullivan argument then gives a grading of $\mathcal{M}$ with the same properties.

More precisely:

Theorem 12 (5.3.4). *There exist gradings of $\mathcal{M}$ compatible with the weight filtration of*

$$H^*(\mathcal{M}) = H^*(X(\mathbb{C}), \mathbb{Q})$$

in the sense that

$$W_n H^*(X(\mathbb{C}), \mathbb{Q}) = \bigoplus_{m \leq n} H^*(X(\mathbb{C}), \mathbb{Q})_m.$$

Any two such gradings are conjugate by an automorphism of $\mathcal{M}$ inducing the identity on

$$\text{Gr}^W H^*(X(\mathbb{C}), \mathbb{Q}).$$

Choose a filtration $(\mathcal{M}_n)$ of $\mathcal{M}$ as in (5.3.1). Gradings of $\mathcal{M}$ compatible with the filtration by the $\mathcal{M}_n$'s are the same as homomorphisms from $\mathbb{G}_m$ into the pro-algebraic group

$$G_1 = \text{Aut}((\mathcal{M}, \mathcal{M}_n)) = \varprojlim_n \text{Aut}((\mathcal{M}_n)),$$

of automorphisms of $\mathcal{M}$ preserving the filtration. Let G_2 be the subgroup of $G_1 \times \mathbb{G}_m$ formed by the couples (g, λ) such that g respects the weight filtration of $H^*(X(\mathbb{C}), \mathbb{Q})$ and acts on $\text{Gr}_i^W H^*$ as multiplication by λ^i . One checks that the kernel of the homomorphism

$$G_2 \longrightarrow \mathbb{G}_m, \quad (g, \lambda) \mapsto \lambda,$$

is unipotent:

$$1 \longrightarrow U \longrightarrow G_2 \longrightarrow \mathbb{G}_m \longrightarrow 1.$$

After extension of scalars to $\mathbb{Q}_\ell$, this morphism has a section: this is the existence of the gradings of (5.3.3). It is therefore surjective as a morphism of schemes over $\mathbb{Q}$. The pro-algebraic group G_2 is thus, over $\mathbb{Q}$, an extension of $\mathbb{G}_m$ by U . Such an extension is always trivial, and any two sections are U -conjugate. This proves existence and uniqueness when only gradings inducing a grading on every $\mathcal{M}_n$ are considered.

To conclude, it remains to prove the following technical lemma, whose somewhat tedious verification is left to the reader.

Lemma 23 (5.3.5). *For every finite set of automorphisms of $\mathcal{M}$, there exists a filtration as in (5.3.1) stable under all of these automorphisms.*

(5.3.6) For X an algebraic variety over an arbitrary algebraically closed field, one can again define a weight filtration on $H^*(X, \mathbb{Q}_\ell)$ in terms of generalized eigenspaces of suitable Frobenius automorphisms. The preceding constructions give gradings of the minimal model of $A(X)$, inducing on cohomology a grading which splits the weight filtration.

Corollary 12 (5.3.7). *If X is proper and smooth, a minimal model of $A(X)$ is a minimal model of $H^*(X, \mathbb{Q}_\ell)$, provided with the DG structure for which $d = 0$.*

Let W be a grading by weights of a minimal model $\mathcal{M}$ of $A(X)$. Since $H^i(X, \mathbb{Q}_\ell)$ is pure of weight i , $\mathcal{M}^n$ is entirely in degrees $> n$, in the sense of (5.3.1). Hence

$$dW_n(\mathcal{M}^n) \subset W_n(\mathcal{M}^{n+1}) = 0.$$

Since $W_n(\mathcal{M}^n)$ consists of cycles, there is a natural projection

$$W_n(\mathcal{M}^n) \longrightarrow H^n(X, \mathbb{Q}_\ell).$$

It is surjective, because H^n is pure of weight n . Let

$$\varphi_n : \mathcal{M}^n \longrightarrow H^n(X, \mathbb{Q}_\ell)$$

be the composite of this projection with the projection of $\mathcal{M}^n$ onto $W_n(\mathcal{M}^n)$, which kills the $W_i(\mathcal{M}^n)$ for $i > n$. It is easy to check that

$$\varphi = (\varphi_n)$$

is a quasi-isomorphism of DG algebras. The corollary follows.

(5.3.8) For X normal, one similarly obtains a theorem analogous to Morgan's theorem for the system of quotients of $\pi_1(X)$ which are unipotent ℓ -adic Lie groups; see [9].

Let us point out some questions left aside.

(5.3.9) For $\mathcal{M}$ a minimal model and $(\mathcal{M}_n)$ a filtration as in (5.3.1), let G be the pro-algebraic group of automorphisms of $\mathcal{M}$ preserving the $\mathcal{M}_n$'s, and let H be the subgroup of those which are homotopic to the identity. Put

$$\mathrm{Aut}_h(\mathcal{M}) = G/H.$$

It is a pro-algebraic group.

If X is an algebraic variety over an algebraically closed field k , obtained by extension of scalars from X_0/k_0 with k_0 countable, the natural action of $\mathrm{Gal}(\bar{k}_0/k_0)$ on $A(X)$ gives a homomorphism

$$\mathrm{Gal}(\bar{k}_0/k_0) \longrightarrow \mathrm{Aut}_h(\mathcal{M})(\mathbb{Q}_\ell). \quad (5.3.9.1)$$

It would be necessary to verify that this homomorphism is continuous, i.e. that for every algebraic quotient Q , of finite type, of $\mathrm{Aut}_h(\mathcal{M})$, the homomorphism from $\mathrm{Gal}(\bar{k}_0/k_0)$ to $Q(\mathbb{Q}_\ell)$ deduced from (5.3.9.1) is continuous when $Q(\mathbb{Q}_\ell)$ is equipped with its usual topology.

One can hope for more precise results, but I do not think there is any reason to expect a reasonable action of $\mathrm{Gal}(\bar{k}_0/k_0)$ on $\mathcal{M}$ itself.

(5.3.10) It would be necessary to verify that the increasing filtrations attached to the gradings by weight of (5.3.3) or (5.3.6) are all conjugate to each other under the group of automorphisms of $\mathcal{M}$, the ℓ -adic minimal model, homotopic to the identity. In the setting of (5.3.3), where $\mathcal{M}$ is the ℓ -adic completion of Sullivan's rational minimal model $\mathcal{M}_\mathbb{Q}$, their conjugacy class should be defined over $\mathbb{Q}$, equivalently should be the orbit of a filtration deduced by extension of scalars from a filtration of $\mathcal{M}_\mathbb{Q}$.

6. The formalism of mixed sheaves

(6.1) Stability

(6.1.1) In this section we establish, in the following contexts, that the category of mixed sheaves is stable under the usual operations.

- a) Equal characteristic p : we consider only schemes of finite type over $\mathbb{F}_p$; the word sheaf means a Weil sheaf.
- b) Equal characteristic 0: more precisely, we consider schemes of finite type over $\mathbb{Z}$, and sheaves in the sense of constructible $\mathbb{Q}_\ell$ -sheaves, while allowing ourselves at every instant to replace X by $X[1/n]$.

As regards schemes, this amounts to considering schemes of finite type over $\mathbb{Q}$: their category is the inductive 2-limit of the categories of schemes of finite type over the $\mathbb{Z}[1/n]$. For sheaves the situation is not the same. If X is of finite type over $\mathbb{Z}$, I do not know whether every constructible sheaf on $X_{\mathbb{Q}}$ comes from a constructible sheaf on some $X[1/n]$. However, if $\mathcal{F}$ and $\mathcal{G}$ are two constructible sheaves on X , one has

$$\mathrm{Hom}_{X_{\mathbb{Q}}}(\mathcal{F}, \mathcal{G}) = \varinjlim_n \mathrm{Hom}_{X[1/n]}(\mathcal{F}, \mathcal{G}).$$

In other words, the inductive 2-limit of the categories of sheaves on the $X[1/n]$ identifies with a full subcategory of the category of sheaves on $X_{\mathbb{Q}}$.

The following terminology gives uniform statements in cases a) and b). If $X_{\mathbb{Q}}$ is a scheme of finite type over $\mathbb{Q}$, obtained by base change from a scheme X of finite type over $\mathbb{Z}$, and if $\mathcal{F}_{\mathbb{Q}}$ is a sheaf on $X_{\mathbb{Q}}$, we say that $\mathcal{F}_{\mathbb{Q}}$ is mixed, pure, etc., if it is obtained from a sheaf $\mathcal{F}$ on some $X[1/n]$, and the restriction of $\mathcal{F}$ to $X[1/nm]$ is mixed, pure, etc., for a suitable m . The results in case b) will be stated as variants of those in case a). One could also give potential variants, in the sense of (3.4.10); we shall not spell them out.

Theorem 13 (6.1.2). *Let $f : X \rightarrow Y$ be a morphism of schemes of finite type over $\mathbb{F}_p$ (respectively over $\mathbb{Q}$), and let $\mathcal{F}$ be a mixed sheaf on X . Then the sheaves $R^i f_* \mathcal{F}$ are mixed.*

In view of the generic base-change theorem (SGA 4 $\frac{1}{2}$, Th. finitude, (1.9)), this follows from the following lemma.

Lemma 24 (6.1.3). *Let S be a scheme of finite type over $\mathbb{Z}[1/\ell]$, let $f : X \rightarrow Y$ be a morphism of S -schemes of finite type, and let $\mathcal{F}$ be a mixed sheaf on X . Then there is a dense open $U \subset S$ over which the sheaves $R^i f_* \mathcal{F}$ are mixed.*

We repeat the proof of SGA 4 $\frac{1}{2}$, Th. finitude, (2.1)–(2.8).

(6.1.4) First suppose that X is smooth over $Y = S$, that $\mathcal{F}$ is lisse, and that the $R^i f_* \mathcal{F}$ are lisse. The sheaves $R^i f_* \mathcal{F}$ are then obtained, by Poincaré duality, from the $R^i f_! \mathcal{F}$, to which (3.3.1) applies.

(6.1.5) If X is smooth over $Y = S$ and $\mathcal{F}$ is lisse, we reduce to (6.1.4) after replacing S by a suitable dense open.

(6.1.6) We prove by induction on n the assertion $(*)_n$: the conclusion of (6.1.3) holds when S is integral, with generic point η , when f is an open immersion with dense image, and when $\dim X_{\eta} < n$. For $n = 0$, after shrinking S , one has $X = Y$, and $(*)_0$ is evident. Suppose $(*)_{n-1}$, and prove $(*)_n$. In $(*)_{n-1}$, one may replace “open immersion” by “immersion”, by factoring an immersion as an open immersion followed by a closed immersion.

Lemma 25 (6.1.7). *After shrinking S , the conclusion of (6.1.3) holds over an open $Y' \subset Y$, whose complement Y_1 is finite over S .*

The assertion is local on Y , and we may suppose Y affine, say $Y \subset \mathbb{A}_S^n$. The induction hypothesis applies to the projections. Thus, for each i , there is a dense open U_i of $\mathbb{A}_S^{n-1}$ such that the conclusion of (6.1.3) holds over $\text{pr}_i^{-1}(U_i)$. It therefore holds over their union, which gives (6.1.7).

(6.1.8) We prove $(*)_n$ when X is smooth over S and $\mathcal{F}$ is lisse. Since the problem is local on Y , we may suppose Y affine, and then projective, by replacing Y by its closure in a projective space. Let

$$i : Y_1 \hookrightarrow Y, \quad j : Y' \hookrightarrow Y$$

be as in (6.1.7), after shrinking S . Formally applying Rb_* to the triangle defined by

$$0 \longrightarrow j_* j^* Rf_* \mathcal{F} \longrightarrow Rf_* \mathcal{F} \longrightarrow i_* i^* Rf_* \mathcal{F} \longrightarrow 0,$$

one obtains a triangle

$$Rb_* j_* j^* Rf_* \mathcal{F} \longrightarrow Ra_* \mathcal{F} \longrightarrow Rb_* i_* i^* Rf_* \mathcal{F} \longrightarrow .$$

The cohomology sheaves of the first two terms are mixed on a dense open of S , by (3.3.1) for the first term and by (6.1.5) for the second. Hence the cohomology sheaves of $b_* i^* Rf_* \mathcal{F}$, then those of $i^* Rf_* \mathcal{F}$, and finally those of $Rf_* \mathcal{F}$, are mixed. To justify this derived-category argument, one writes $\mathcal{F}$ as coming from the projective limit of a projective system of locally constant sheaves of free R/m^n -modules, compatible by reduction, as in (1.1.1), and repeats the preceding argument for that system.

(6.1.9) We prove $(*)_n$ in general. The use of a derived category is justified as above. We first reduce to the case where X contains a dense open V smooth over S . When S is the spectrum of a perfect field, it suffices to replace X by X_{red} . In general, after shrinking S , making a finite radicial surjective base change $S' \rightarrow S$, and replacing X and Y by their reductions, nothing is changed for the étale topology. Shrinking V , we may also suppose $\mathcal{F}$ lisse on V . Apply (6.1.8) to

$$V \xrightarrow{j} X \xrightarrow{f} Y.$$

Define A by the triangle

$$\mathcal{F} \longrightarrow Rj_* j^* \mathcal{F} \longrightarrow A \longrightarrow .$$

The cohomology sheaves of A have support in $X - V$, and $\dim(X - V)_\eta < n$. By induction we may suppose $Rf_* A$ mixed. Applying Rf_* to the triangle above, one obtains

$$Rf_* \mathcal{F} \longrightarrow R(fj)_* j^* \mathcal{F} \longrightarrow Rf_* A \longrightarrow ,$$

and two of the three terms are mixed. The third is therefore mixed.

(6.1.10) To prove (6.1.2), the problem is local on Y , which we may suppose affine. Taking an affine covering of X and using the Leray spectral sequence, we reduce to the case where X is also affine. We then factor f as an open immersion followed by a proper morphism, $f = gj$. Thus $Rf_* = Rg_* Rj_*$. The open immersion is covered by some $(*)_n$, and the proper morphism by (3.3.1).

Corollary 13 (6.1.11). *The four operations $R\text{Hom}$, Rf_* , f^* , and $Rf^!$ transform mixed sheaves into mixed sheaves. The same stability holds for local Ext sheaves.*

(6.1.12) We next deduce from (6.1.2) an analogous stability for sheaves of vanishing cycles. Let S be a smooth curve over $\mathbb{F}_q$ (respectively over $\mathbb{Q}$), let s be a closed point of S , let X be an S -scheme of finite type, and let $\mathcal{F}$ be a sheaf on X . After the base change $S(s) \rightarrow S$, one obtains $(X', \mathcal{F}')$ over the henselian trait $S(s)$. Let η be the generic point of $S(s)$, let $\bar{s}$ be a geometric point above s , and let $\bar{\eta}$ be a geometric generic point of the strict localization $S_{(\bar{s})}$. The sheaves of vanishing cycles $R^i\Psi(\mathcal{F})$ are sheaves on $X_{\bar{s}}$, endowed with a continuous action of $\text{Gal}(\bar{\eta}/\eta)$ above its action on $X_{\bar{s}}$, via $\text{Gal}(\bar{s}/s)$. The restriction of this action to inertia is automatically quasi-unipotent.

For a sheaf $\mathcal{G}$ with such an action, and with quasi-unipotent inertia, we define purity and mixedness by passing to the unipotent part of the inertia action and to the corresponding nilpotent logarithm. With this convention, the vanishing-cycle sheaves of a mixed sheaf are mixed. This follows from (6.1.2), by the standard description of nearby cycles after replacing the trait by finite extensions and using the stability of mixed sheaves under direct image.

(6.2) Pure complexes

In this section we place ourselves in the situation of (6.1.1) a).

(6.2.1) For every scheme X of finite type over $\mathbb{F}_q$, let

$$a : X \longrightarrow \text{Spec}(\mathbb{F}_q)$$

and put

$$K_X = Ra^!\mathbb{Q}_\ell.$$

This is the dualizing complex. If X is smooth and purely of dimension n , then

$$K_X \simeq \mathbb{Q}_\ell(n)[2n].$$

For $K \in D_c^b(X)$, define

$$DK = R\mathcal{H}om(K, K_X).$$

One has the usual formalism of duality: D exchanges $Rf_!$ and Rf_* , and for proper f ,

$$DRf_*K \simeq Rf_*DK.$$

(6.2.2) Let $K \in D_c^b(X)$. We say that K is *punctually pure of weight n* if, for every i , the cohomology sheaf $\mathcal{H}^i K$ is punctually pure of weight $n + i$. We say that K is *pure of weight n* if both K and DK are punctually pure of weight n .

(6.2.3) If K is punctually pure of weight n , then the usual truncations show that the spectral sequence of cohomology sheaves is compatible with weights. In particular, extensions between pieces of distinct forced weights split in the cases needed below, and the duality condition in the definition of purity is stable under the operations already considered.

(6.2.4) This definition extends the one for lisse sheaves. If X is smooth of pure dimension n , and if $\mathcal{F}$ is a lisse sheaf punctually pure of weight m , then $\mathcal{F}$ placed in degree 0 is pure of weight m , for

$$D\mathcal{F} \simeq \mathcal{F}^\vee(n)[2n],$$

and the shift and Tate twist have the expected effect on weights.

(6.2.5) Examples and elementary consequences are as follows.

- a) If K is pure of weight n , then $K[r](s)$ is pure of weight $n + r - 2s$.
- b) If $j : U \hookrightarrow X$ is an open immersion and $\mathcal{F}$ is lisse, punctually pure of weight n on U , then in favorable cases $j_!\mathcal{F}$ and $j_*\mathcal{F}$ are pure of the expected weight, by duality.
- c) For X smooth, $j : U \hookrightarrow X$ the complement of a smooth divisor D , and $\mathcal{F}$ lisse, punctually pure of weight n on U , tamely ramified along D , the sheaf $j_*\mathcal{F}$ is again pure of weight n .

Proposition 8 (6.2.6). *If $f : X \rightarrow Y$ is proper and $K \in D_c^b(X)$ is pure, then Rf_*K is pure of the same weight.*

Apply (6.2.3) to K and to DK , using

$$DRf_*K = Rf_*DK$$

because f is proper.

Variant 3 (6.2.7). *For every scheme X of finite type over $\mathbb{Z}[1/\ell]$, with*

$$a : X \longrightarrow \mathrm{Spec}(\mathbb{Z}[1/\ell]),$$

put

$$K_X = Ra^!\mathbb{Q}_\ell.$$

Again one has a duality functor

$$D'K = R\mathcal{H}om(K, K_X)$$

and a definition of purity parallel to (6.2.4). If X is of finite type over $\mathbb{F}_p$, the two duality functors differ only by a shift and a twist, and the preceding definition of purity generalizes (6.2.4). Proposition (6.2.6) remains valid.

(6.2.8) Let k be an algebraically closed field, let S be the spectrum of the henselization of $k[T]$ at (T) , let s be its closed point, η its generic point, and $\bar{\eta}$ a geometric generic point. Let $f : X \rightarrow S$ be proper.

A complex $K \in D_c^b(X)$ is called *potentially pure* if $(X/S, K)$ comes from an arithmetic situation as follows:

- a) $\mathrm{Spec}(k)$ is a geometric generic point of an integral scheme A_0 of finite type over $\mathbb{Z}[1/\ell]$;
- b) S comes, by extension of scalars to k and henselization, from a smooth curve S_0 over A_0 with a section;
- c) f comes from a proper morphism $f_0 : X_0 \rightarrow S_0$;
- d) K comes from a pure complex $K_0 \in D_c^b(X_0)$.

The following theorem generalizes (3.6.1).

Theorem 14 (6.2.9, local invariant-cycle theorem for pure complexes). *With the preceding notation, let K be a potentially pure complex on X . Then the specialization morphism*

$$\mathrm{sp}^* : H^*(X_{\bar{s}}, K) \longrightarrow H^*(X_{\bar{\eta}}, K)^{\mathrm{Gal}(\bar{\eta}/\eta)}$$

is surjective.

The proof is parallel to that of (3.6.1). One has the same cross diagram, with terms

$$H^{i-1}(X_{\bar{s}}, K)(-1), \quad H^i(X_{\bar{\eta}}, K), \quad H^i(X_{\bar{s}}, K),$$

and duality identifies the relevant upper and lower terms. The analogues of (3.6.2) and (3.6.3) are proved by replacing the references to (3.3.9) and (3.3.4) by references to (6.2.6) and (6.2.3). One concludes as in (3.6).

(6.2.10) In the remainder of this section the arguments are geometric, except for references to (6.2.9) and to (3.4). Let k be algebraically closed. For every scheme X of finite type over k , with $a : X \rightarrow \operatorname{Spec}(k)$, define

$$K_X = Ra^!\mathbb{Q}_\ell, \quad D(K) = R\mathcal{H}om(K, K_X).$$

Let $X \subset P$ be projective. Hyperplane sections of X are parametrized by the dual projective space $\check{P}$, and each line in $\check{P}$ parametrizes a pencil of hyperplane sections:

$$\begin{array}{ccc} & Z & \\ q \swarrow & & \searrow f \\ X & & \check{P}. \end{array}$$

For $K \in D_c^b(X)$, the sheaves $R^i f_* q^* K$ are lisse on a dense open $U \subset \check{P}$. A hyperplane section Y_u , for $u \in U$, will be called general. The group $\pi_1(U, u)$ acts on $H^*(Y_u, K)$. If a line D is sufficiently general and $u \in V = U \cap D$, the image of $\pi_1(V, u)$ in $\operatorname{GL} H^*(Y_u, K)$ coincides with that of $\pi_1(U, u)$. In particular,

$$H^*(Y_u, K)^{\pi_1(U, u)} = H^*(Y_u, K)^{\pi_1(V, u)}. \quad (6.2.10.2)$$

Proposition 9 (6.2.11). *With the preceding notation, let n be an integer, and suppose that for every i , the support of $\mathcal{H}^i(DK[-2n-2])$ has dimension $< n+1-i$. Then:*

i) For every hyperplane section Y , the map

$$H^i(X, K) \longrightarrow H^i(Y, K)$$

is an isomorphism for $i < n$, and

$$H^n(X, K) \longrightarrow H^n(Y, K)$$

is injective.

ii) For a general pencil and for $u \in D$, the image of $H^n(X, K)$ in $H^n(Y_u, K)$ coincides with the image of the compactly supported cohomology of the base of the pencil with coefficients in the sheaf $R^n f_(i^* q^* K)$.*

The proof is parallel to that of (4.1.6). One first checks weak Lefschetz,

$$H_Y^i(X, K) = 0 \quad (i < n).$$

By duality this vanishing is reduced to the assumed dimension condition on the supports of the cohomology sheaves of $DK[-2n-2]$. Assertion i) follows from the long exact cohomology sequence. For a sufficiently general hyperplane section Y , the restriction of K to Y again satisfies the hypothesis, with n replaced by $n-1$. This follows from generic acyclicity applied to $Z/\check{P}$ and $q^* K$, using

$$D(K|Y) \simeq (DK)|Y(-1)[-2].$$

One then applies (4.3.7) as in (4.3.8), and concludes as in (4.3.9).

Corollary 14 (6.2.12, global invariant-cycle theorem for pure complexes). *Under the hypotheses of (6.2.11), if K is potentially pure, then, for Y_u a general hyperplane section of X , $H^n(X, K)$ identifies with the subgroup*

$$H^n(Y_u, K)^{\pi_1(U, u)}$$

of monodromy invariants.

It suffices to verify this for a sufficiently general hyperplane section. For a sufficiently general pencil D , the complex i^*q^*K on the total space of the pencil is potentially pure. Applying (6.2.9) at each point of $D - V$ gives the surjectivity onto the invariant part. By (6.2.11), the image of $H^n(X, K)$ is exactly that invariant part.

Theorem 15 (6.2.13). *Let $X \subset P$ be a projective scheme, let η be the first Chern class of $\mathcal{O}(1)$, and let $K \in D_c^b(X)$ be a potentially pure complex. Identify $\mathbb{Z}_\ell$ with $\mathbb{Z}_\ell(1)$ over k , so that η may be regarded as an element of $H^2(X, \mathbb{Z}_\ell)$. Let n be an integer, and suppose that K and $DK[-2n]$ satisfy the following condition: for every i , the support of $\mathcal{H}^i$ has dimension $\leq n - i$ (with $\dim \emptyset = -\infty$). Then, for every $i \geq 0$, the iterated cup product*

$$\eta^i : H^{n-i}(X, K) \longrightarrow H^{n+i}(X, K)$$

is an isomorphism.

The proof is parallel to that of (4.1.1), by induction on $\dim X$. If X has dimension zero and $\mathcal{H}^i K \neq 0$, the hypothesis forces $i = n$, so the assertion is immediate. If Y is a sufficiently general hyperplane section, the restriction of K to Y satisfies the same hypotheses with n replaced by $n - 1$. The induction reduces the problem to the morphism

$$H^{n-1}(X, K) \longrightarrow H^{n-1}(Y, K) \longrightarrow H^{n+1}(X, K).$$

The second arrow is transposed to the restriction morphism for $DK[-2n]$. It remains only to check that the perfect duality between the two cohomology groups on Y induces a perfect duality between the invariant parts. This follows from (4.1.4), since the representation of $\pi_1(U, u)$ on $H^*(Y_u, K)$ is semisimple by (6.2.6) and (3.4.1)(iii).

Added in proof. Let X be a complex algebraic variety, assumed irreducible and normal. The intermediate cohomology of MacPherson and Goresky can be defined as the hypercohomology of X with coefficients in a certain complex of sheaves I_X . The construction of I_X has an analogue in étale cohomology, over an arbitrary base field. O. Gabber has just proved the purity of I_X . His theorem allows one to apply the results of this section to MacPherson–Goresky cohomology.

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